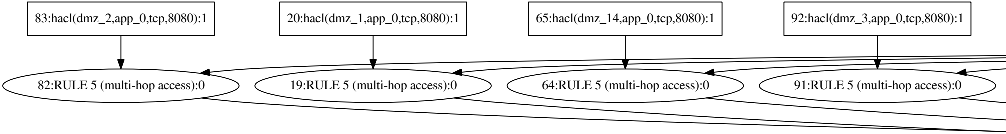
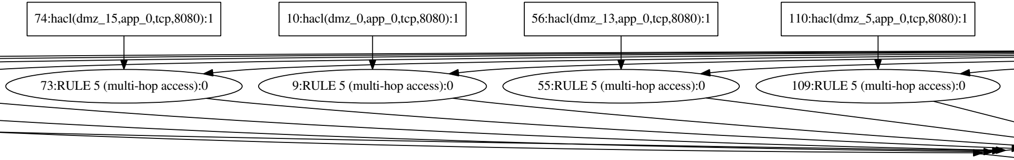
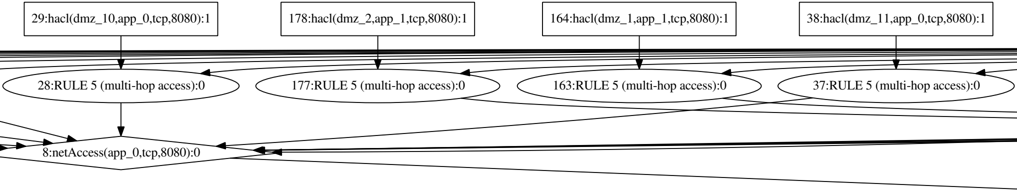
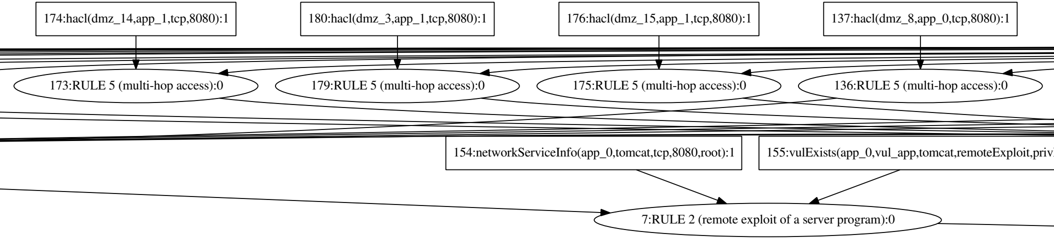
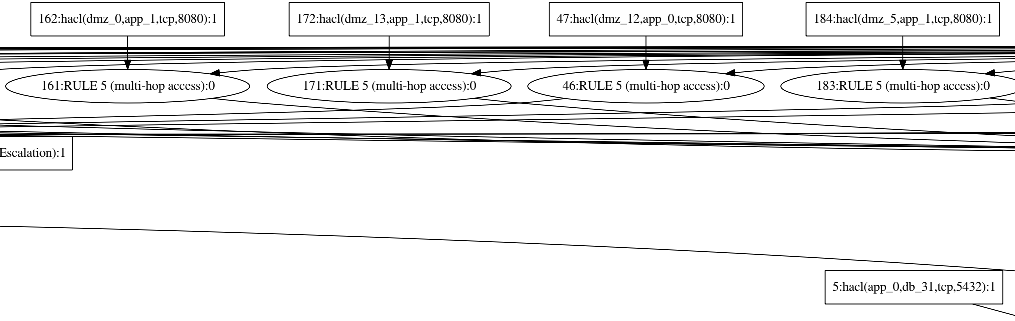
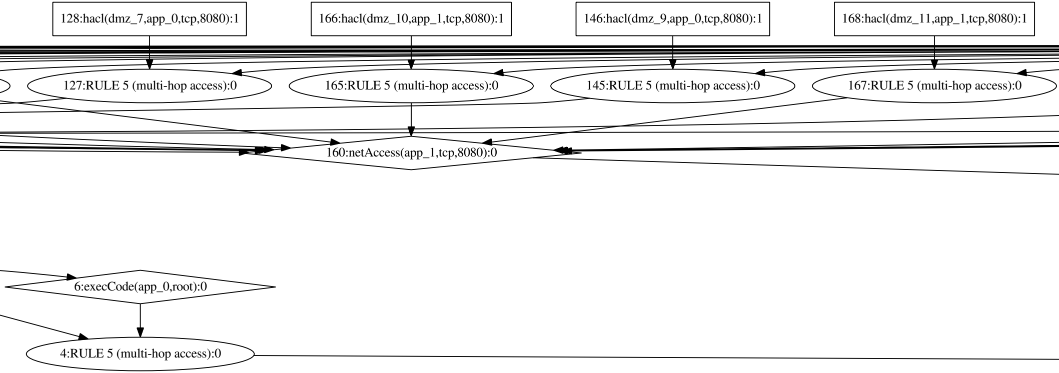

83:hacl(dmz_2,app_0,tcp,8080):1
20:hacl(dmz_1,app_0,tcp,8080):1
65:hacl(dmz_14,app_0,tcp,8080):1
92:hacl(dmz_3,app_0,tcp,8080):1
82:RULE 5 (multi-hop access):0
19:RULE 5 (multi-hop access):0
64:RULE 5 (multi-hop access):0
91:RULE 5 (multi-hop access):0

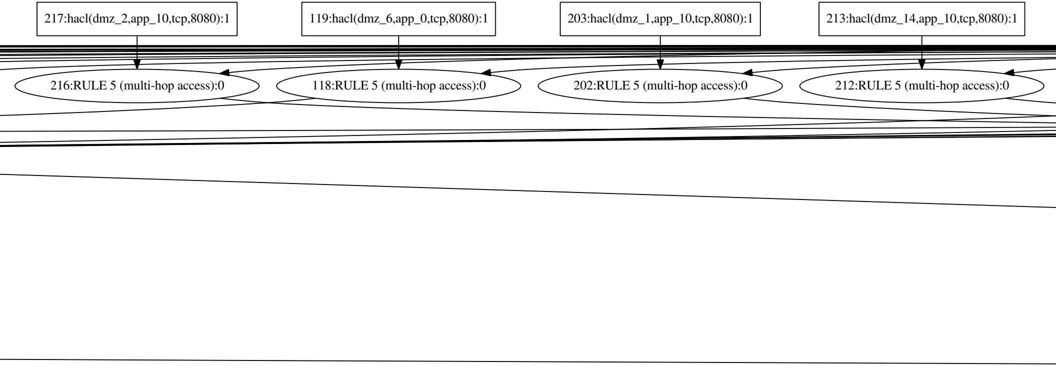

74:hacl(dmz_15,app_0,tcp,8080):1
10:hacl(dmz_0,app_0,tcp,8080):1
56:hacl(dmz_13,app_0,tcp,8080):1
110:hacl(dmz_5,app_0,tcp,8080):1
73:RULE 5 (multi-hop access):0
9:RULE 5 (multi-hop access):0
55:RULE 5 (multi-hop access):0
109:RULE 5 (multi-hop access):0

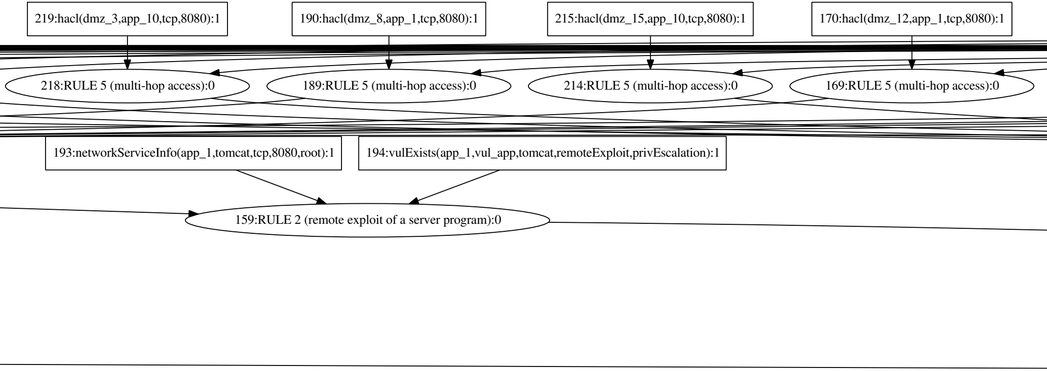

29:hacl(dmz_10,app_0,tcp,8080):1
178:hacl(dmz_2,app_1,tcp,8080):1
164:hacl(dmz_1,app_1,tcp,8080):1
38:hacl(dmz_11,app_0,tcp,8080):1
28:RULE 5 (multi-hop access):0
177:RULE 5 (multi-hop access):0
163:RULE 5 (multi-hop access):0
37:RULE 5 (multi-hop access):0
8:netAccess(app_0,tcp,8080):0

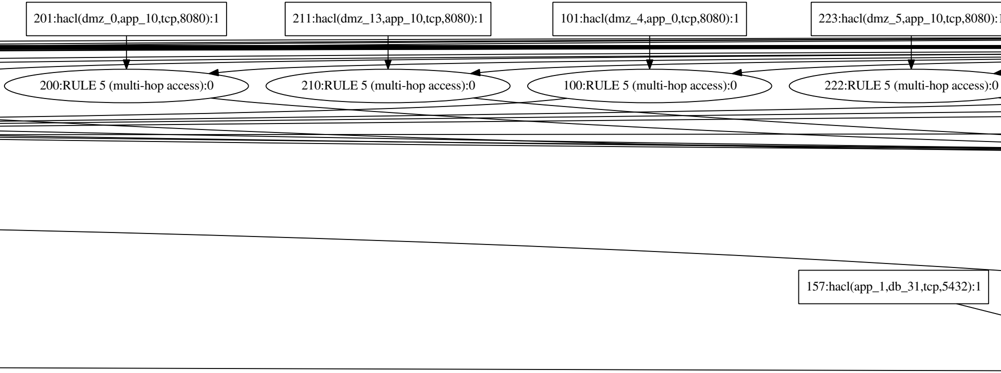

174:hacl(dmz_14,app_1,tcp,8080):1
180:hacl(dmz_3,app_1,tcp,8080):1
176:hacl(dmz_15,app_1,tcp,8080):1
137:hacl(dmz_8,app_0,tcp,8080):1
173:RULE 5 (multi-hop access):0
179:RULE 5 (multi-hop access):0
175:RULE 5 (multi-hop access):0
136:RULE 5 (multi-hop access):0
154:networkServiceInfo(app_0,tomcat,tcp,8080,root):1
155:vulExists(app_0,vul_app,tomcat,remoteExploit,priv
7:RULE 2 (remote exploit of a server program):0

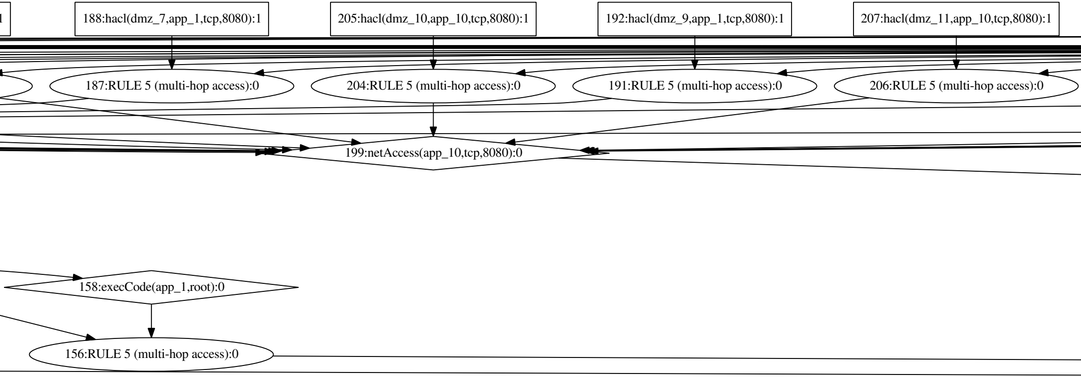

162:hacl(dmz_0,app_1,tcp,8080):1
172:hacl(dmz_13,app_1,tcp,8080):1
47:hacl(dmz_12,app_0,tcp,8080):1
184:hacl(dmz_5,app_1,tcp,8080):1
161:RULE 5 (multi-hop access):0
171:RULE 5 (multi-hop access):0
46:RULE 5 (multi-hop access):0
183:RULE 5 (multi-hop access):0
Escalation):1
5:hacl(app_0,db_31,tcp,5432):1

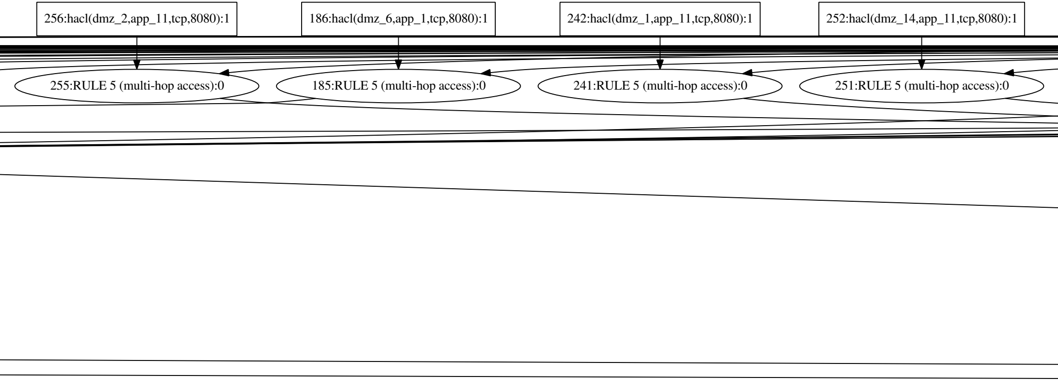

128:hacl(dmz_7,app_0,tcp,8080):1
166:hacl(dmz_10,app_1,tcp,8080):1
146:hacl(dmz_9,app_0,tcp,8080):1
168:hacl(dmz_11,app_1,tcp,8080):1
127:RULE 5 (multi-hop access):0
165:RULE 5 (multi-hop access):0
145:RULE 5 (multi-hop access):0
167:RULE 5 (multi-hop access):0
160:netAccess(app_1,tcp,8080):0
6:execCode(app_0,root):0
4:RULE 5 (multi-hop access):0

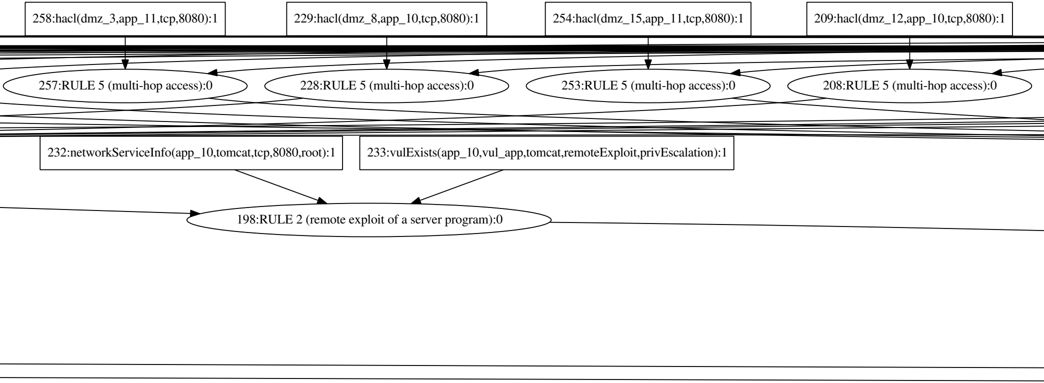

217:hacl(dmz_2,app_10,tcp,8080):1
119:hacl(dmz_6,app_0,tcp,8080):1
203:hacl(dmz_1,app_10,tcp,8080):1
213:hacl(dmz_14,app_10,tcp,8080):1
216:RULE 5 (multi-hop access):0
118:RULE 5 (multi-hop access):0
202:RULE 5 (multi-hop access):0
212:RULE 5 (multi-hop access):0

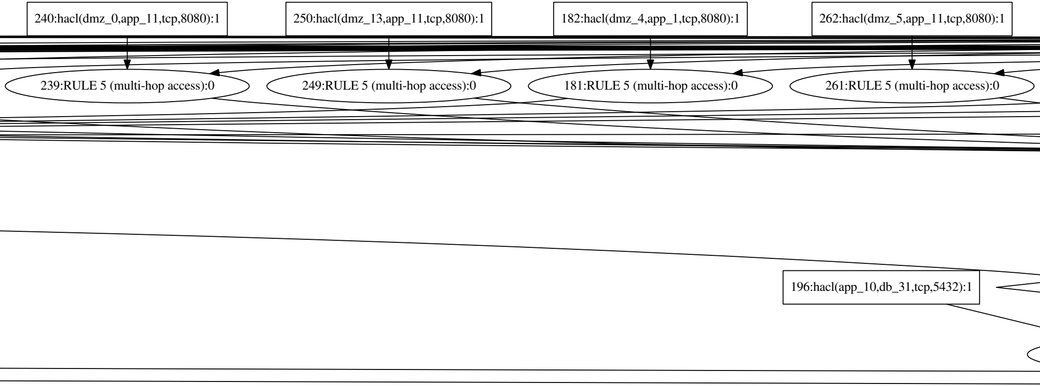

219:hacl(dmz_3,app_10,tcp,8080):1
190:hacl(dmz_8,app_1,tcp,8080):1
215:hacl(dmz_15,app_10,tcp,8080):1
170:hacl(dmz_12,app_1,tcp,8080):1
218:RULE 5 (multi-hop access):0
189:RULE 5 (multi-hop access):0
214:RULE 5 (multi-hop access):0
169:RULE 5 (multi-hop access):0
193:networkServiceInfo(app_1,tomcat,tcp,8080,root):1
194:vulExists(app_1,vul_app,tomcat,remoteExploit,privEscalation):1
159:RULE 2 (remote exploit of a server program):0

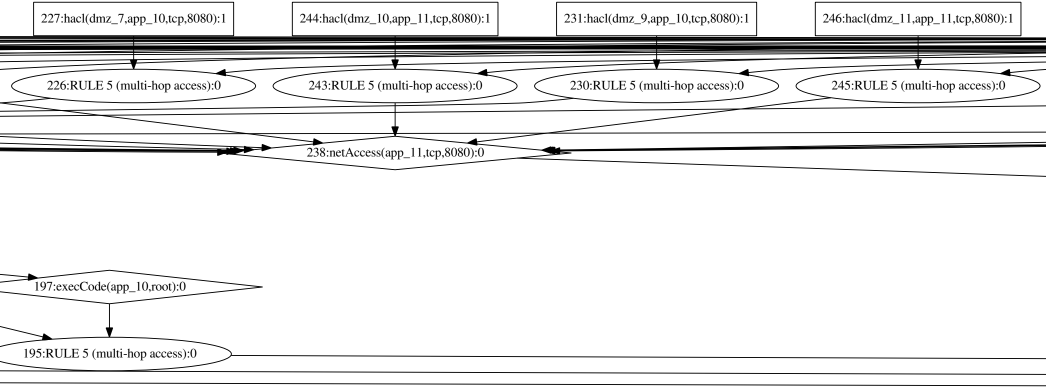

201:hacl(dmz_0,app_10,tcp,8080):1
211:hacl(dmz_13,app_10,tcp,8080):1
101:hacl(dmz_4,app_0,tcp,8080):1
223:hacl(dmz_5,app_10,tcp,8080):
200:RULE 5 (multi-hop access):0
210:RULE 5 (multi-hop access):0
100:RULE 5 (multi-hop access):0
222:RULE 5 (multi-hop access):0
157:hacl(app_1,db_31,tcp,5432):1

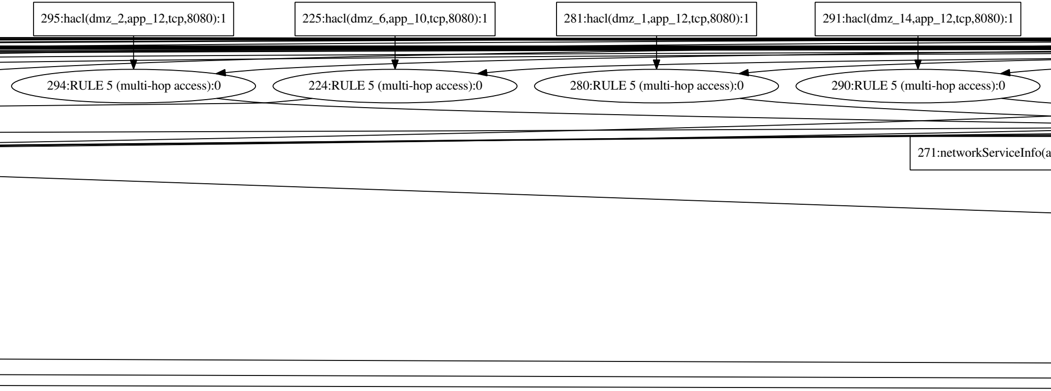

188:hacl(dmz_7,app_1,tcp,8080):1
205:hacl(dmz_10,app_10,tcp,8080):1
192:hacl(dmz_9,app_1,tcp,8080):1
207:hacl(dmz_11,app_10,tcp,8080):1
187:RULE 5 (multi-hop access):0
204:RULE 5 (multi-hop access):0
191:RULE 5 (multi-hop access):0
206:RULE 5 (multi-hop access):0
199:netAccess(app_10,tcp,8080):0
158:execCode(app_1,root):0
156:RULE 5 (multi-hop access):0

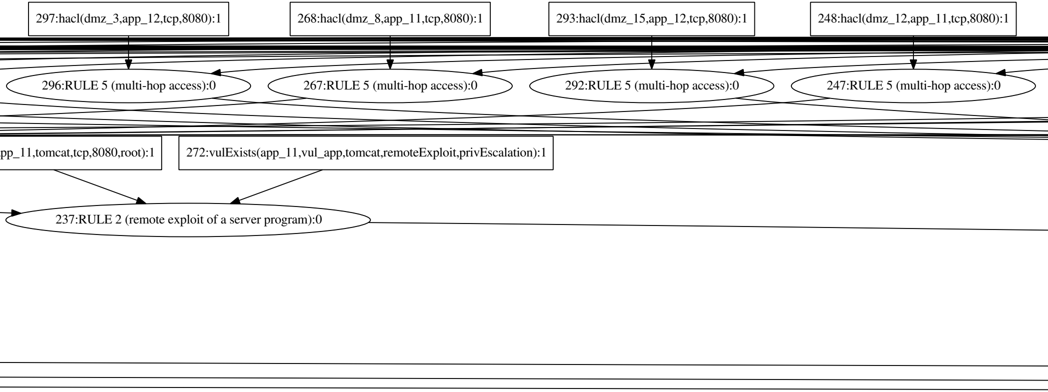

256:hacl(dmz_2,app_11,tcp,8080):1
186:hacl(dmz_6,app_1,tcp,8080):1
242:hacl(dmz_1,app_11,tcp,8080):1
252:hacl(dmz_14,app_11,tcp,8080):1
255:RULE 5 (multi-hop access):0
185:RULE 5 (multi-hop access):0
241:RULE 5 (multi-hop access):0
251:RULE 5 (multi-hop access):0

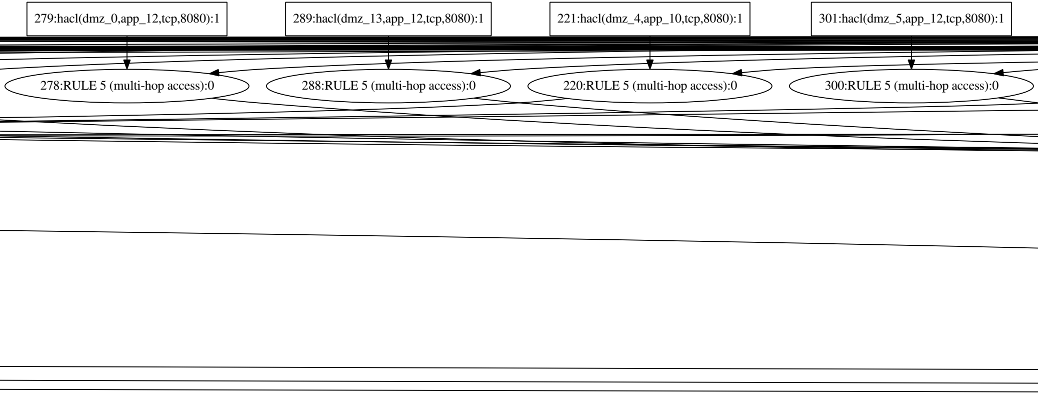

258:hacl(dmz_3,app_11,tcp,8080):1
229:hacl(dmz_8,app_10,tcp,8080):1
254:hacl(dmz_15,app_11,tcp,8080):1
209:hacl(dmz_12,app_10,tcp,8080):1
257:RULE 5 (multi-hop access):0
228:RULE 5 (multi-hop access):0
253:RULE 5 (multi-hop access):0
208:RULE 5 (multi-hop access):0
232:networkServiceInfo(app_10,tomcat,tcp,8080,root):1
233:vulExists(app_10,vul_app,tomcat,remoteExploit,privEscalation):1
198:RULE 2 (remote exploit of a server program):0

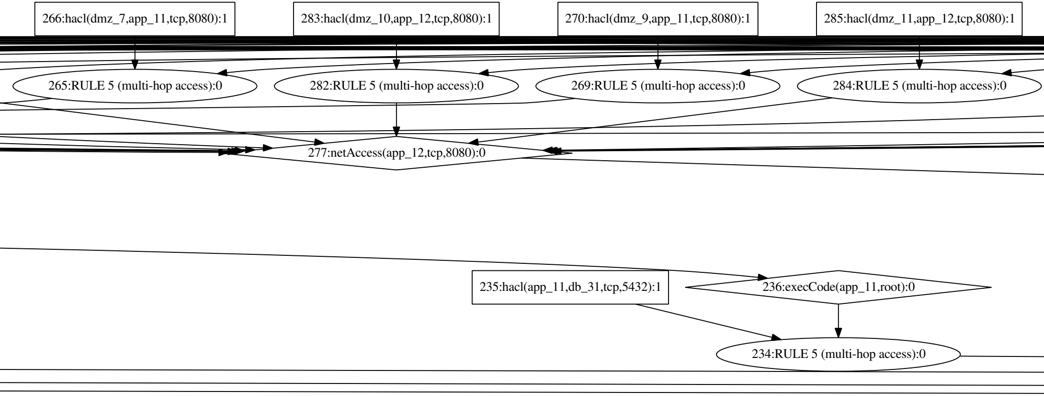

240:hacl(dmz_0,app_11,tcp,8080):1
250:hacl(dmz_13,app_11,tcp,8080):1
182:hacl(dmz_4,app_1,tcp,8080):1
262:hacl(dmz_5,app_11,tcp,8080):1
239:RULE 5 (multi-hop access):0
249:RULE 5 (multi-hop access):0
181:RULE 5 (multi-hop access):0
261:RULE 5 (multi-hop access):0
196:hacl(app_10,db_31,tcp,5432):1

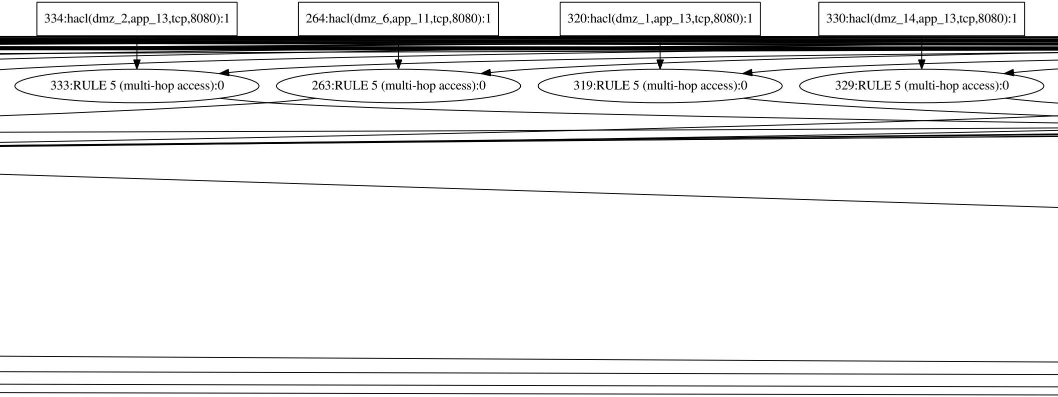

227:hacl(dmz_7,app_10,tcp,8080):1
244:hacl(dmz_10,app_11,tcp,8080):1
231:hacl(dmz_9,app_10,tcp,8080):1
246:hacl(dmz_11,app_11,tcp,8080):1
226:RULE 5 (multi-hop access):0
243:RULE 5 (multi-hop access):0
230:RULE 5 (multi-hop access):0
245:RULE 5 (multi-hop access):0
238:netAccess(app_11,tcp,8080):0
197:execCode(app_10,root):0
195:RULE 5 (multi-hop access):0

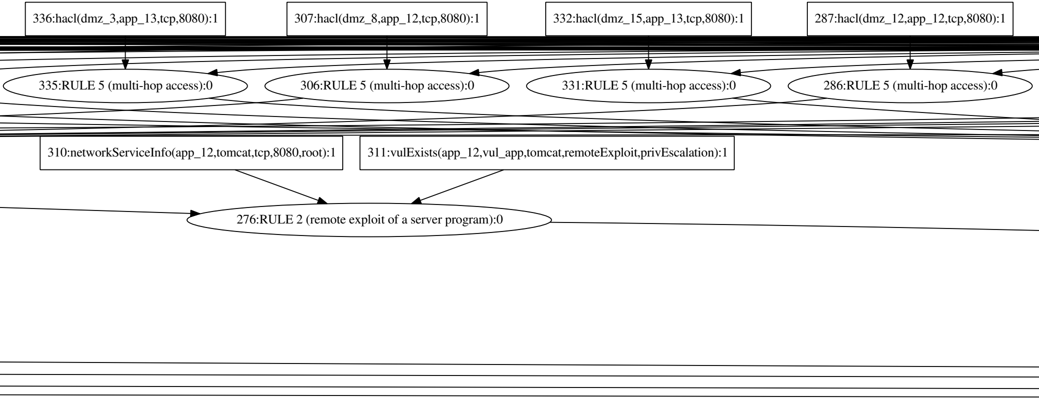

295:hacl(dmz_2,app_12,tcp,8080):1
225:hacl(dmz_6,app_10,tcp,8080):1
281:hacl(dmz_1,app_12,tcp,8080):1
291:hacl(dmz_14,app_12,tcp,8080):1
294:RULE 5 (multi-hop access):0
224:RULE 5 (multi-hop access):0
280:RULE 5 (multi-hop access):0
290:RULE 5 (multi-hop access):0
271:networkServiceInfo(a

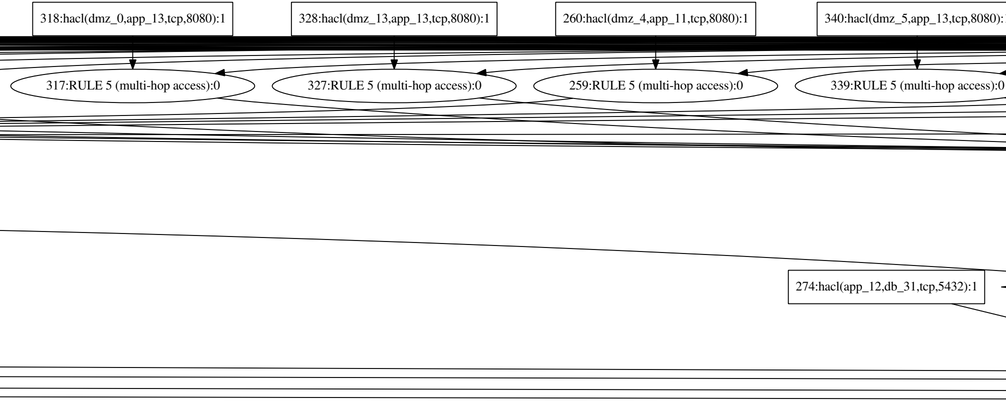

297:hacl(dmz_3,app_12,tcp,8080):1
268:hacl(dmz_8,app_11,tcp,8080):1
293:hacl(dmz_15,app_12,tcp,8080):1
248:hacl(dmz_12,app_11,tcp,8080):1
296:RULE 5 (multi-hop access):0
267:RULE 5 (multi-hop access):0
292:RULE 5 (multi-hop access):0
247:RULE 5 (multi-hop access):0
pp_11,tomcat,tcp,8080,root):1
272:vulExists(app_11,vul_app,tomcat,remoteExploit,privEscalation):1
237:RULE 2 (remote exploit of a server program):0

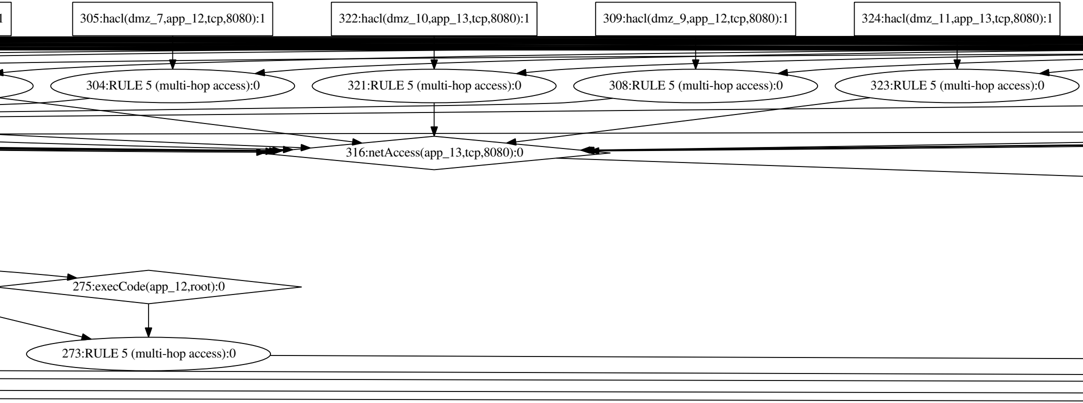

279:hacl(dmz_0,app_12,tcp,8080):1
289:hacl(dmz_13,app_12,tcp,8080):1
221:hacl(dmz_4,app_10,tcp,8080):1
301:hacl(dmz_5,app_12,tcp,8080):1
278:RULE 5 (multi-hop access):0
288:RULE 5 (multi-hop access):0
220:RULE 5 (multi-hop access):0
300:RULE 5 (multi-hop access):0

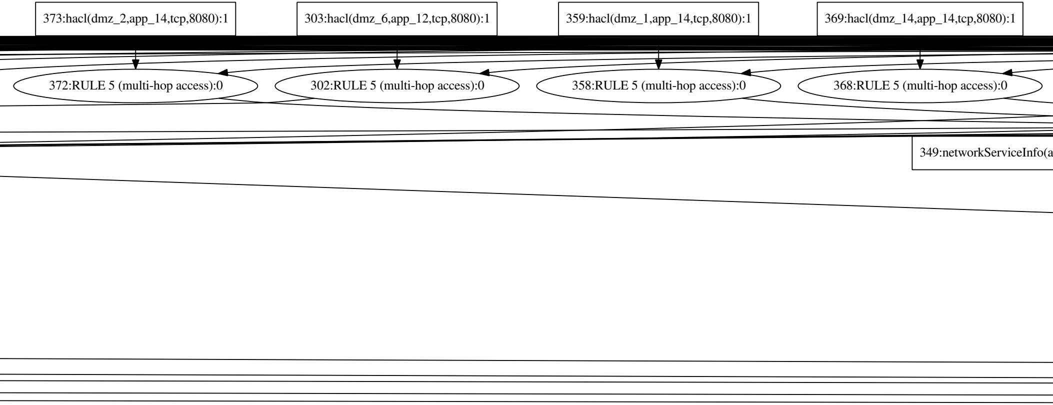

266:hacl(dmz_7,app_11,tcp,8080):1
283:hacl(dmz_10,app_12,tcp,8080):1
270:hacl(dmz_9,app_11,tcp,8080):1
285:hacl(dmz_11,app_12,tcp,8080):1
265:RULE 5 (multi-hop access):0
282:RULE 5 (multi-hop access):0
269:RULE 5 (multi-hop access):0
284:RULE 5 (multi-hop access):0
277:netAccess(app_12,tcp,8080):0
235:hacl(app_11,db_31,tcp,5432):1
236:execCode(app_11,root):0
234:RULE 5 (multi-hop access):0

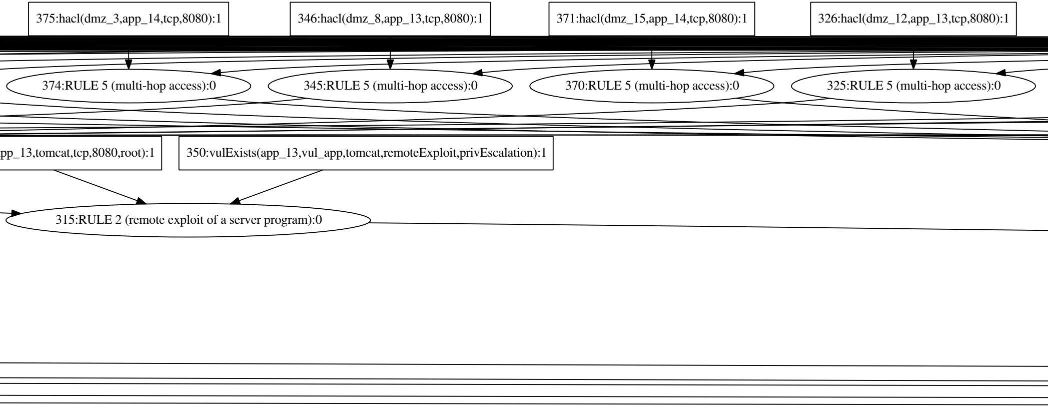

334:hacl(dmz_2,app_13,tcp,8080):1
264:hacl(dmz_6,app_11,tcp,8080):1
320:hacl(dmz_1,app_13,tcp,8080):1
330:hacl(dmz_14,app_13,tcp,8080):1
333:RULE 5 (multi-hop access):0
263:RULE 5 (multi-hop access):0
319:RULE 5 (multi-hop access):0
329:RULE 5 (multi-hop access):0

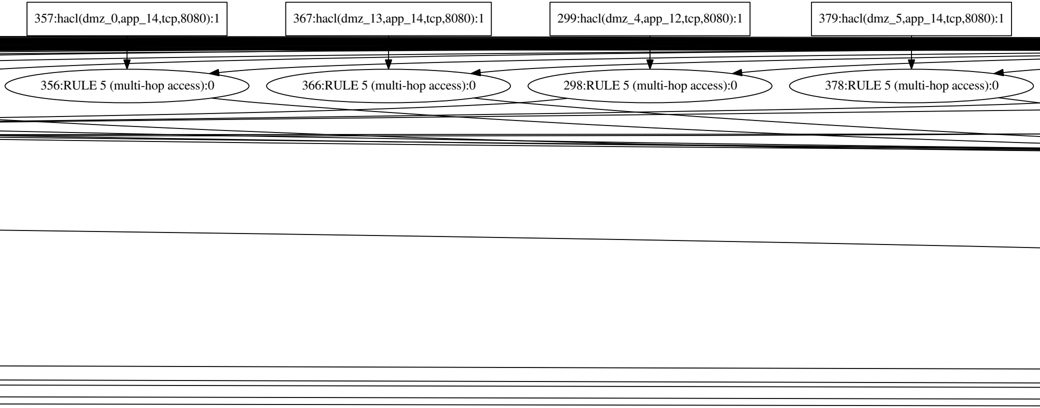

336:hacl(dmz_3,app_13,tcp,8080):1
307:hacl(dmz_8,app_12,tcp,8080):1
332:hacl(dmz_15,app_13,tcp,8080):1
287:hacl(dmz_12,app_12,tcp,8080):1
335:RULE 5 (multi-hop access):0
306:RULE 5 (multi-hop access):0
331:RULE 5 (multi-hop access):0
286:RULE 5 (multi-hop access):0
310:networkServiceInfo(app_12,tomcat,tcp,8080,root):1
311:vulExists(app_12,vul_app,tomcat,remoteExploit,privEscalation):1
276:RULE 2 (remote exploit of a server program):0

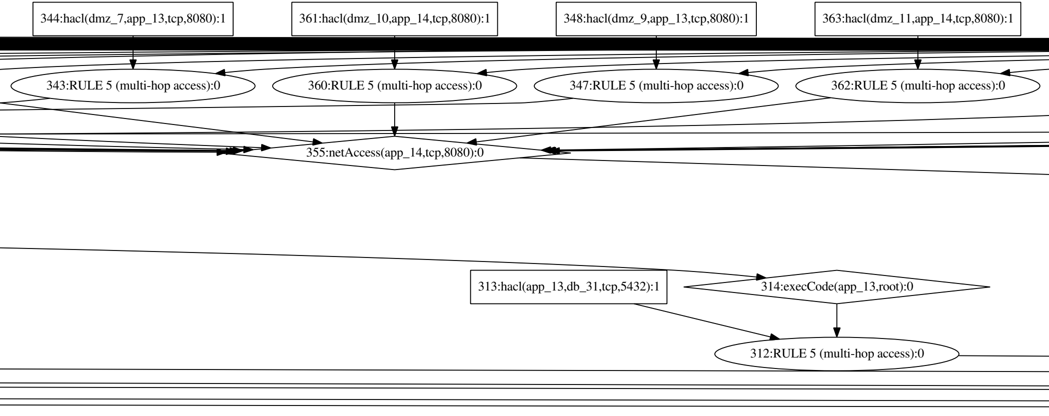

318:hacl(dmz_0,app_13,tcp,8080):1
328:hacl(dmz_13,app_13,tcp,8080):1
260:hacl(dmz_4,app_11,tcp,8080):1
340:hacl(dmz_5,app_13,tcp,8080):
317:RULE 5 (multi-hop access):0
327:RULE 5 (multi-hop access):0
259:RULE 5 (multi-hop access):0
339:RULE 5 (multi-hop access):0
274:hacl(app_12,db_31,tcp,5432):1

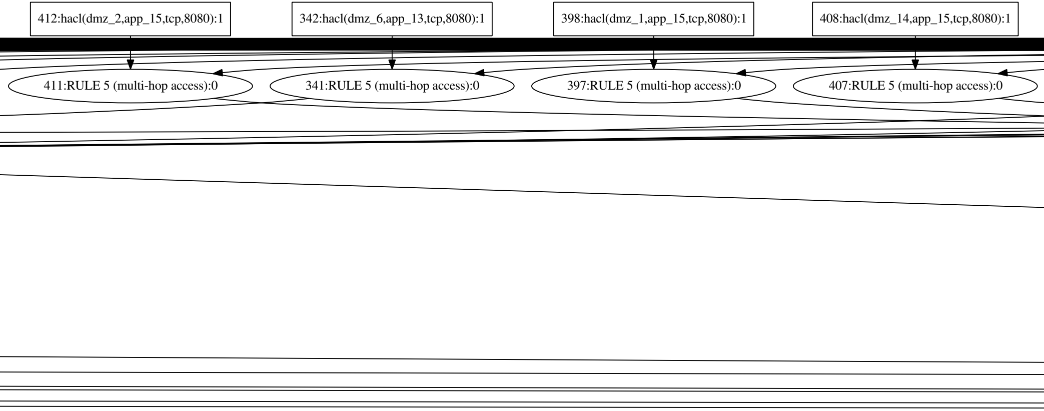

305:hacl(dmz_7,app_12,tcp,8080):1
322:hacl(dmz_10,app_13,tcp,8080):1
309:hacl(dmz_9,app_12,tcp,8080):1
324:hacl(dmz_11,app_13,tcp,8080):1
304:RULE 5 (multi-hop access):0
321:RULE 5 (multi-hop access):0
308:RULE 5 (multi-hop access):0
323:RULE 5 (multi-hop access):0
316:netAccess(app_13,tcp,8080):0
275:execCode(app_12,root):0
273:RULE 5 (multi-hop access):0

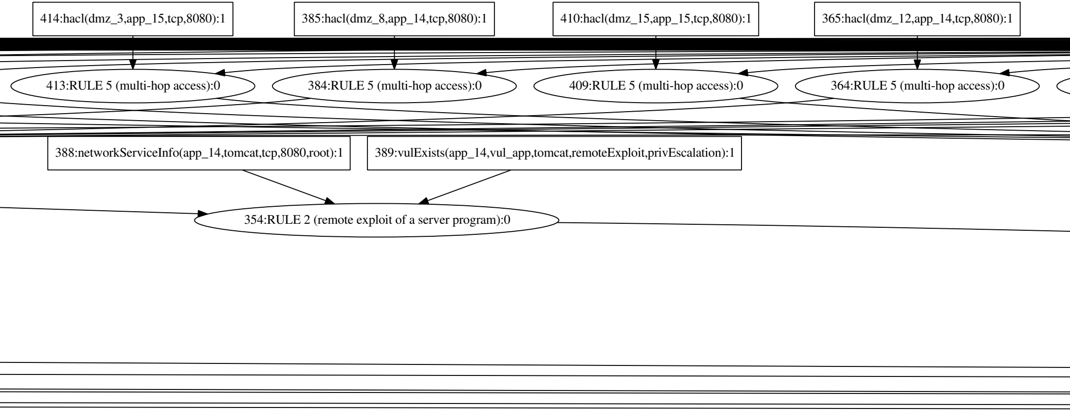

373:hacl(dmz_2,app_14,tcp,8080):1
303:hacl(dmz_6,app_12,tcp,8080):1
359:hacl(dmz_1,app_14,tcp,8080):1
369:hacl(dmz_14,app_14,tcp,8080):1
372:RULE 5 (multi-hop access):0
302:RULE 5 (multi-hop access):0
358:RULE 5 (multi-hop access):0
368:RULE 5 (multi-hop access):0
349:networkServiceInfo(a

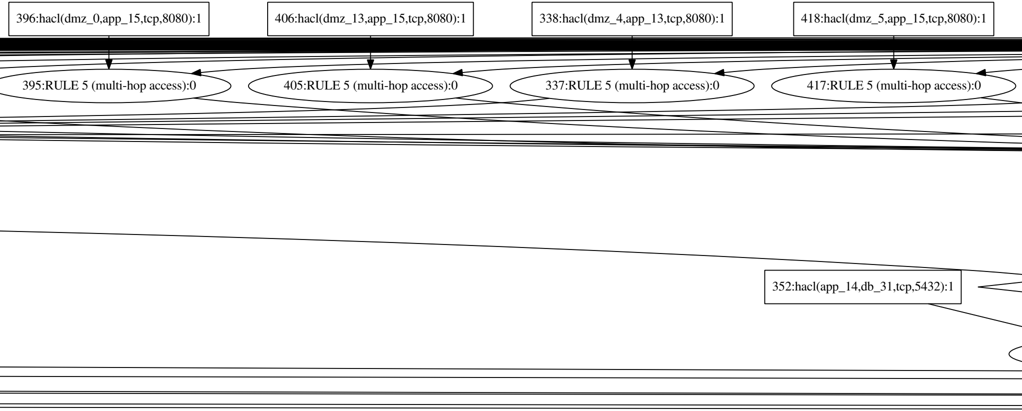

375:hacl(dmz_3,app_14,tcp,8080):1
346:hacl(dmz_8,app_13,tcp,8080):1
371:hacl(dmz_15,app_14,tcp,8080):1
326:hacl(dmz_12,app_13,tcp,8080):1
374:RULE 5 (multi-hop access):0
345:RULE 5 (multi-hop access):0
370:RULE 5 (multi-hop access):0
325:RULE 5 (multi-hop access):0
pp_13,tomcat,tcp,8080,root):1
350:vulExists(app_13,vul_app,tomcat,remoteExploit,privEscalation):1
315:RULE 2 (remote exploit of a server program):0

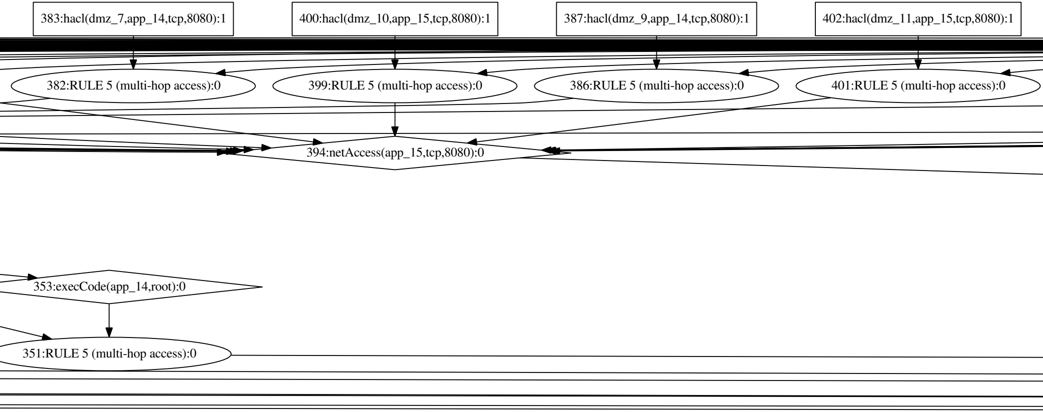

357:hacl(dmz_0,app_14,tcp,8080):1
367:hacl(dmz_13,app_14,tcp,8080):1
299:hacl(dmz_4,app_12,tcp,8080):1
379:hacl(dmz_5,app_14,tcp,8080):1
356:RULE 5 (multi-hop access):0
366:RULE 5 (multi-hop access):0
298:RULE 5 (multi-hop access):0
378:RULE 5 (multi-hop access):0

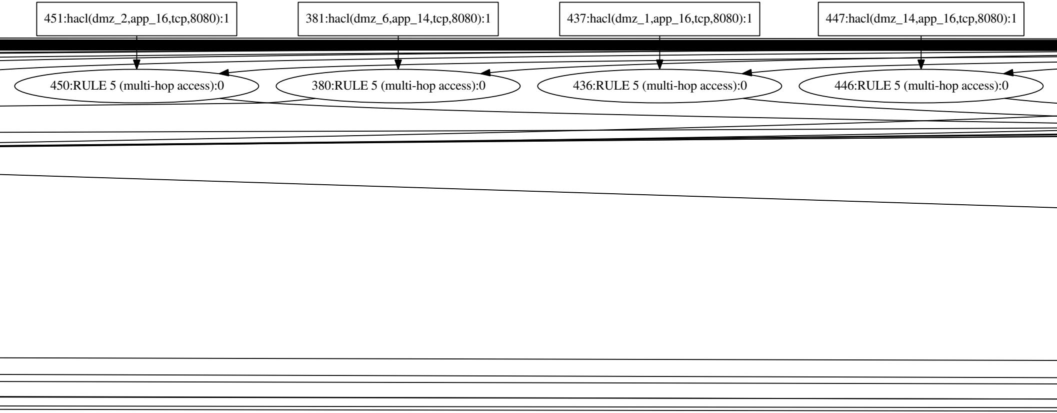

344:hacl(dmz_7,app_13,tcp,8080):1
361:hacl(dmz_10,app_14,tcp,8080):1
348:hacl(dmz_9,app_13,tcp,8080):1
363:hacl(dmz_11,app_14,tcp,8080):1
343:RULE 5 (multi-hop access):0
360:RULE 5 (multi-hop access):0
347:RULE 5 (multi-hop access):0
362:RULE 5 (multi-hop access):0
355:netAccess(app_14,tcp,8080):0
313:hacl(app_13,db_31,tcp,5432):1
314:execCode(app_13,root):0
312:RULE 5 (multi-hop access):0

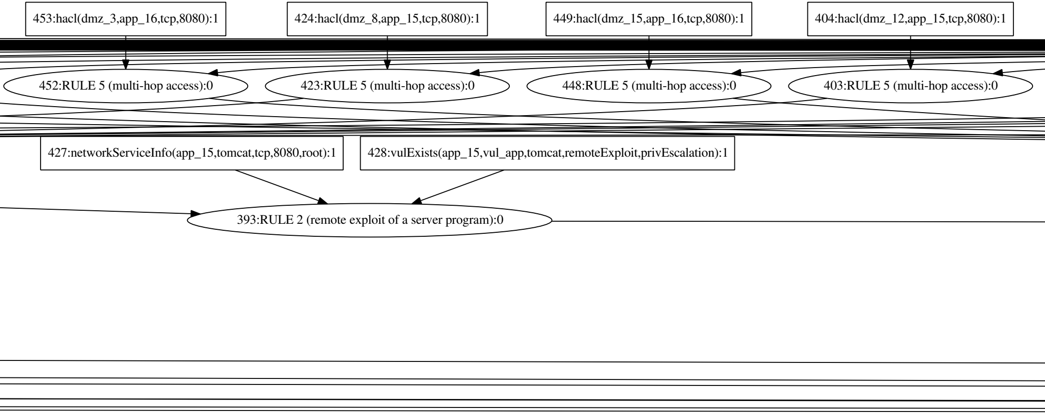

412:hacl(dmz_2,app_15,tcp,8080):1
342:hacl(dmz_6,app_13,tcp,8080):1
398:hacl(dmz_1,app_15,tcp,8080):1
408:hacl(dmz_14,app_15,tcp,8080):1
411:RULE 5 (multi-hop access):0
341:RULE 5 (multi-hop access):0
397:RULE 5 (multi-hop access):0
407:RULE 5 (multi-hop access):0

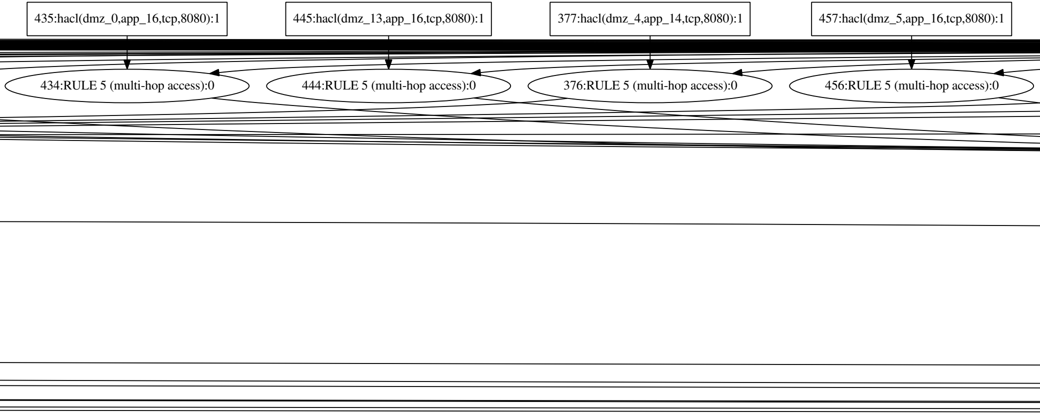

414:hacl(dmz_3,app_15,tcp,8080):1
385:hacl(dmz_8,app_14,tcp,8080):1
410:hacl(dmz_15,app_15,tcp,8080):1
365:hacl(dmz_12,app_14,tcp,8080):1
413:RULE 5 (multi-hop access):0
384:RULE 5 (multi-hop access):0
409:RULE 5 (multi-hop access):0
364:RULE 5 (multi-hop access):0
388:networkServiceInfo(app_14,tomcat,tcp,8080,root):1
389:vulExists(app_14,vul_app,tomcat,remoteExploit,privEscalation):1
354:RULE 2 (remote exploit of a server program):0

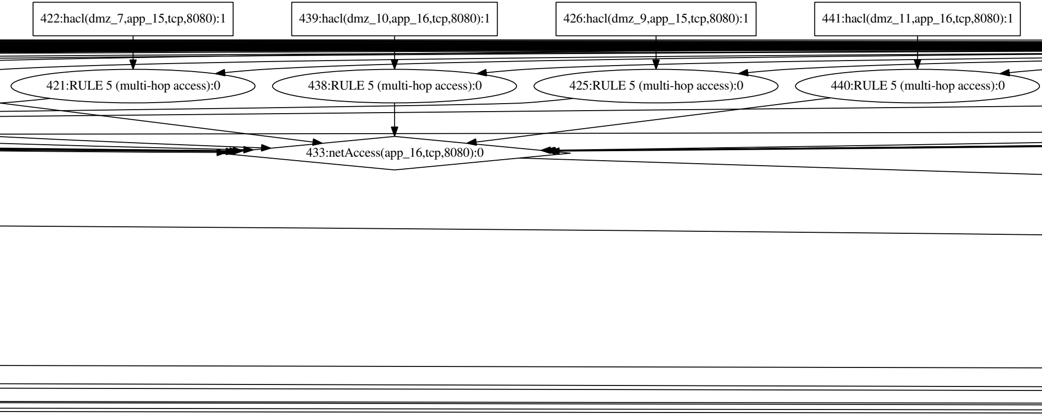

396:hacl(dmz_0,app_15,tcp,8080):1
406:hacl(dmz_13,app_15,tcp,8080):1
338:hacl(dmz_4,app_13,tcp,8080):1
418:hacl(dmz_5,app_15,tcp,8080):1
395:RULE 5 (multi-hop access):0
405:RULE 5 (multi-hop access):0
337:RULE 5 (multi-hop access):0
417:RULE 5 (multi-hop access):0
352:hacl(app_14,db_31,tcp,5432):1

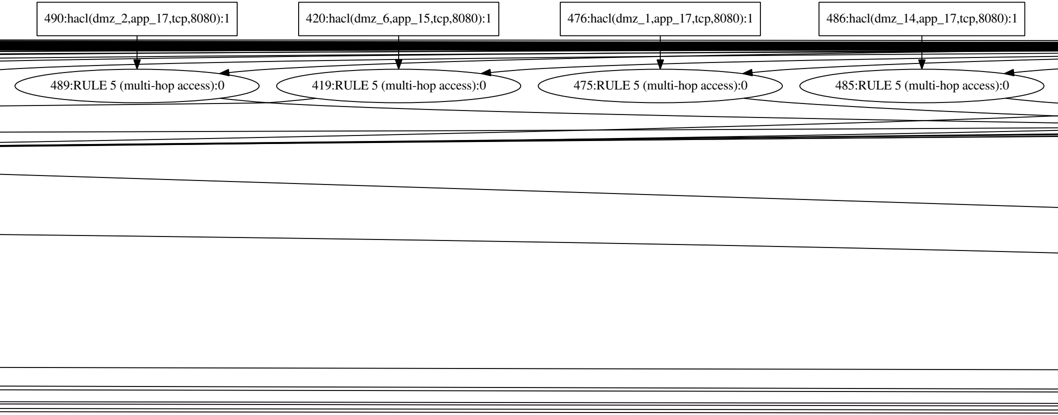

383:hacl(dmz_7,app_14,tcp,8080):1
400:hacl(dmz_10,app_15,tcp,8080):1
387:hacl(dmz_9,app_14,tcp,8080):1
402:hacl(dmz_11,app_15,tcp,8080):1
382:RULE 5 (multi-hop access):0
399:RULE 5 (multi-hop access):0
386:RULE 5 (multi-hop access):0
401:RULE 5 (multi-hop access):0
394:netAccess(app_15,tcp,8080):0
353:execCode(app_14,root):0
351:RULE 5 (multi-hop access):0

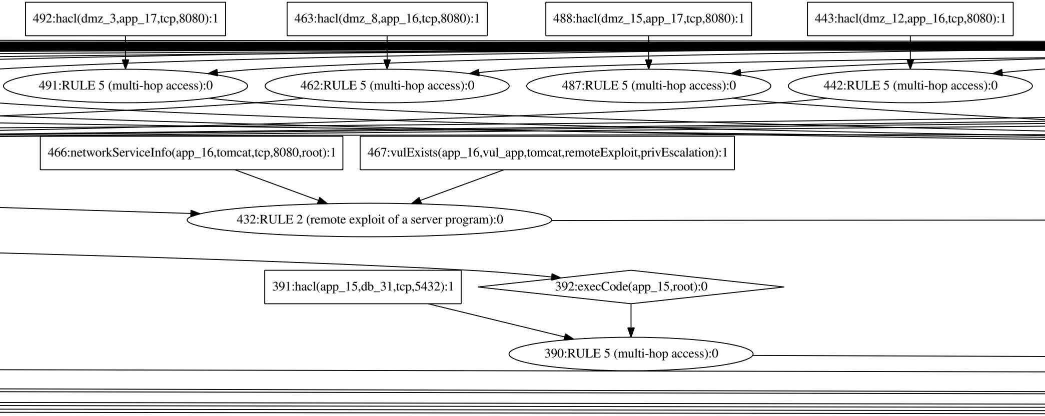

451:hacl(dmz_2,app_16,tcp,8080):1
381:hacl(dmz_6,app_14,tcp,8080):1
437:hacl(dmz_1,app_16,tcp,8080):1
447:hacl(dmz_14,app_16,tcp,8080):1
450:RULE 5 (multi-hop access):0
380:RULE 5 (multi-hop access):0
436:RULE 5 (multi-hop access):0
446:RULE 5 (multi-hop access):0

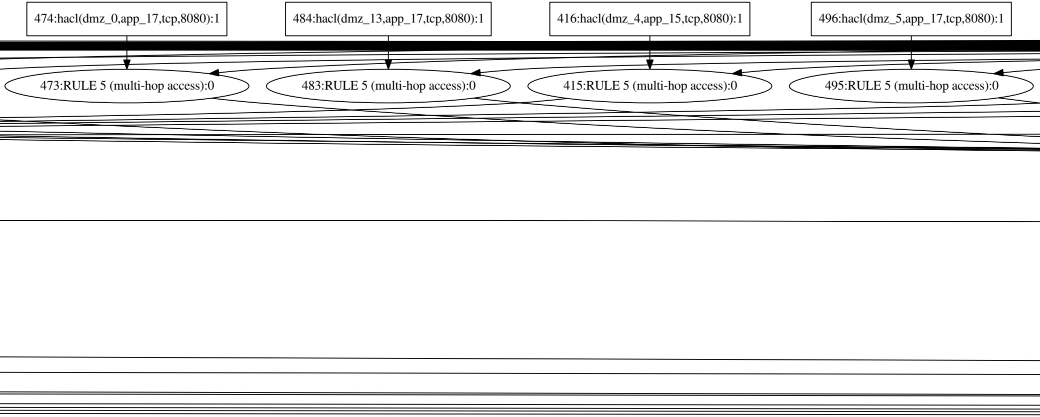

453:hacl(dmz_3,app_16,tcp,8080):1
424:hacl(dmz_8,app_15,tcp,8080):1
449:hacl(dmz_15,app_16,tcp,8080):1
404:hacl(dmz_12,app_15,tcp,8080):1
452:RULE 5 (multi-hop access):0
423:RULE 5 (multi-hop access):0
448:RULE 5 (multi-hop access):0
403:RULE 5 (multi-hop access):0
427:networkServiceInfo(app_15,tomcat,tcp,8080,root):1
428:vulExists(app_15,vul_app,tomcat,remoteExploit,privEscalation):1
393:RULE 2 (remote exploit of a server program):0

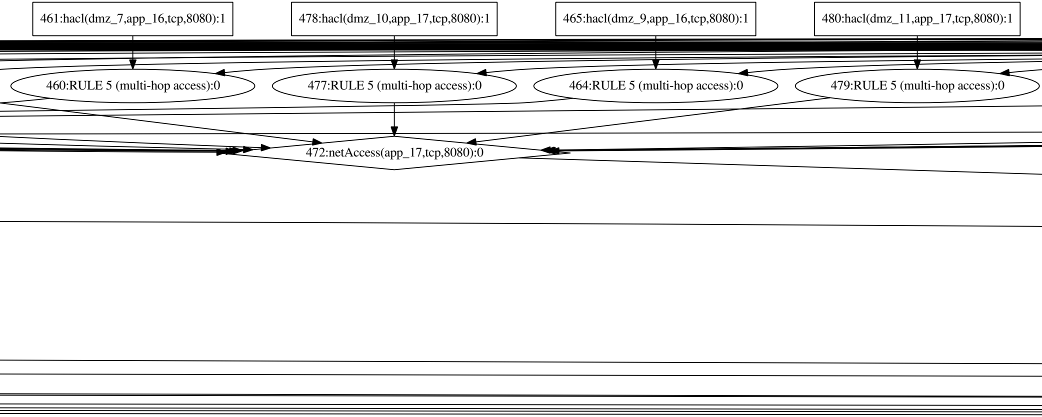

435:hacl(dmz_0,app_16,tcp,8080):1
445:hacl(dmz_13,app_16,tcp,8080):1
377:hacl(dmz_4,app_14,tcp,8080):1
457:hacl(dmz_5,app_16,tcp,8080):1
434:RULE 5 (multi-hop access):0
444:RULE 5 (multi-hop access):0
376:RULE 5 (multi-hop access):0
456:RULE 5 (multi-hop access):0

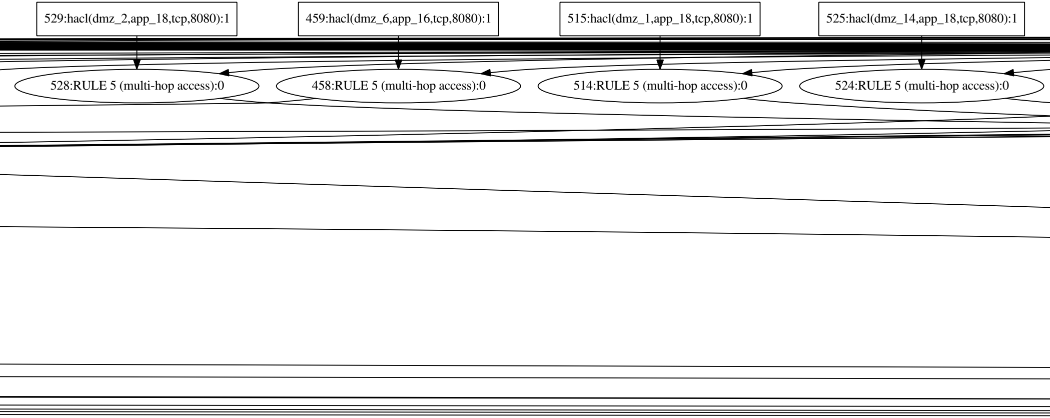

422:hacl(dmz_7,app_15,tcp,8080):1
439:hacl(dmz_10,app_16,tcp,8080):1
426:hacl(dmz_9,app_15,tcp,8080):1
441:hacl(dmz_11,app_16,tcp,8080):1
421:RULE 5 (multi-hop access):0
438:RULE 5 (multi-hop access):0
425:RULE 5 (multi-hop access):0
440:RULE 5 (multi-hop access):0
433:netAccess(app_16,tcp,8080):0

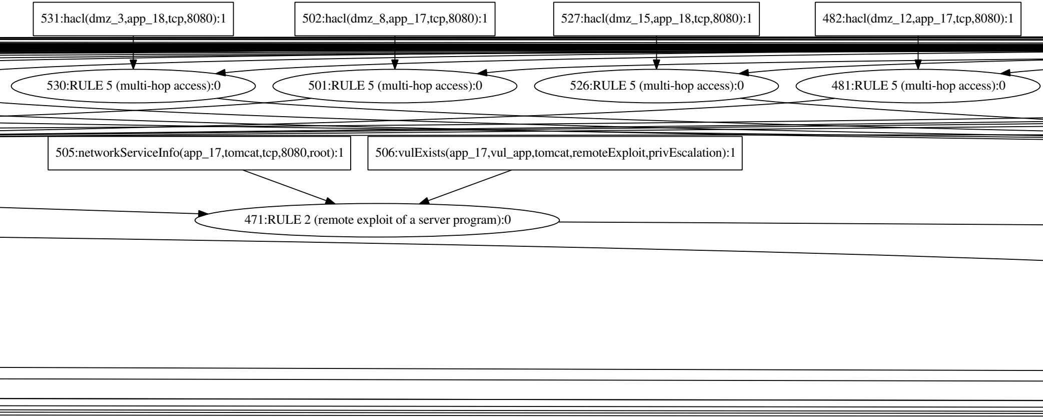

490:hacl(dmz_2,app_17,tcp,8080):1
420:hacl(dmz_6,app_15,tcp,8080):1
476:hacl(dmz_1,app_17,tcp,8080):1
486:hacl(dmz_14,app_17,tcp,8080):1
489:RULE 5 (multi-hop access):0
419:RULE 5 (multi-hop access):0
475:RULE 5 (multi-hop access):0
485:RULE 5 (multi-hop access):0

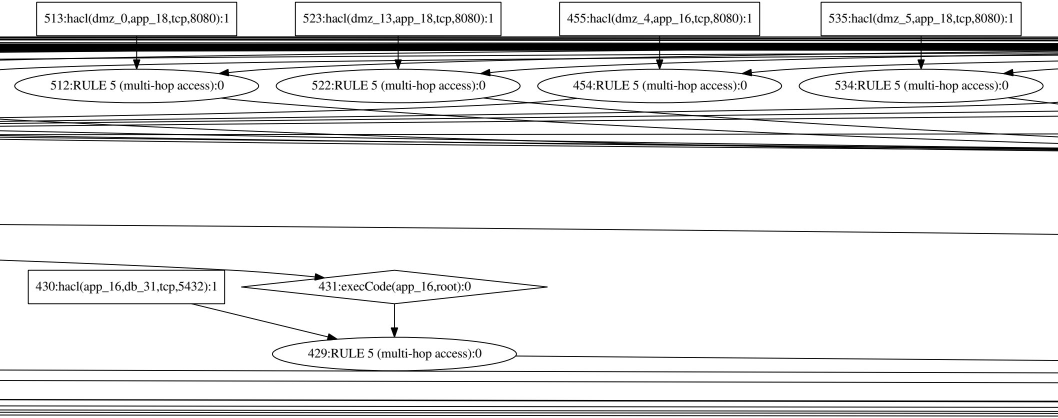

492:hacl(dmz_3,app_17,tcp,8080):1
463:hacl(dmz_8,app_16,tcp,8080):1
488:hacl(dmz_15,app_17,tcp,8080):1
443:hacl(dmz_12,app_16,tcp,8080):1
491:RULE 5 (multi-hop access):0
462:RULE 5 (multi-hop access):0
487:RULE 5 (multi-hop access):0
442:RULE 5 (multi-hop access):0
466:networkServiceInfo(app_16,tomcat,tcp,8080,root):1
467:vulExists(app_16,vul_app,tomcat,remoteExploit,privEscalation):1
432:RULE 2 (remote exploit of a server program):0
391:hacl(app_15,db_31,tcp,5432):1
392:execCode(app_15,root):0
390:RULE 5 (multi-hop access):0

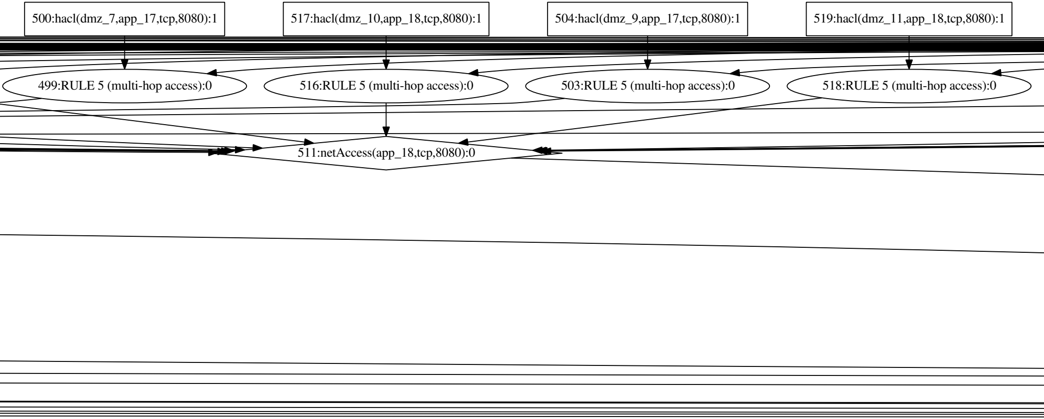

474:hacl(dmz_0,app_17,tcp,8080):1
484:hacl(dmz_13,app_17,tcp,8080):1
416:hacl(dmz_4,app_15,tcp,8080):1
496:hacl(dmz_5,app_17,tcp,8080):1
473:RULE 5 (multi-hop access):0
483:RULE 5 (multi-hop access):0
415:RULE 5 (multi-hop access):0
495:RULE 5 (multi-hop access):0

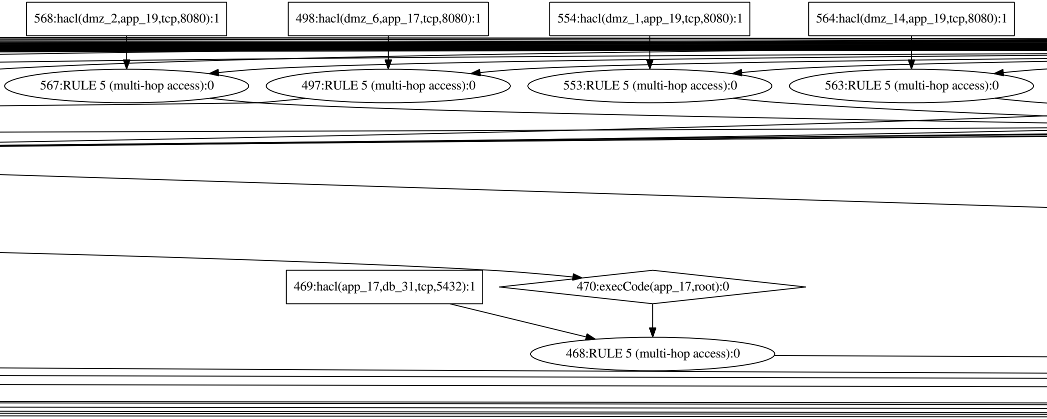

461:hacl(dmz_7,app_16,tcp,8080):1
478:hacl(dmz_10,app_17,tcp,8080):1
465:hacl(dmz_9,app_16,tcp,8080):1
480:hacl(dmz_11,app_17,tcp,8080):1
460:RULE 5 (multi-hop access):0
477:RULE 5 (multi-hop access):0
464:RULE 5 (multi-hop access):0
479:RULE 5 (multi-hop access):0
472:netAccess(app_17,tcp,8080):0

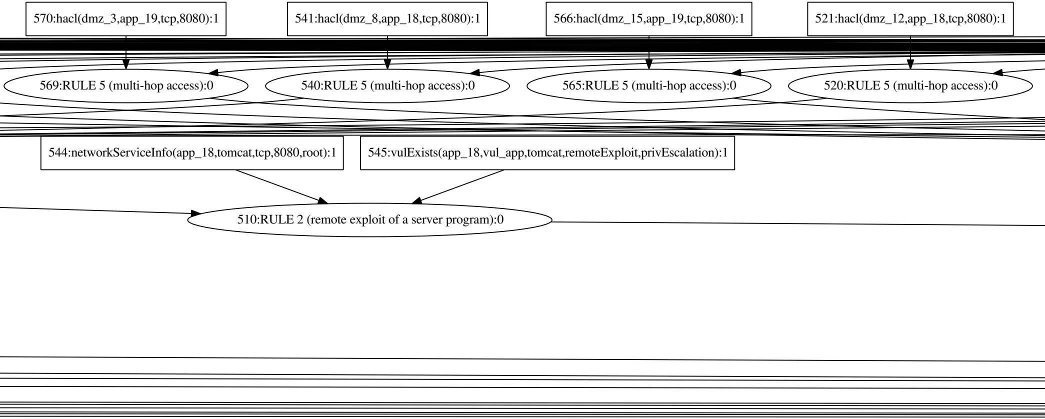

529:hacl(dmz_2,app_18,tcp,8080):1
459:hacl(dmz_6,app_16,tcp,8080):1
515:hacl(dmz_1,app_18,tcp,8080):1
525:hacl(dmz_14,app_18,tcp,8080):1
528:RULE 5 (multi-hop access):0
458:RULE 5 (multi-hop access):0
514:RULE 5 (multi-hop access):0
524:RULE 5 (multi-hop access):0

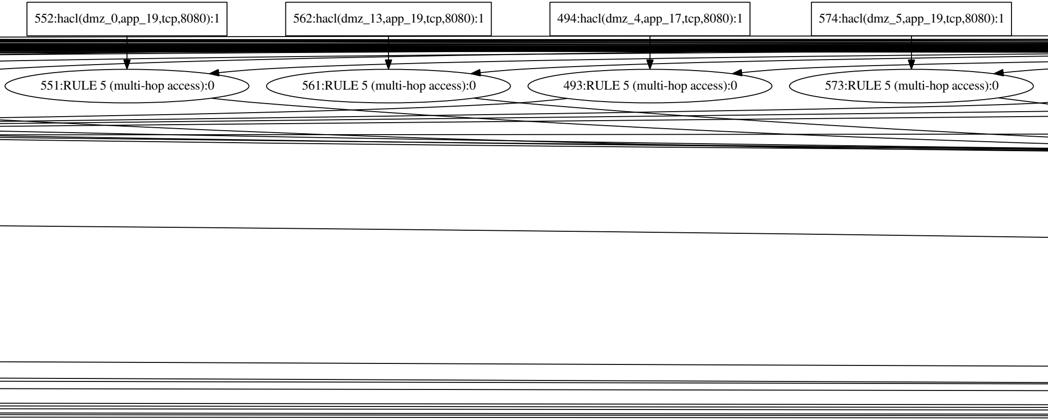

531:hacl(dmz_3,app_18,tcp,8080):1
502:hacl(dmz_8,app_17,tcp,8080):1
527:hacl(dmz_15,app_18,tcp,8080):1
482:hacl(dmz_12,app_17,tcp,8080):1
530:RULE 5 (multi-hop access):0
501:RULE 5 (multi-hop access):0
526:RULE 5 (multi-hop access):0
481:RULE 5 (multi-hop access):0
505:networkServiceInfo(app_17,tomcat,tcp,8080,root):1
506:vulExists(app_17,vul_app,tomcat,remoteExploit,privEscalation):1
471:RULE 2 (remote exploit of a server program):0

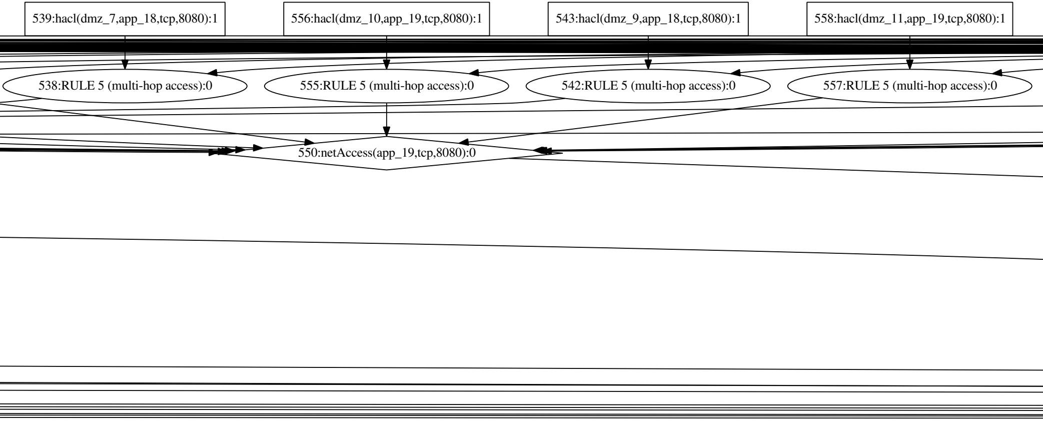

513:hacl(dmz_0,app_18,tcp,8080):1
523:hacl(dmz_13,app_18,tcp,8080):1
455:hacl(dmz_4,app_16,tcp,8080):1
535:hacl(dmz_5,app_18,tcp,8080):1
512:RULE 5 (multi-hop access):0
522:RULE 5 (multi-hop access):0
454:RULE 5 (multi-hop access):0
534:RULE 5 (multi-hop access):0
430:hacl(app_16,db_31,tcp,5432):1
431:execCode(app_16,root):0
429:RULE 5 (multi-hop access):0

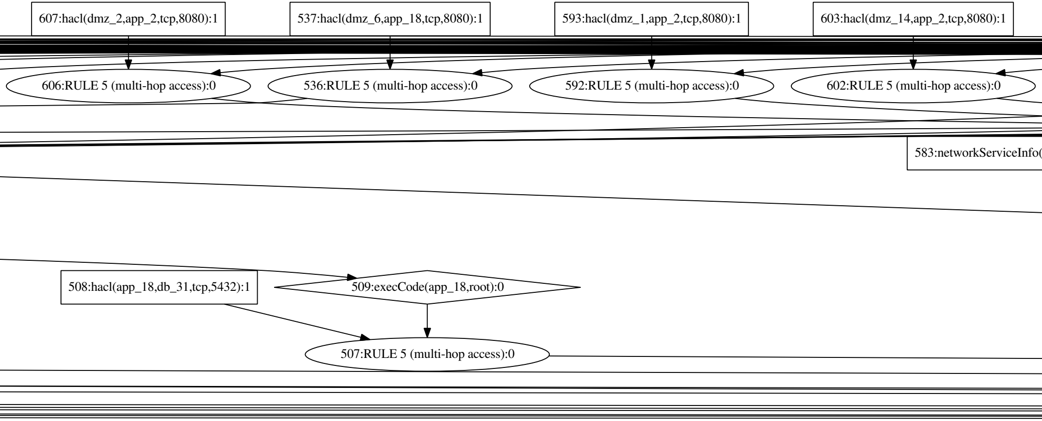

500:hacl(dmz_7,app_17,tcp,8080):1
517:hacl(dmz_10,app_18,tcp,8080):1
504:hacl(dmz_9,app_17,tcp,8080):1
519:hacl(dmz_11,app_18,tcp,8080):1
499:RULE 5 (multi-hop access):0
516:RULE 5 (multi-hop access):0
503:RULE 5 (multi-hop access):0
518:RULE 5 (multi-hop access):0
511:netAccess(app_18,tcp,8080):0

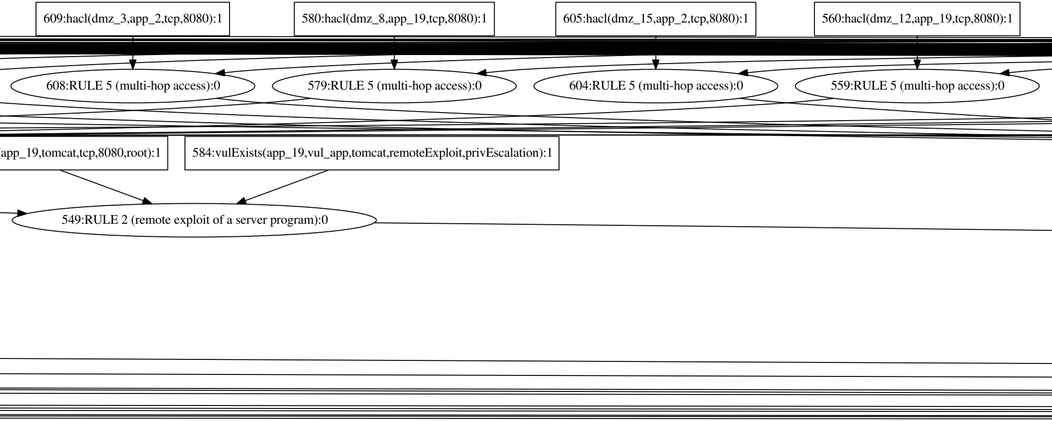

568:hacl(dmz_2,app_19,tcp,8080):1
498:hacl(dmz_6,app_17,tcp,8080):1
554:hacl(dmz_1,app_19,tcp,8080):1
564:hacl(dmz_14,app_19,tcp,8080):1
567:RULE 5 (multi-hop access):0
497:RULE 5 (multi-hop access):0
553:RULE 5 (multi-hop access):0
563:RULE 5 (multi-hop access):0
469:hacl(app_17,db_31,tcp,5432):1
470:execCode(app_17,root):0
468:RULE 5 (multi-hop access):0

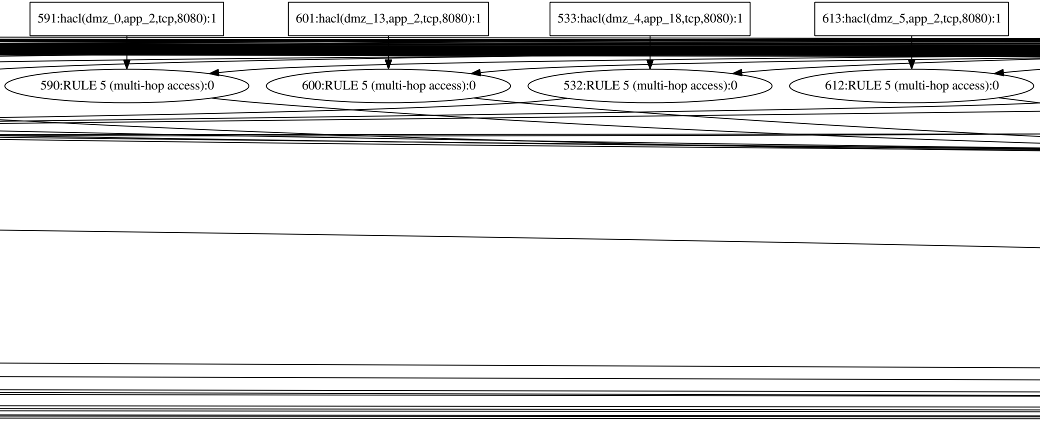

570:hacl(dmz_3,app_19,tcp,8080):1
541:hacl(dmz_8,app_18,tcp,8080):1
566:hacl(dmz_15,app_19,tcp,8080):1
521:hacl(dmz_12,app_18,tcp,8080):1
569:RULE 5 (multi-hop access):0
540:RULE 5 (multi-hop access):0
565:RULE 5 (multi-hop access):0
520:RULE 5 (multi-hop access):0
544:networkServiceInfo(app_18,tomcat,tcp,8080,root):1
545:vulExists(app_18,vul_app,tomcat,remoteExploit,privEscalation):1
510:RULE 2 (remote exploit of a server program):0

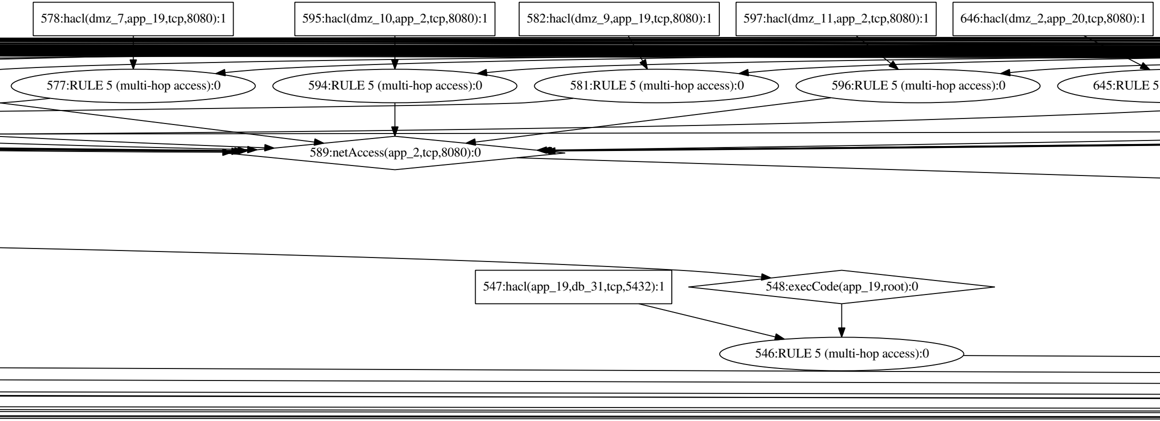

552:hacl(dmz_0,app_19,tcp,8080):1
562:hacl(dmz_13,app_19,tcp,8080):1
494:hacl(dmz_4,app_17,tcp,8080):1
574:hacl(dmz_5,app_19,tcp,8080):1
551:RULE 5 (multi-hop access):0
561:RULE 5 (multi-hop access):0
493:RULE 5 (multi-hop access):0
573:RULE 5 (multi-hop access):0

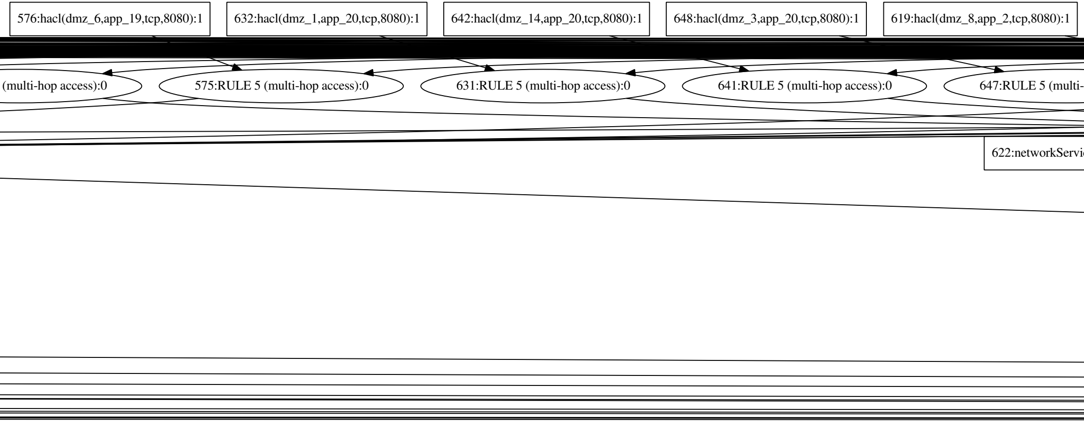

539:hacl(dmz_7,app_18,tcp,8080):1
556:hacl(dmz_10,app_19,tcp,8080):1
543:hacl(dmz_9,app_18,tcp,8080):1
558:hacl(dmz_11,app_19,tcp,8080):1
538:RULE 5 (multi-hop access):0
555:RULE 5 (multi-hop access):0
542:RULE 5 (multi-hop access):0
557:RULE 5 (multi-hop access):0
550:netAccess(app_19,tcp,8080):0

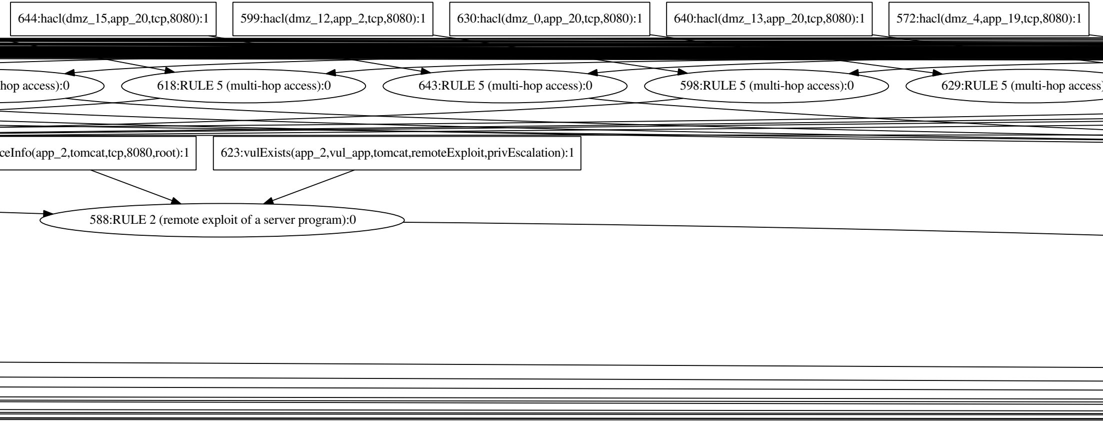

607:hacl(dmz_2,app_2,tcp,8080):1
537:hacl(dmz_6,app_18,tcp,8080):1
593:hacl(dmz_1,app_2,tcp,8080):1
603:hacl(dmz_14,app_2,tcp,8080):1
606:RULE 5 (multi-hop access):0
536:RULE 5 (multi-hop access):0
592:RULE 5 (multi-hop access):0
602:RULE 5 (multi-hop access):0
583:networkServiceInfo(
508:hacl(app_18,db_31,tcp,5432):1
509:execCode(app_18,root):0
507:RULE 5 (multi-hop access):0

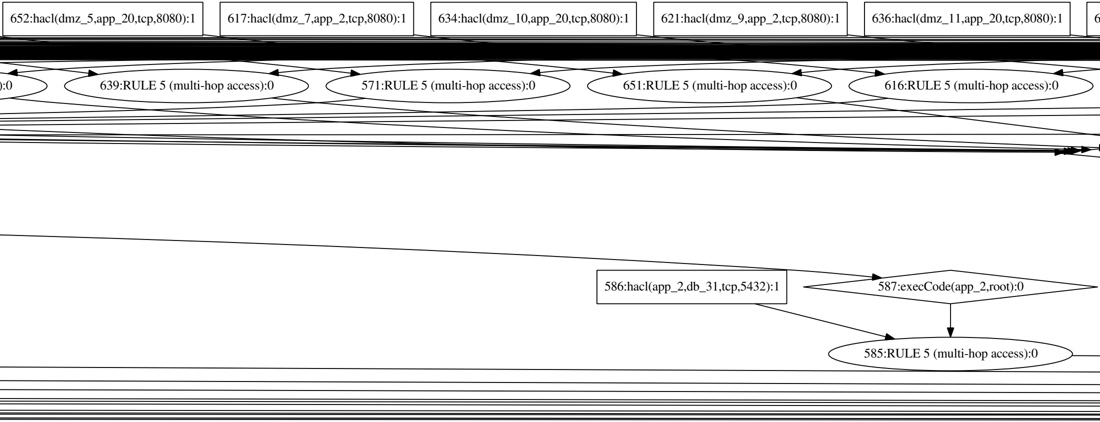

609:hacl(dmz_3,app_2,tcp,8080):1
580:hacl(dmz_8,app_19,tcp,8080):1
605:hacl(dmz_15,app_2,tcp,8080):1
560:hacl(dmz_12,app_19,tcp,8080):1
608:RULE 5 (multi-hop access):0
579:RULE 5 (multi-hop access):0
604:RULE 5 (multi-hop access):0
559:RULE 5 (multi-hop access):0
(app_19,tomcat,tcp,8080,root):1
584:vulExists(app_19,vul_app,tomcat,remoteExploit,privEscalation):1
549:RULE 2 (remote exploit of a server program):0

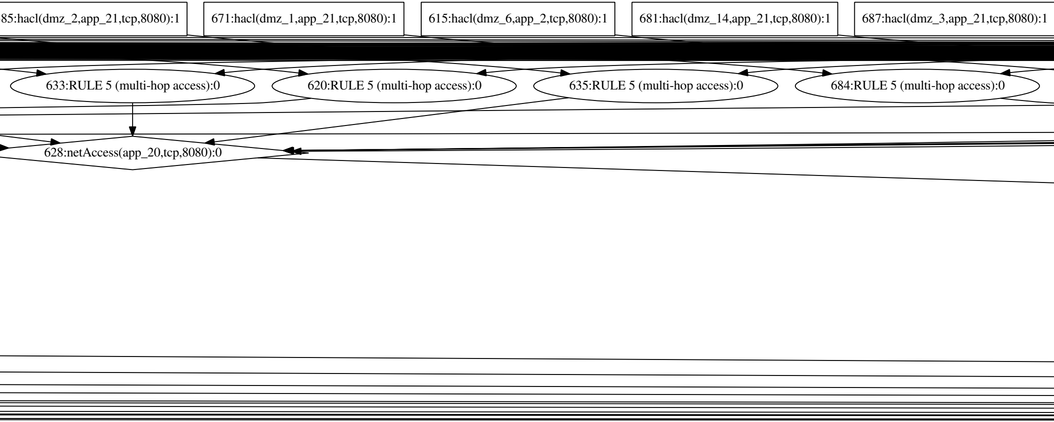

591:hacl(dmz_0,app_2,tcp,8080):1
601:hacl(dmz_13,app_2,tcp,8080):1
533:hacl(dmz_4,app_18,tcp,8080):1
613:hacl(dmz_5,app_2,tcp,8080):1
590:RULE 5 (multi-hop access):0
600:RULE 5 (multi-hop access):0
532:RULE 5 (multi-hop access):0
612:RULE 5 (multi-hop access):0

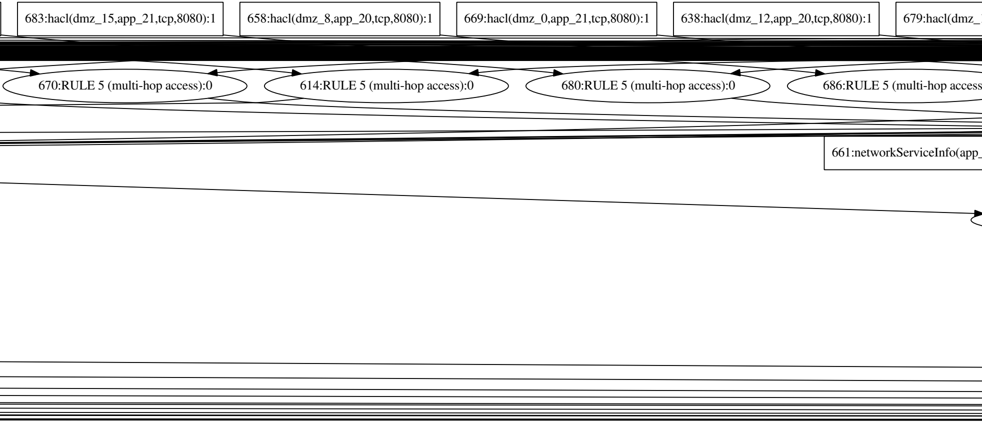

578:hacl(dmz_7,app_19,tcp,8080):1
595:hacl(dmz_10,app_2,tcp,8080):1
582:hacl(dmz_9,app_19,tcp,8080):1
597:hacl(dmz_11,app_2,tcp,8080):1
646:hacl(dmz_2,app_20,tcp,8080):1
577:RULE 5 (multi-hop access):0
594:RULE 5 (multi-hop access):0
581:RULE 5 (multi-hop access):0
596:RULE 5 (multi-hop access):0
645:RULE 5
589:netAccess(app_2,tcp,8080):0
547:hacl(app_19,db_31,tcp,5432):1
548:execCode(app_19,root):0
546:RULE 5 (multi-hop access):0

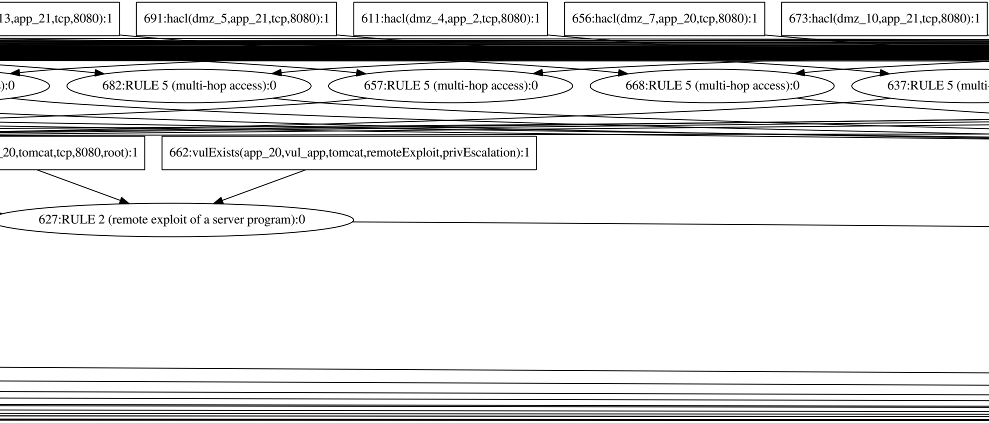

576:hacl(dmz_6,app_19,tcp,8080):1
632:hacl(dmz_1,app_20,tcp,8080):1
642:hacl(dmz_14,app_20,tcp,8080):1
648:hacl(dmz_3,app_20,tcp,8080):1
619:hacl(dmz_8,app_2,tcp,8080):1
(multi-hop access):0
575:RULE 5 (multi-hop access):0
631:RULE 5 (multi-hop access):0
641:RULE 5 (multi-hop access):0
647:RULE 5 (multi-
622:networkServi

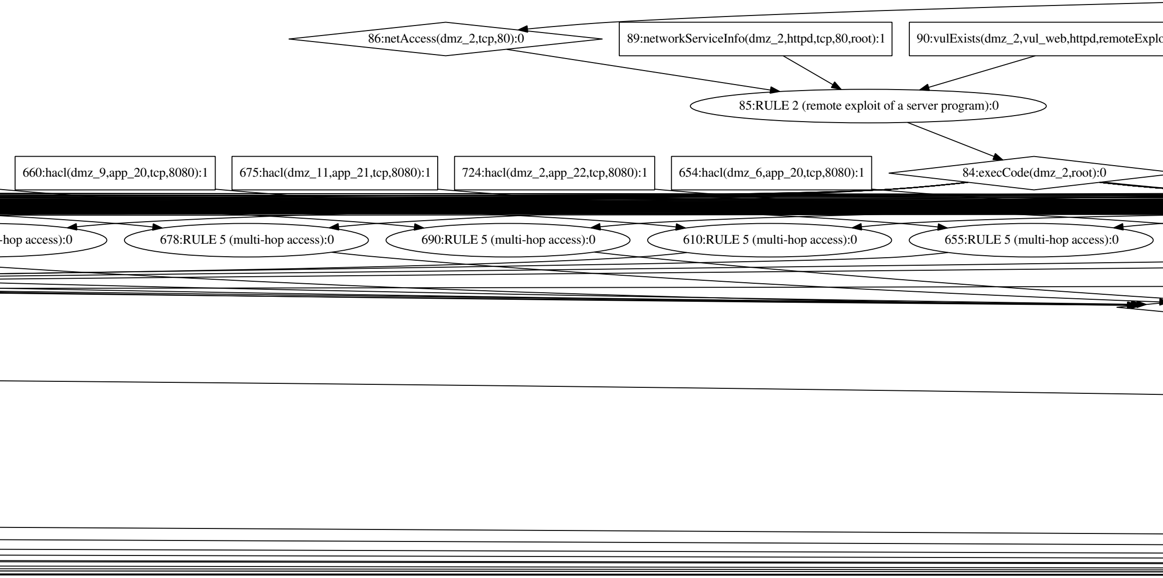

644:hacl(dmz_15,app_20,tcp,8080):1
599:hacl(dmz_12,app_2,tcp,8080):1
630:hacl(dmz_0,app_20,tcp,8080):1
640:hacl(dmz_13,app_20,tcp,8080):1
572:hacl(dmz_4,app_19,tcp,8080):1
hop access):0
618:RULE 5 (multi-hop access):0
643:RULE 5 (multi-hop access):0
598:RULE 5 (multi-hop access):0
629:RULE 5 (multi-hop access)
ceInfo(app_2,tomcat,tcp,8080,root):1
623:vulExists(app_2,vul_app,tomcat,remoteExploit,privEscalation):1
588:RULE 2 (remote exploit of a server program):0

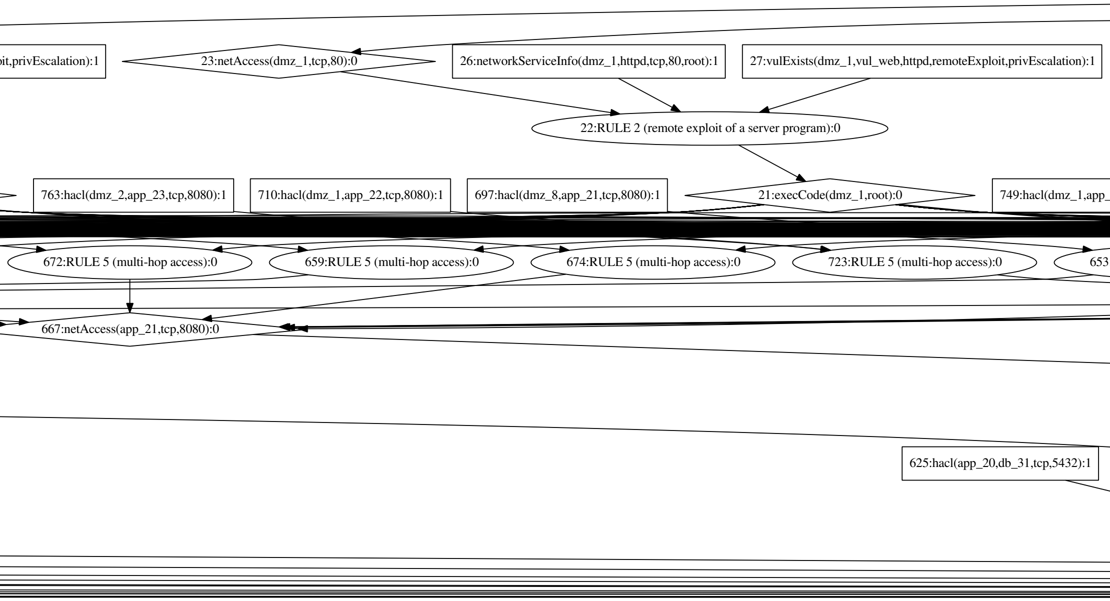

652:hacl(dmz_5,app_20,tcp,8080):1
617:hacl(dmz_7,app_2,tcp,8080):1
634:hacl(dmz_10,app_20,tcp,8080):1
621:hacl(dmz_9,app_2,tcp,8080):1
636:hacl(dmz_11,app_20,tcp,8080):1
639:RULE 5 (multi-hop access):0
571:RULE 5 (multi-hop access):0
651:RULE 5 (multi-hop access):0
616:RULE 5 (multi-hop access):0
586:hacl(app_2,db_31,tcp,5432):1
587:execCode(app_2,root):0
585:RULE 5 (multi-hop access):0


85:hacl(dmz_2,app_21,tcp,8080):1
671:hacl(dmz_1,app_21,tcp,8080):1
615:hacl(dmz_6,app_2,tcp,8080):1
681:hacl(dmz_14,app_21,tcp,8080):1
687:hacl(dmz_3,app_21,tcp,8080):1
633:RULE 5 (multi-hop access):0
620:RULE 5 (multi-hop access):0
635:RULE 5 (multi-hop access):0
684:RULE 5 (multi-hop access):0
628:netAccess(app_20,tcp,8080):0

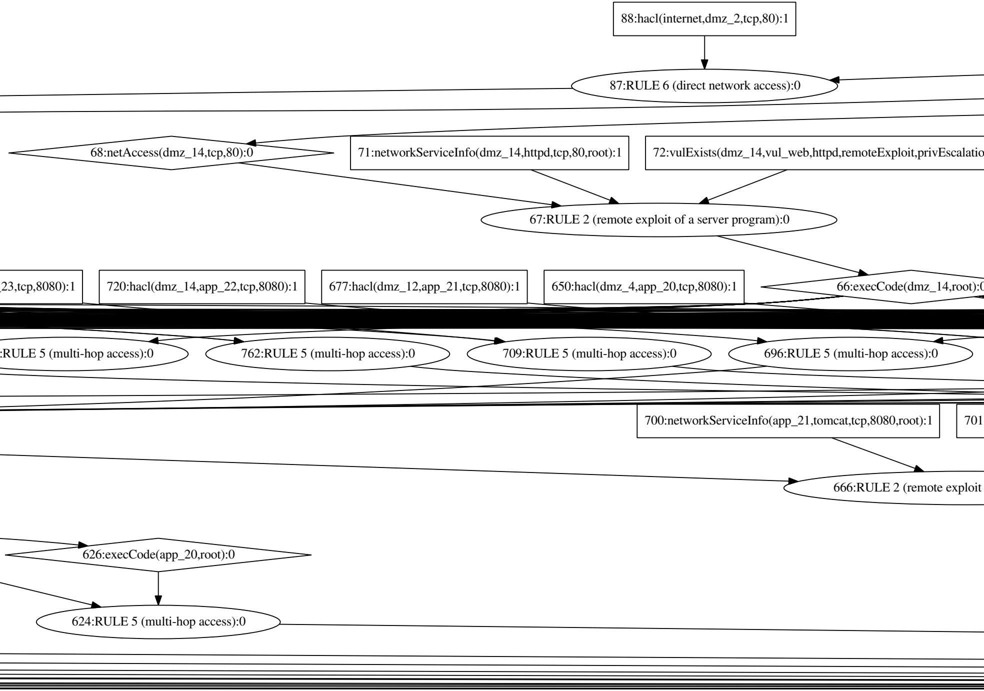

683:hacl(dmz_15,app_21,tcp,8080):1
658:hacl(dmz_8,app_20,tcp,8080):1
669:hacl(dmz_0,app_21,tcp,8080):1
638:hacl(dmz_12,app_20,tcp,8080):1
679:hacl(dmz_
670:RULE 5 (multi-hop access):0
614:RULE 5 (multi-hop access):0
680:RULE 5 (multi-hop access):0
686:RULE 5 (multi-hop access
661:networkServiceInfo(app_

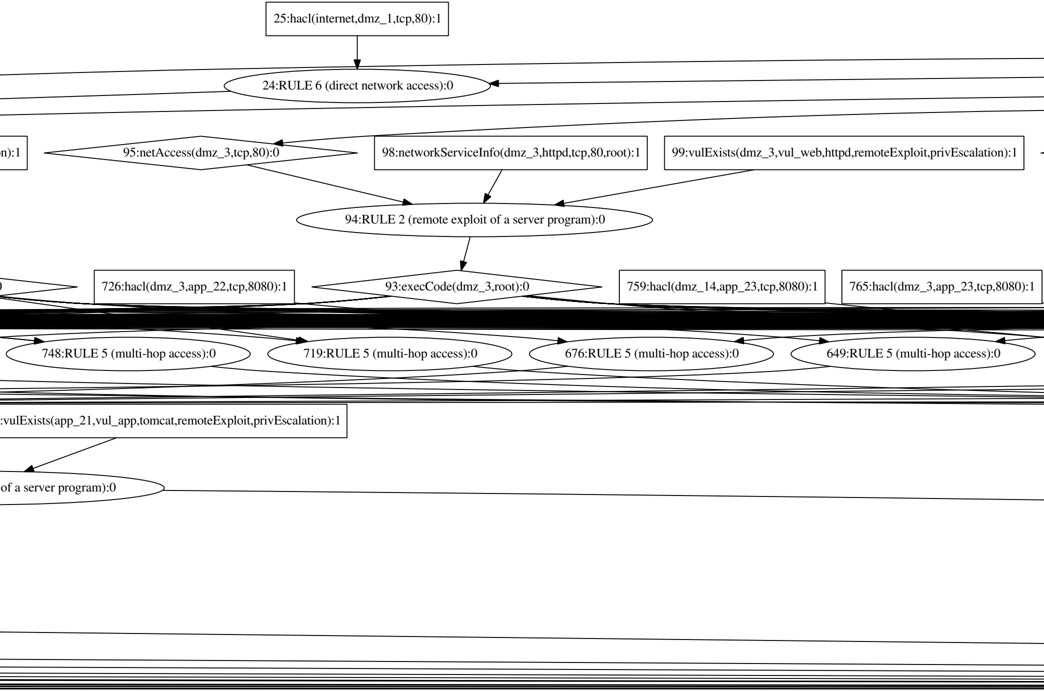

13,app_21,tcp,8080):1
691:hacl(dmz_5,app_21,tcp,8080):1
611:hacl(dmz_4,app_2,tcp,8080):1
656:hacl(dmz_7,app_20,tcp,8080):1
673:hacl(dmz_10,app_21,tcp,8080):1
):0
682:RULE 5 (multi-hop access):0
657:RULE 5 (multi-hop access):0
668:RULE 5 (multi-hop access):0
637:RULE 5 (multi-
_20,tomcat,tcp,8080,root):1
662:vulExists(app_20,vul_app,tomcat,remoteExploit,privEscalation):1
627:RULE 2 (remote exploit of a server program):0

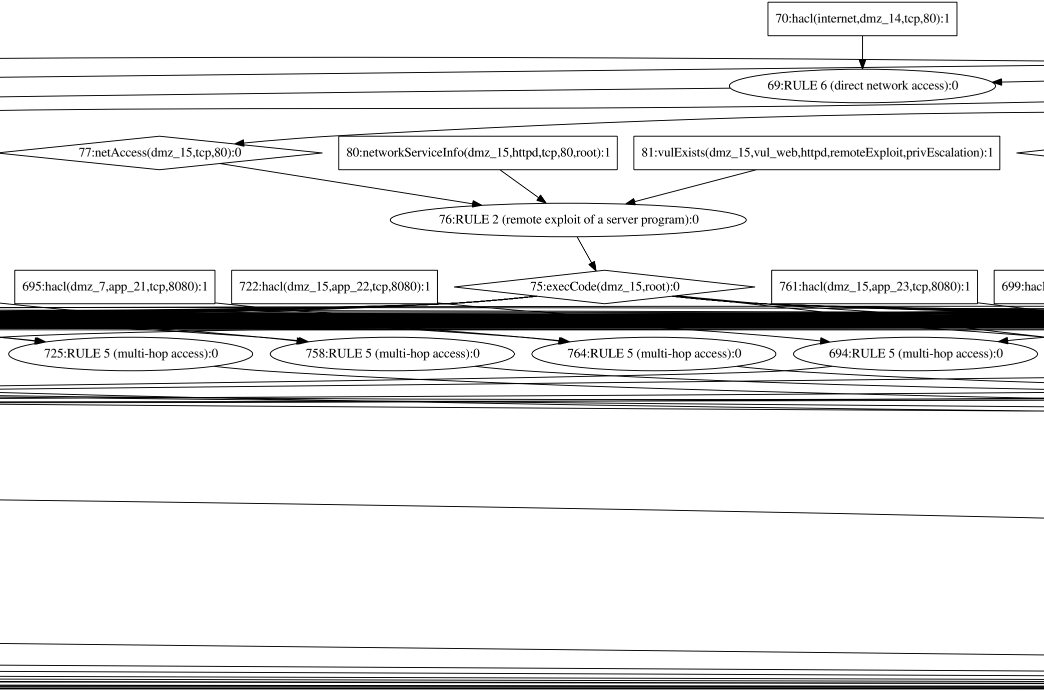

86:netAccess(dmz_2,tcp,80):0
89:networkServiceInfo(dmz_2,httpd,tcp,80,root):1
90:vulExists(dmz_2,vul_web,httpd,remoteExplo
85:RULE 2 (remote exploit of a server program):0
660:hacl(dmz_9,app_20,tcp,8080):1
675:hacl(dmz_11,app_21,tcp,8080):1
724:hacl(dmz_2,app_22,tcp,8080):1
654:hacl(dmz_6,app_20,tcp,8080):1
84:execCode(dmz_2,root):0
-hop access):0
678:RULE 5 (multi-hop access):0
690:RULE 5 (multi-hop access):0
610:RULE 5 (multi-hop access):0
655:RULE 5 (multi-hop access):0

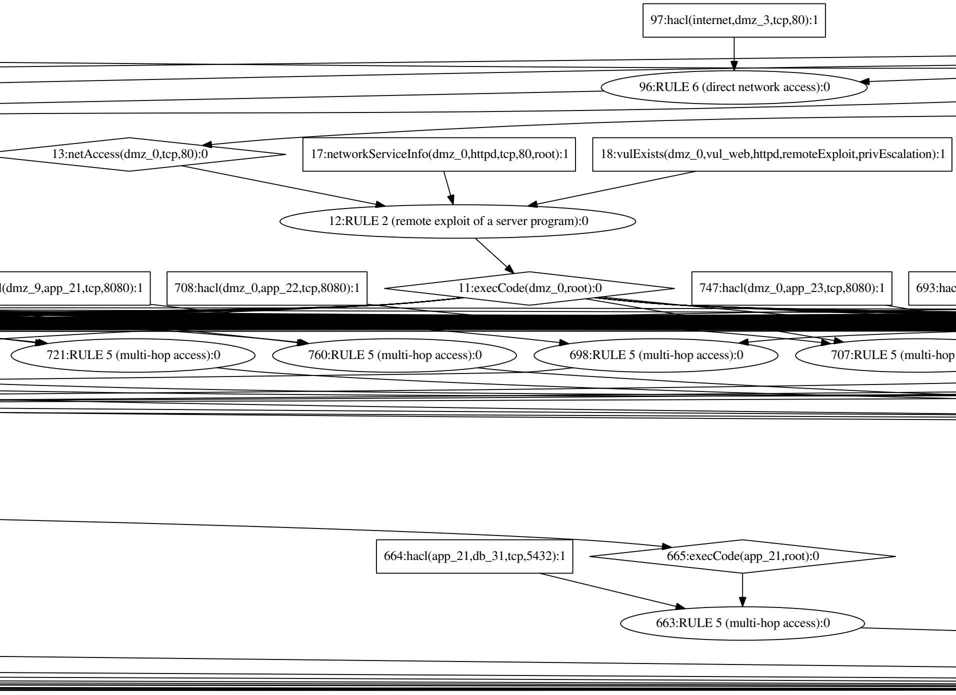


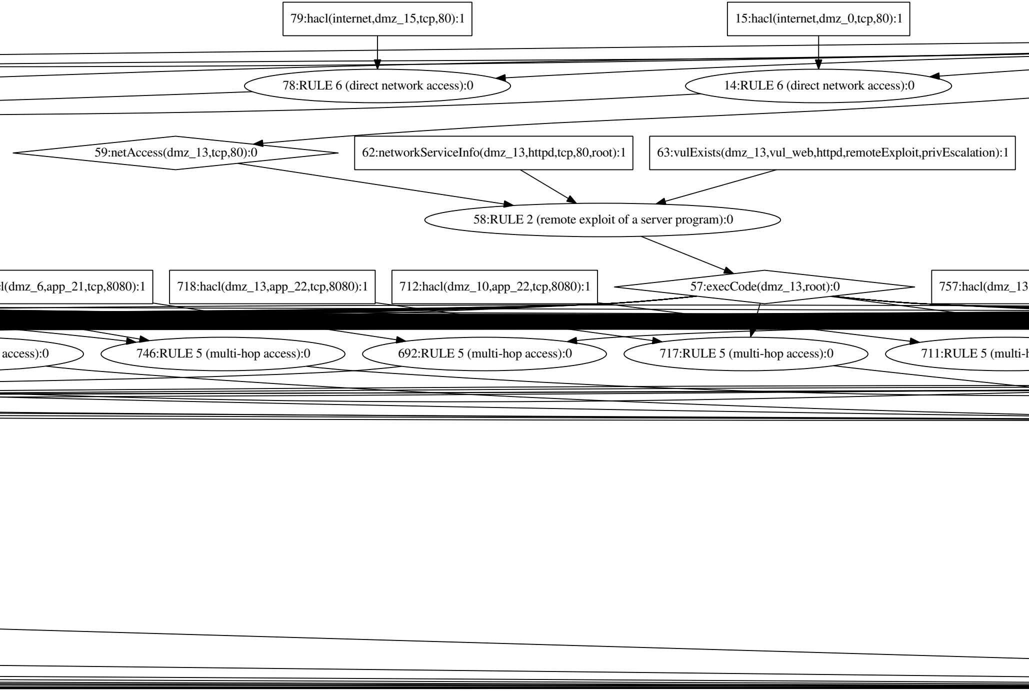
it,privEscalation):1
23:netAccess(dmz_1,tcp,80):0
26:networkServiceInfo(dmz_1,httpd,tcp,80,root):1
27:vulExists(dmz_1,vul_web,httpd,remoteExploit,privEscalation):1
22:RULE 2 (remote exploit of a server program):0
763:hacl(dmz_2,app_23,tcp,8080):1
710:hacl(dmz_1,app_22,tcp,8080):1
697:hacl(dmz_8,app_21,tcp,8080):1
21:execCode(dmz_1,root):0
749:hacl(dmz_1,app_
672:RULE 5 (multi-hop access):0
659:RULE 5 (multi-hop access):0
674:RULE 5 (multi-hop access):0
723:RULE 5 (multi-hop access):0
653
667:netAccess(app_21,tcp,8080):0
625:hacl(app_20,db_31,tcp,5432):1

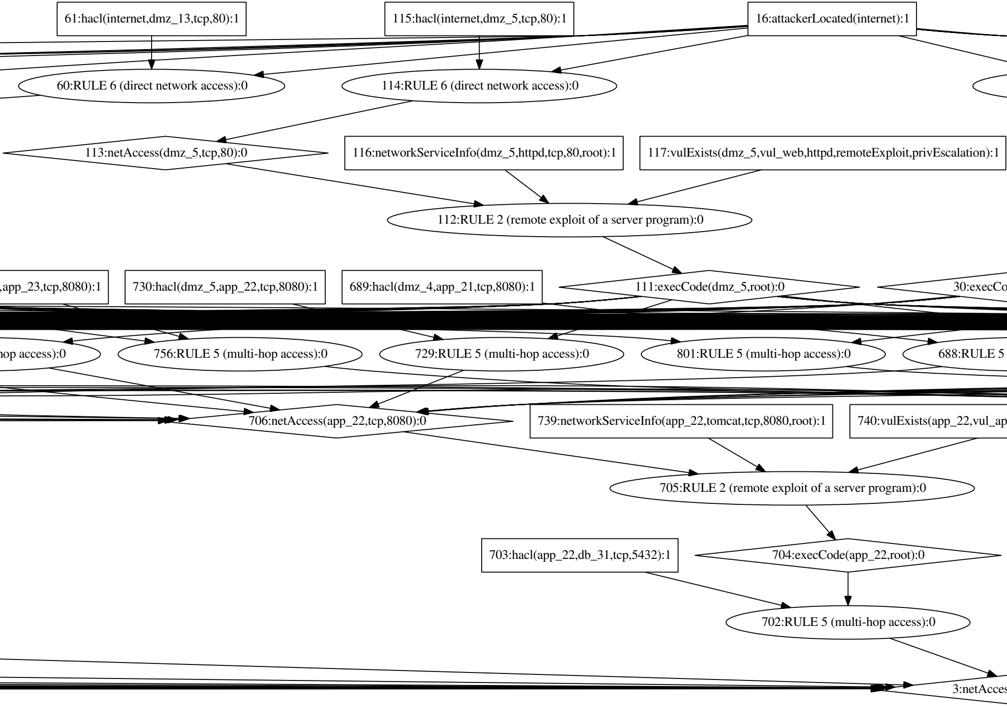

88:hacl(internet,dmz_2,tcp,80):1
87:RULE 6 (direct network access):0
68:netAccess(dmz_14,tcp,80):0
71:networkServiceInfo(dmz_14,httpd,tcp,80,root):1
72:vulExists(dmz_14,vul_web,httpd,remoteExploit,privEscalatio
67:RULE 2 (remote exploit of a server program):0
23,tcp,8080):1
720:hacl(dmz_14,app_22,tcp,8080):1
677:hacl(dmz_12,app_21,tcp,8080):1
650:hacl(dmz_4,app_20,tcp,8080):1
66:execCode(dmz_14,root):0
:RULE 5 (multi-hop access):0
762:RULE 5 (multi-hop access):0
709:RULE 5 (multi-hop access):0
696:RULE 5 (multi-hop access):0
700:networkServiceInfo(app_21,tomcat,tcp,8080,root):1
701
666:RULE 2 (remote exploit
626:execCode(app_20,root):0
624:RULE 5 (multi-hop access):0

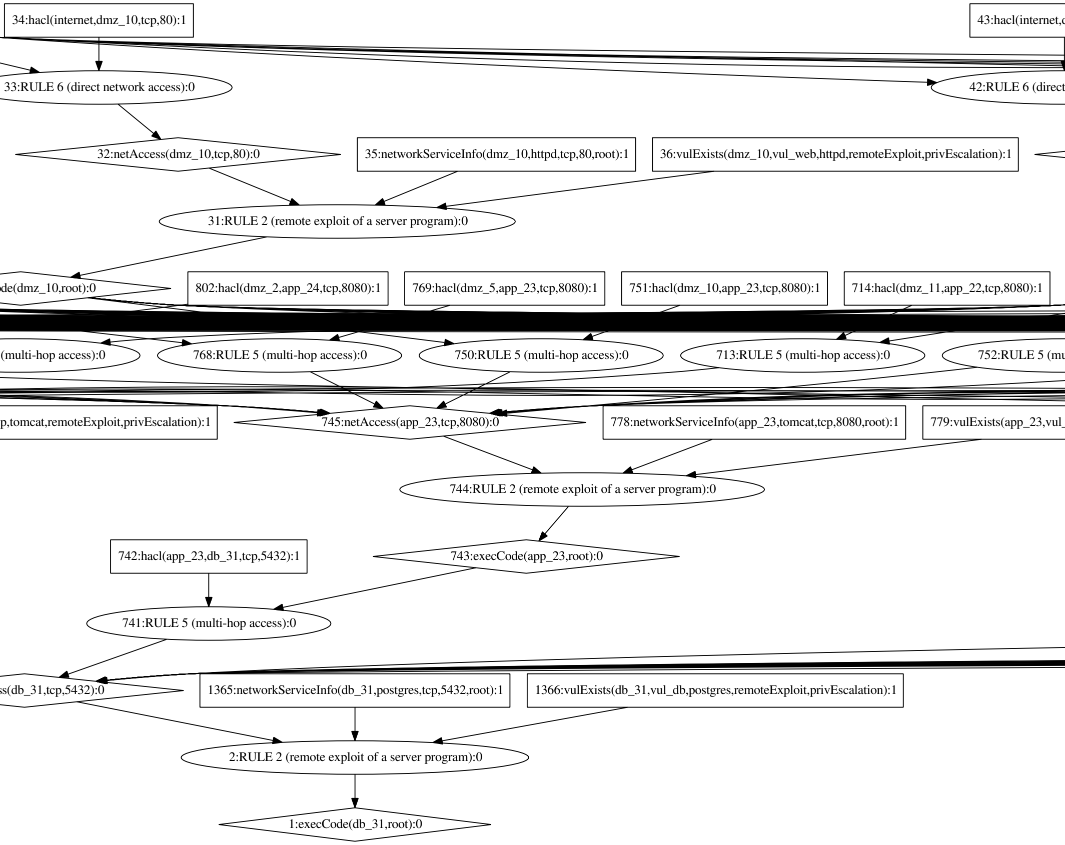

25:hacl(internet,dmz_1,tcp,80):1
24:RULE 6 (direct network access):0
n):1
95:netAccess(dmz_3,tcp,80):0
98:networkServiceInfo(dmz_3,httpd,tcp,80,root):1
99:vulExists(dmz_3,vul_web,httpd,remoteExploit,privEscalation):1
94:RULE 2 (remote exploit of a server program):0
0
726:hacl(dmz_3,app_22,tcp,8080):1
93:execCode(dmz_3,root):0
759:hacl(dmz_14,app_23,tcp,8080):1
765:hacl(dmz_3,app_23,tcp,8080):1
748:RULE 5 (multi-hop access):0
719:RULE 5 (multi-hop access):0
676:RULE 5 (multi-hop access):0
649:RULE 5 (multi-hop access):0
:vulExists(app_21,vul_app,tomcat,remoteExploit,privEscalation):1
of a server program):0

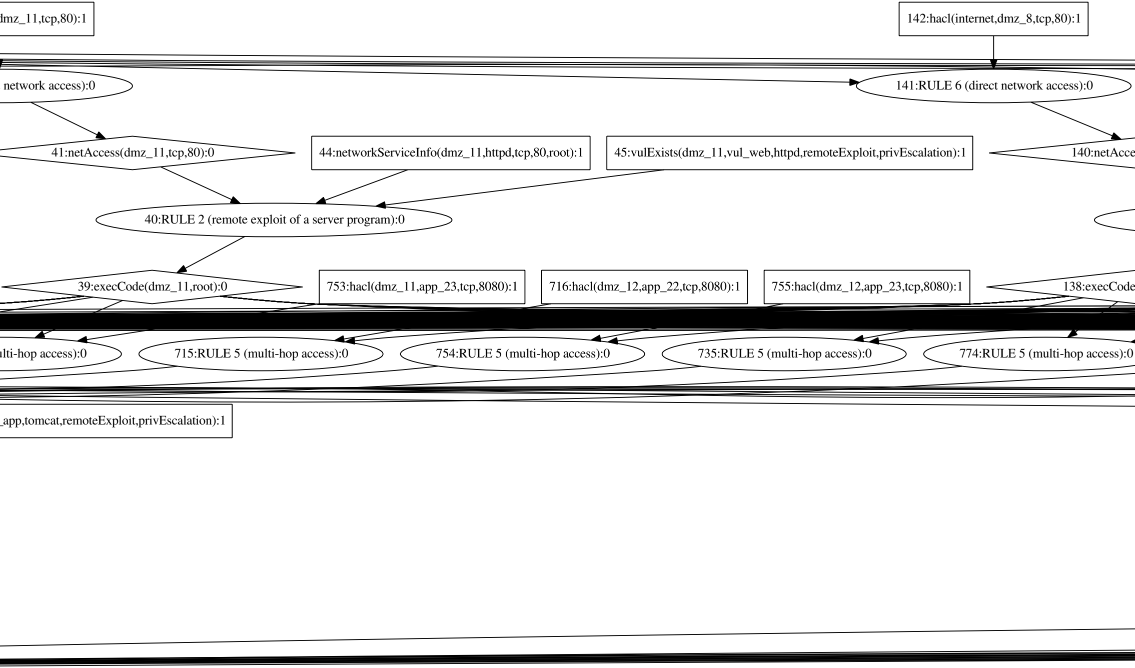

70:hacl(internet,dmz_14,tcp,80):1
69:RULE 6 (direct network access):0
77:netAccess(dmz_15,tcp,80):0
80:networkServiceInfo(dmz_15,httpd,tcp,80,root):1
81:vulExists(dmz_15,vul_web,httpd,remoteExploit,privEscalation):1
76:RULE 2 (remote exploit of a server program):0
695:hacl(dmz_7,app_21,tcp,8080):1
722:hacl(dmz_15,app_22,tcp,8080):1
75:execCode(dmz_15,root):0
761:hacl(dmz_15,app_23,tcp,8080):1
699:hacl
725:RULE 5 (multi-hop access):0
758:RULE 5 (multi-hop access):0
764:RULE 5 (multi-hop access):0
694:RULE 5 (multi-hop access):0

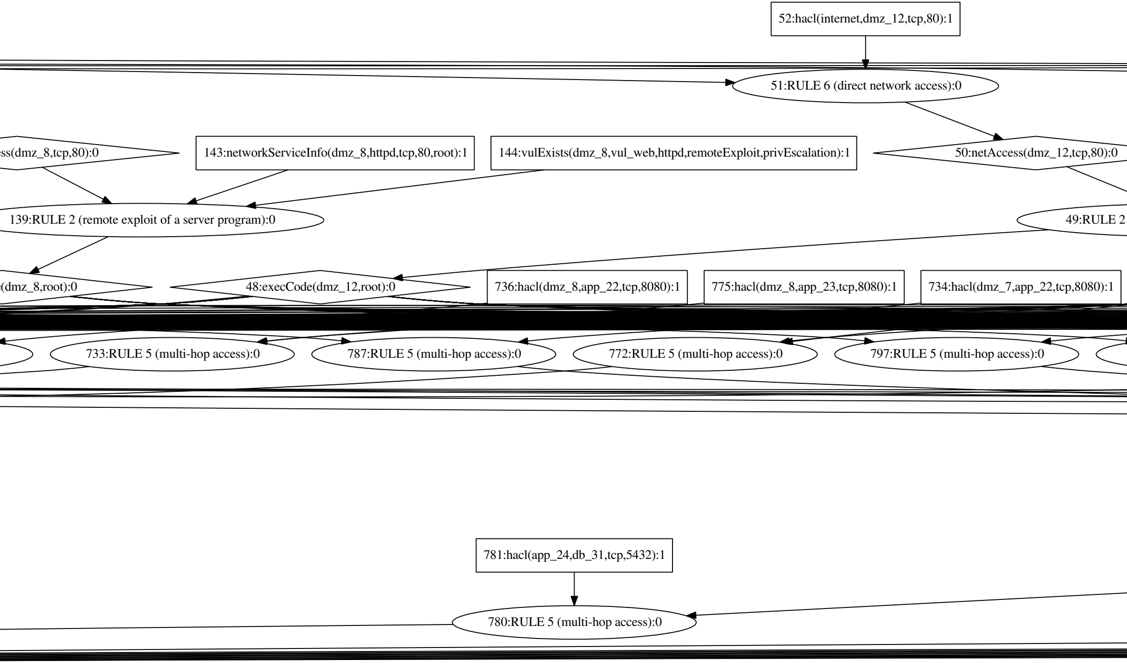

97:hacl(internet,dmz_3,tcp,80):1
96:RULE 6 (direct network access):0
13:netAccess(dmz_0,tcp,80):0
17:networkServiceInfo(dmz_0,httpd,tcp,80,root):1
18:vulExists(dmz_0,vul_web,httpd,remoteExploit,privEscalation):1
12:RULE 2 (remote exploit of a server program):0
l(dmz_9,app_21,tcp,8080):1
708:hacl(dmz_0,app_22,tcp,8080):1
11:execCode(dmz_0,root):0
747:hacl(dmz_0,app_23,tcp,8080):1
693:hac
721:RULE 5 (multi-hop access):0
760:RULE 5 (multi-hop access):0
698:RULE 5 (multi-hop access):0
707:RULE 5 (multi-hop
664:hacl(app_21,db_31,tcp,5432):1
665:execCode(app_21,root):0
663:RULE 5 (multi-hop access):0

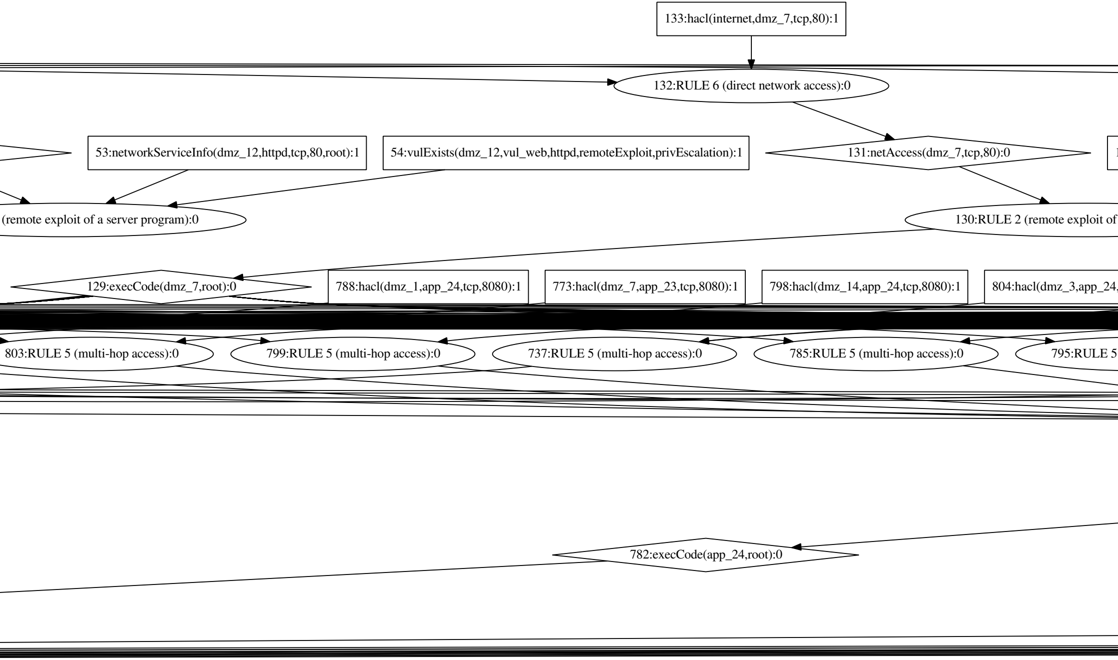

79:hacl(internet,dmz_15,tcp,80):1
15:hacl(internet,dmz_0,tcp,80):1
78:RULE 6 (direct network access):0
14:RULE 6 (direct network access):0
59:netAccess(dmz_13,tcp,80):0
62:networkServiceInfo(dmz_13,httpd,tcp,80,root):1
63:vulExists(dmz_13,vul_web,httpd,remoteExploit,privEscalation):1
58:RULE 2 (remote exploit of a server program):0
l(dmz_6,app_21,tcp,8080):1
718:hacl(dmz_13,app_22,tcp,8080):1
712:hacl(dmz_10,app_22,tcp,8080):1
57:execCode(dmz_13,root):0
757:hacl(dmz_13
access):0
746:RULE 5 (multi-hop access):0
692:RULE 5 (multi-hop access):0
717:RULE 5 (multi-hop access):0
711:RULE 5 (multi-h

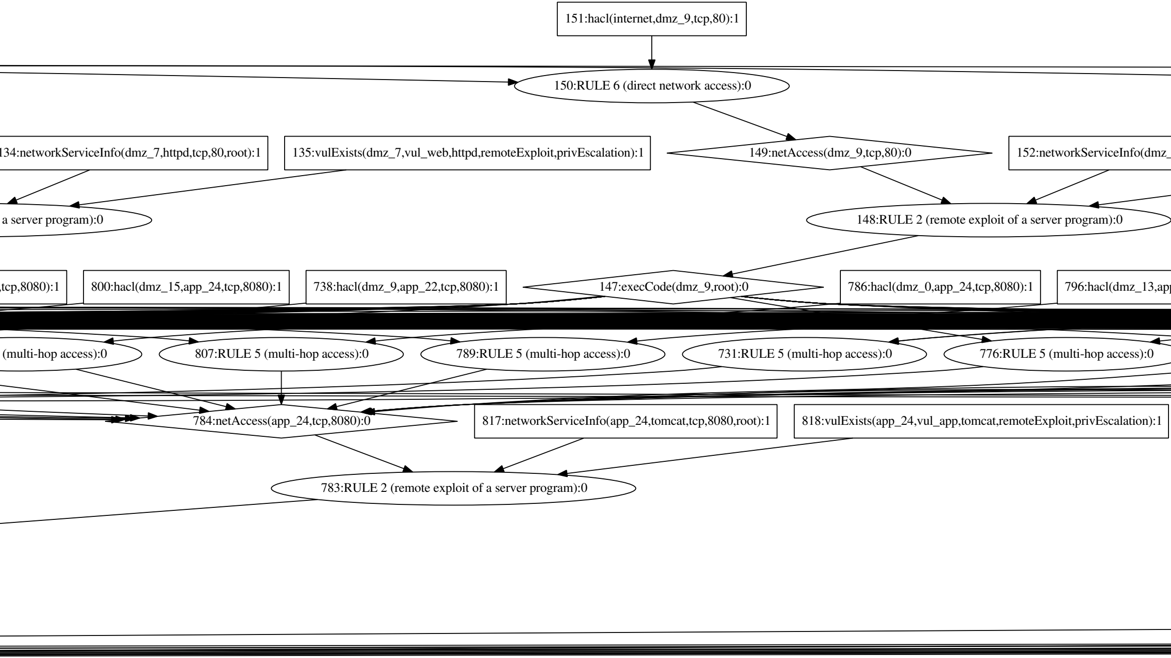

61:hacl(internet,dmz_13,tcp,80):1
115:hacl(internet,dmz_5,tcp,80):1
16:attackerLocated(internet):1
60:RULE 6 (direct network access):0
114:RULE 6 (direct network access):0
113:netAccess(dmz_5,tcp,80):0
116:networkServiceInfo(dmz_5,httpd,tcp,80,root):1
117:vulExists(dmz_5,vul_web,httpd,remoteExploit,privEscalation):1
112:RULE 2 (remote exploit of a server program):0
,app_23,tcp,8080):1
730:hacl(dmz_5,app_22,tcp,8080):1
689:hacl(dmz_4,app_21,tcp,8080):1
111:execCode(dmz_5,root):0
30:execCo
hop access):0
756:RULE 5 (multi-hop access):0
729:RULE 5 (multi-hop access):0
801:RULE 5 (multi-hop access):0
688:RULE 5
706:netAccess(app_22,tcp,8080):0
739:networkServiceInfo(app_22,tomcat,tcp,8080,root):1
740:vulExists(app_22,vul_ap
705:RULE 2 (remote exploit of a server program):0
703:hacl(app_22,db_31,tcp,5432):1
704:execCode(app_22,root):0
702:RULE 5 (multi-hop access):0
3:netAcces

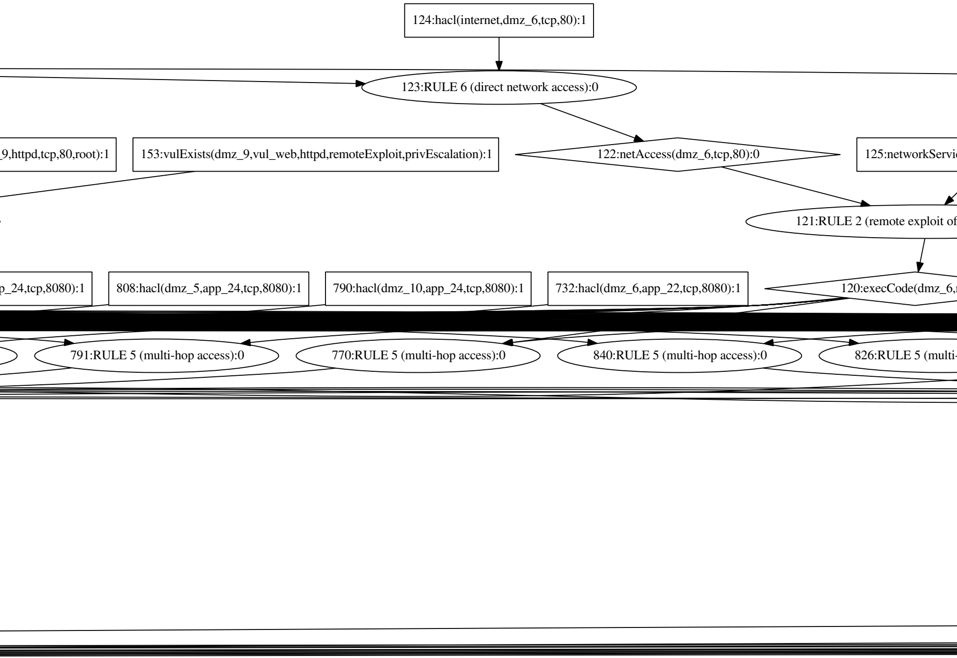

34:hacl(internet,dmz_10,tcp,80):1
43:hacl(internet,d
33:RULE 6 (direct network access):0
42:RULE 6 (direct
32:netAccess(dmz_10,tcp,80):0
35:networkServiceInfo(dmz_10,httpd,tcp,80,root):1
36:vulExists(dmz_10,vul_web,httpd,remoteExploit,privEscalation):1
31:RULE 2 (remote exploit of a server program):0
de(dmz_10,root):0
802:hacl(dmz_2,app_24,tcp,8080):1
769:hacl(dmz_5,app_23,tcp,8080):1
751:hacl(dmz_10,app_23,tcp,8080):1
714:hacl(dmz_11,app_22,tcp,8080):1
(multi-hop access):0
768:RULE 5 (multi-hop access):0
750:RULE 5 (multi-hop access):0
713:RULE 5 (multi-hop access):0
752:RULE 5 (mu
p,tomcat,remoteExploit,privEscalation):1
745:netAccess(app_23,tcp,8080):0
778:networkServiceInfo(app_23,tomcat,tcp,8080,root):1
779:vulExists(app_23,vul_
744:RULE 2 (remote exploit of a server program):0
742:hacl(app_23,db_31,tcp,5432):1
743:execCode(app_23,root):0
741:RULE 5 (multi-hop access):0
ss(db_31,tcp,5432):0
1365:networkServiceInfo(db_31,postgres,tcp,5432,root):1
1366:vulExists(db_31,vul_db,postgres,remoteExploit,privEscalation):1
2:RULE 2 (remote exploit of a server program):0
1:execCode(db_31,root):0

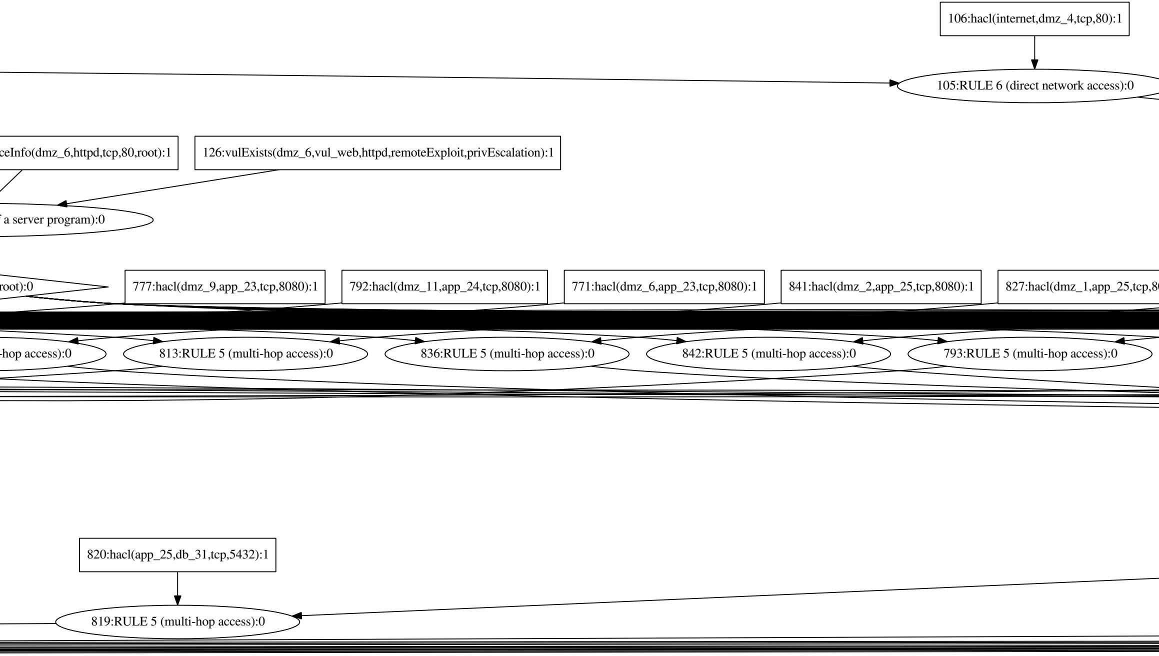

dmz_11,tcp,80):1
142:hacl(internet,dmz_8,tcp,80):1
network access):0
141:RULE 6 (direct network access):0
41:netAccess(dmz_11,tcp,80):0
44:networkServiceInfo(dmz_11,httpd,tcp,80,root):1
45:vulExists(dmz_11,vul_web,httpd,remoteExploit,privEscalation):1
140:netAcce
40:RULE 2 (remote exploit of a server program):0
39:execCode(dmz_11,root):0
753:hacl(dmz_11,app_23,tcp,8080):1
716:hacl(dmz_12,app_22,tcp,8080):1
755:hacl(dmz_12,app_23,tcp,8080):1
138:execCode
lti-hop access):0
715:RULE 5 (multi-hop access):0
754:RULE 5 (multi-hop access):0
735:RULE 5 (multi-hop access):0
774:RULE 5 (multi-hop access):0
_app,tomcat,remoteExploit,privEscalation):1

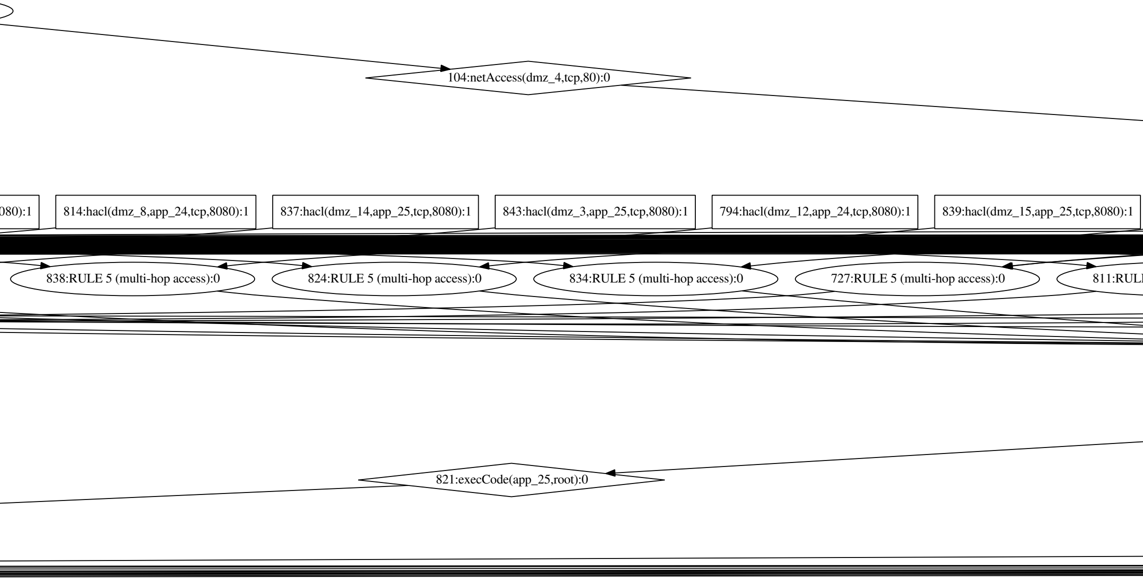

52:hacl(internet,dmz_12,tcp,80):1
51:RULE 6 (direct network access):0
ss(dmz_8,tcp,80):0
143:networkServiceInfo(dmz_8,httpd,tcp,80,root):1
144:vulExists(dmz_8,vul_web,httpd,remoteExploit,privEscalation):1
50:netAccess(dmz_12,tcp,80):0
139:RULE 2 (remote exploit of a server program):0
49:RULE 2
(dmz_8,root):0
48:execCode(dmz_12,root):0
736:hacl(dmz_8,app_22,tcp,8080):1
775:hacl(dmz_8,app_23,tcp,8080):1
734:hacl(dmz_7,app_22,tcp,8080):1
733:RULE 5 (multi-hop access):0
787:RULE 5 (multi-hop access):0
772:RULE 5 (multi-hop access):0
797:RULE 5 (multi-hop access):0
781:hacl(app_24,db_31,tcp,5432):1
780:RULE 5 (multi-hop access):0

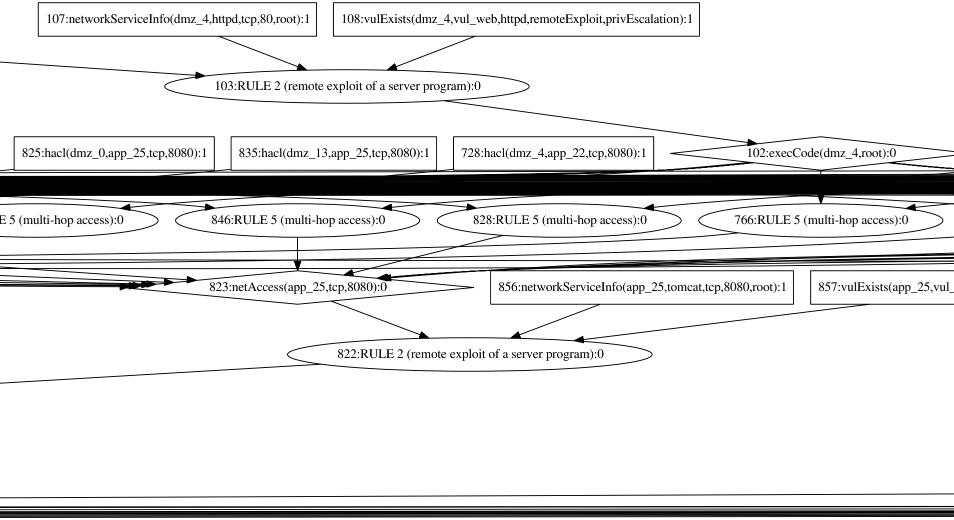

133:hacl(internet,dmz_7,tcp,80):1
132:RULE 6 (direct network access):0
53:networkServiceInfo(dmz_12,httpd,tcp,80,root):1
54:vulExists(dmz_12,vul_web,httpd,remoteExploit,privEscalation):1
131:netAccess(dmz_7,tcp,80):0
1
(remote exploit of a server program):0
130:RULE 2 (remote exploit of
129:execCode(dmz_7,root):0
788:hacl(dmz_1,app_24,tcp,8080):1
773:hacl(dmz_7,app_23,tcp,8080):1
798:hacl(dmz_14,app_24,tcp,8080):1
804:hacl(dmz_3,app_24,
803:RULE 5 (multi-hop access):0
799:RULE 5 (multi-hop access):0
737:RULE 5 (multi-hop access):0
785:RULE 5 (multi-hop access):0
795:RULE 5
782:execCode(app_24,root):0

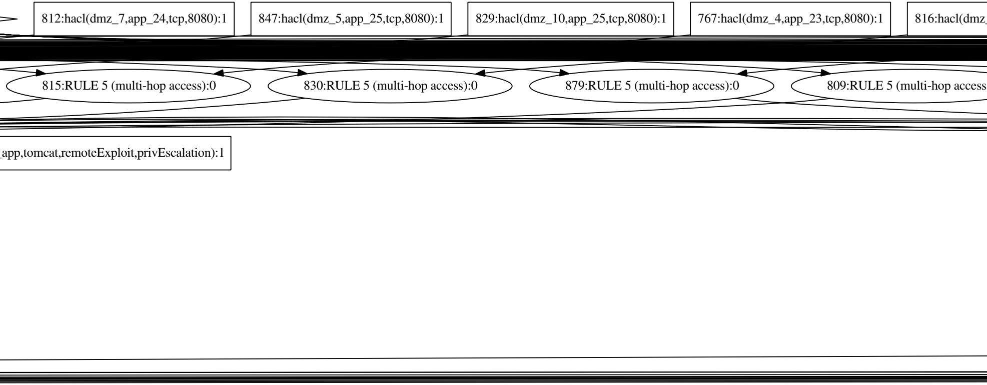

151:hacl(internet,dmz_9,tcp,80):1
150:RULE 6 (direct network access):0
134:networkServiceInfo(dmz_7,httpd,tcp,80,root):1
135:vulExists(dmz_7,vul_web,httpd,remoteExploit,privEscalation):1
149:netAccess(dmz_9,tcp,80):0
152:networkServiceInfo(dmz_
a server program):0
148:RULE 2 (remote exploit of a server program):0
tcp,8080):1
800:hacl(dmz_15,app_24,tcp,8080):1
738:hacl(dmz_9,app_22,tcp,8080):1
147:execCode(dmz_9,root):0
786:hacl(dmz_0,app_24,tcp,8080):1
796:hacl(dmz_13,app
(multi-hop access):0
807:RULE 5 (multi-hop access):0
789:RULE 5 (multi-hop access):0
731:RULE 5 (multi-hop access):0
776:RULE 5 (multi-hop access):0
784:netAccess(app_24,tcp,8080):0
817:networkServiceInfo(app_24,tomcat,tcp,8080,root):1
818:vulExists(app_24,vul_app,tomcat,remoteExploit,privEscalation):1
783:RULE 2 (remote exploit of a server program):0

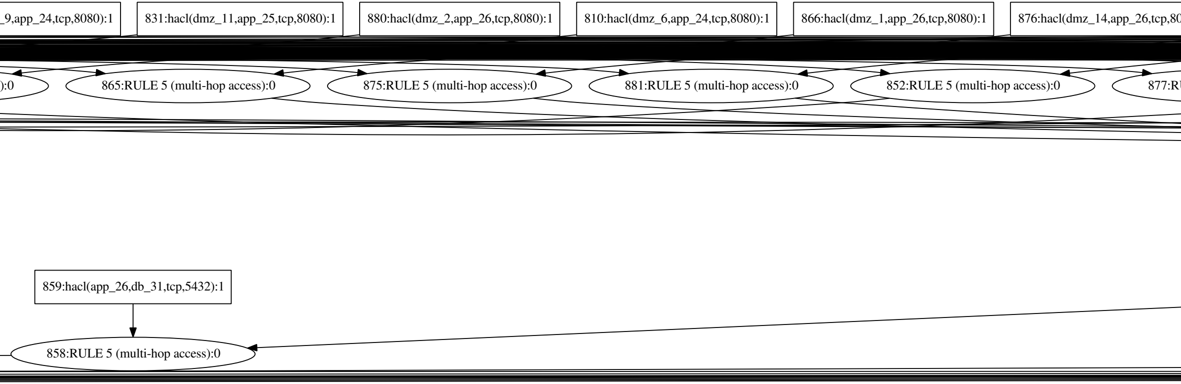

124:hacl(internet,dmz_6,tcp,80):1
123:RULE 6 (direct network access):0
9,httpd,tcp,80,root):1
153:vulExists(dmz_9,vul_web,httpd,remoteExploit,privEscalation):1
122:netAccess(dmz_6,tcp,80):0
125:networkServi
121:RULE 2 (remote exploit of
p_24,tcp,8080):1
808:hacl(dmz_5,app_24,tcp,8080):1
790:hacl(dmz_10,app_24,tcp,8080):1
732:hacl(dmz_6,app_22,tcp,8080):1
120:execCode(dmz_6,
791:RULE 5 (multi-hop access):0
770:RULE 5 (multi-hop access):0
840:RULE 5 (multi-hop access):0
826:RULE 5 (multi-

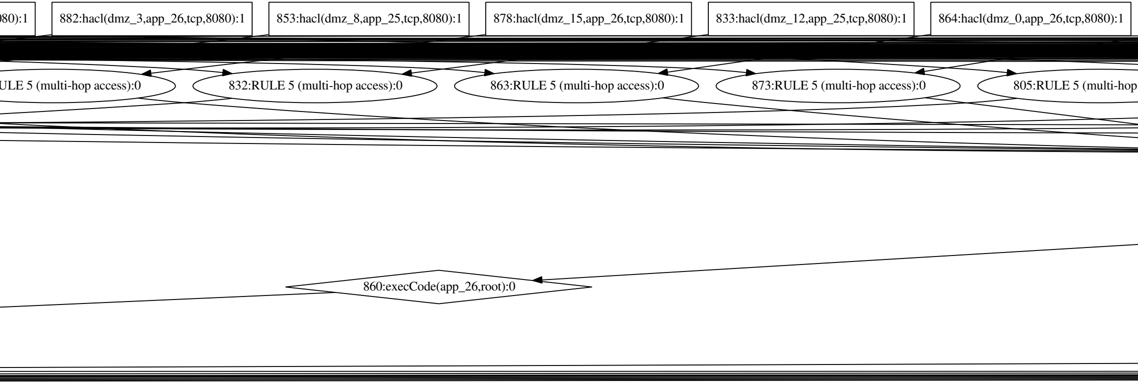

106:hacl(internet,dmz_4,tcp,80):1
105:RULE 6 (direct network access):0
ceInfo(dmz_6,httpd,tcp,80,root):1
126:vulExists(dmz_6,vul_web,httpd,remoteExploit,privEscalation):1
f a server program):0
root):0
777:hacl(dmz_9,app_23,tcp,8080):1
792:hacl(dmz_11,app_24,tcp,8080):1
771:hacl(dmz_6,app_23,tcp,8080):1
841:hacl(dmz_2,app_25,tcp,8080):1
827:hacl(dmz_1,app_25,tcp,8
-hop access):0
813:RULE 5 (multi-hop access):0
836:RULE 5 (multi-hop access):0
842:RULE 5 (multi-hop access):0
793:RULE 5 (multi-hop access):0
820:hacl(app_25,db_31,tcp,5432):1
819:RULE 5 (multi-hop access):0

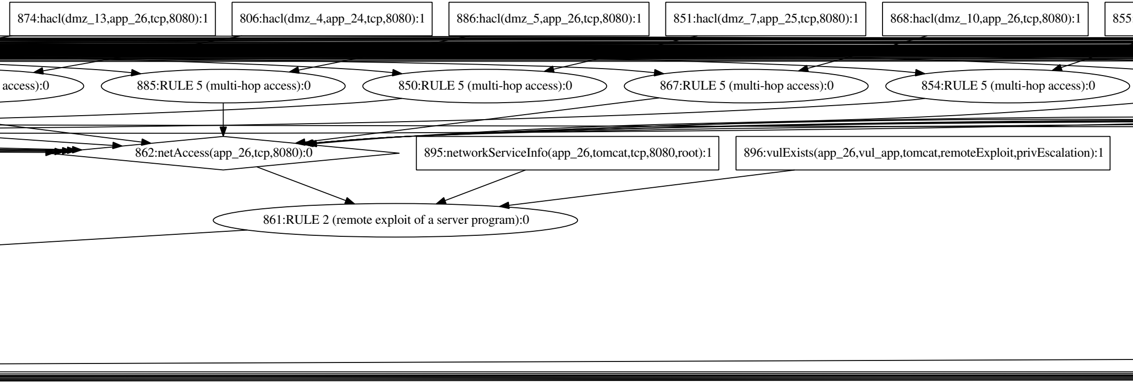

104:netAccess(dmz_4,tcp,80):0
080):1
814:hacl(dmz_8,app_24,tcp,8080):1
837:hacl(dmz_14,app_25,tcp,8080):1
843:hacl(dmz_3,app_25,tcp,8080):1
794:hacl(dmz_12,app_24,tcp,8080):1
839:hacl(dmz_15,app_25,tcp,8080):1
838:RULE 5 (multi-hop access):0
824:RULE 5 (multi-hop access):0
834:RULE 5 (multi-hop access):0
727:RULE 5 (multi-hop access):0
811:RUL
821:execCode(app_25,root):0

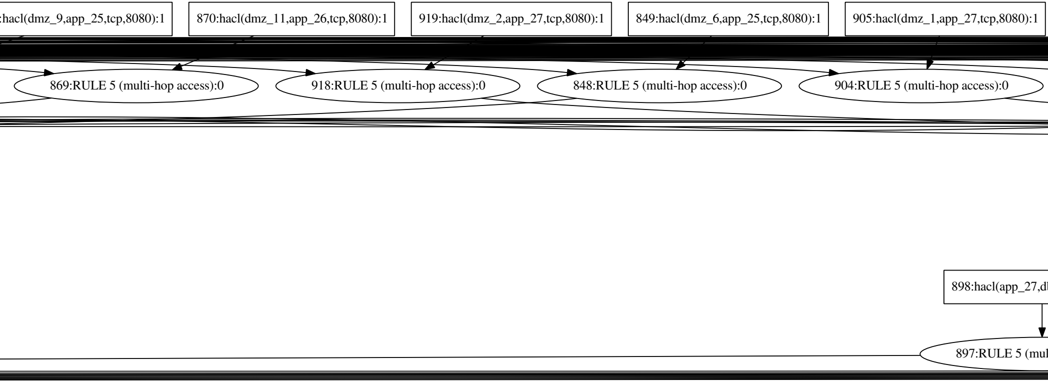

107:networkServiceInfo(dmz_4,httpd,tcp,80,root):1
108:vulExists(dmz_4,vul_web,httpd,remoteExploit,privEscalation):1
103:RULE 2 (remote exploit of a server program):0
825:hacl(dmz_0,app_25,tcp,8080):1
835:hacl(dmz_13,app_25,tcp,8080):1
728:hacl(dmz_4,app_22,tcp,8080):1
102:execCode(dmz_4,root):0
E 5 (multi-hop access):0
846:RULE 5 (multi-hop access):0
828:RULE 5 (multi-hop access):0
766:RULE 5 (multi-hop access):0
823:netAccess(app_25,tcp,8080):0
856:networkServiceInfo(app_25,tomcat,tcp,8080,root):1
857:vulExists(app_25,vul_
822:RULE 2 (remote exploit of a server program):0

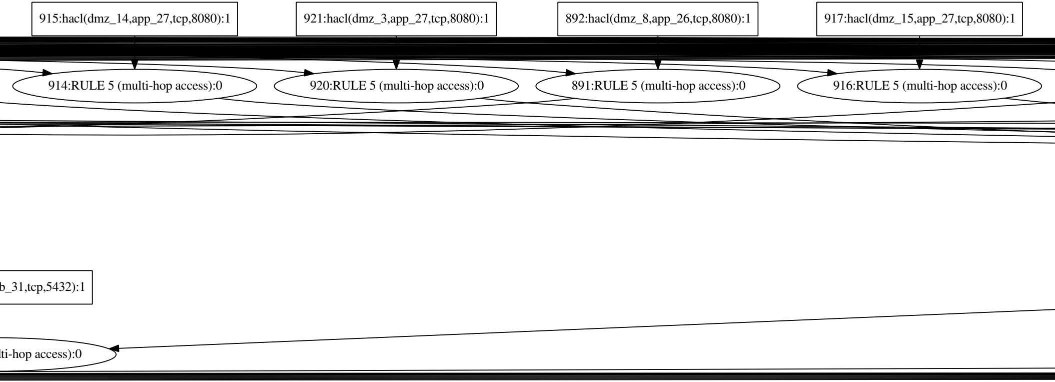

812:hacl(dmz_7,app_24,tcp,8080):1
847:hacl(dmz_5,app_25,tcp,8080):1
829:hacl(dmz_10,app_25,tcp,8080):1
767:hacl(dmz_4,app_23,tcp,8080):1
816:hacl(dmz_
815:RULE 5 (multi-hop access):0
830:RULE 5 (multi-hop access):0
879:RULE 5 (multi-hop access):0
809:RULE 5 (multi-hop access
_app,tomcat,remoteExploit,privEscalation):1

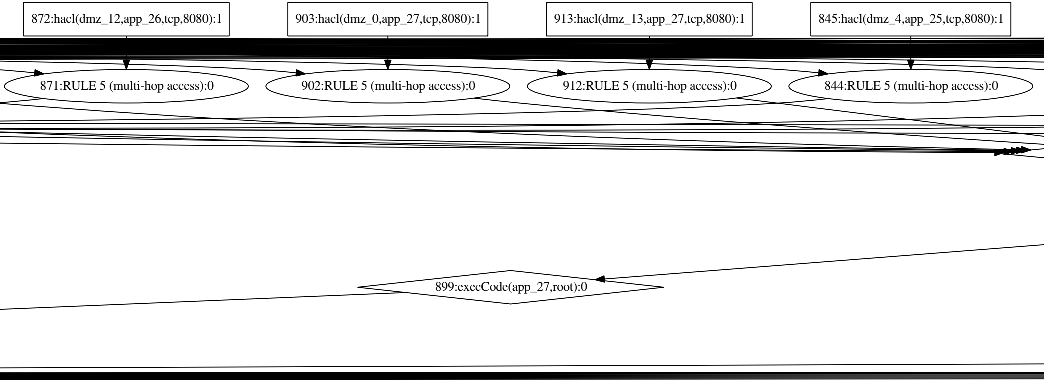

_9,app_24,tcp,8080):1
831:hacl(dmz_11,app_25,tcp,8080):1
880:hacl(dmz_2,app_26,tcp,8080):1
810:hacl(dmz_6,app_24,tcp,8080):1
866:hacl(dmz_1,app_26,tcp,8080):1
876:hacl(dmz_14,app_26,tcp,80
):0
865:RULE 5 (multi-hop access):0
875:RULE 5 (multi-hop access):0
881:RULE 5 (multi-hop access):0
852:RULE 5 (multi-hop access):0
877:RU
859:hacl(app_26,db_31,tcp,5432):1
858:RULE 5 (multi-hop access):0

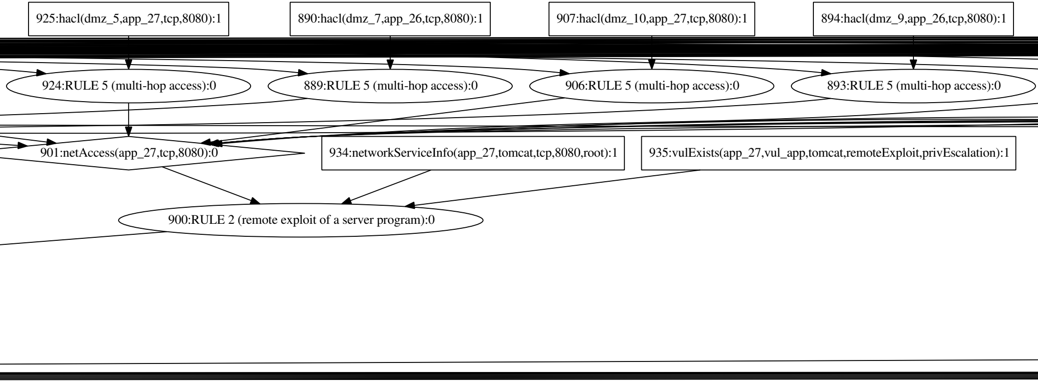

080):1
882:hacl(dmz_3,app_26,tcp,8080):1
853:hacl(dmz_8,app_25,tcp,8080):1
878:hacl(dmz_15,app_26,tcp,8080):1
833:hacl(dmz_12,app_25,tcp,8080):1
864:hacl(dmz_0,app_26,tcp,8080):1
ULE 5 (multi-hop access):0
832:RULE 5 (multi-hop access):0
863:RULE 5 (multi-hop access):0
873:RULE 5 (multi-hop access):0
805:RULE 5 (multi-hop
860:execCode(app_26,root):0

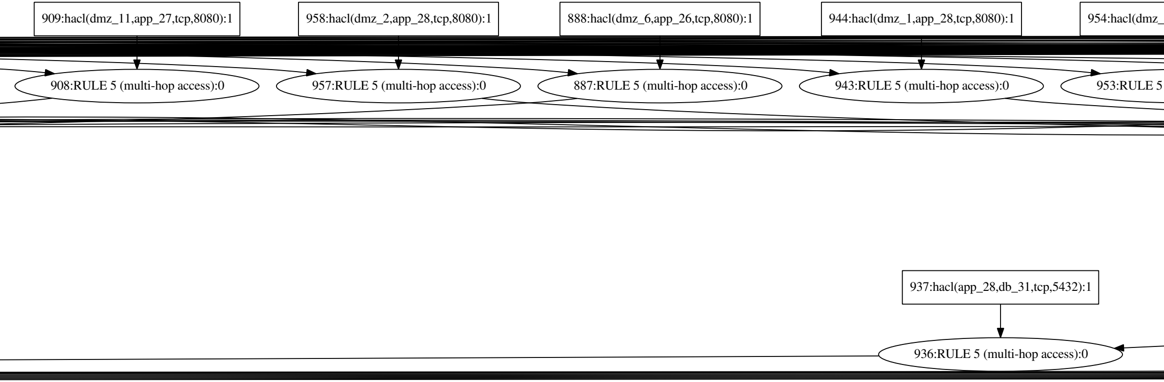

874:hacl(dmz_13,app_26,tcp,8080):1
806:hacl(dmz_4,app_24,tcp,8080):1
886:hacl(dmz_5,app_26,tcp,8080):1
851:hacl(dmz_7,app_25,tcp,8080):1
868:hacl(dmz_10,app_26,tcp,8080):1
855:
access):0
885:RULE 5 (multi-hop access):0
850:RULE 5 (multi-hop access):0
867:RULE 5 (multi-hop access):0
854:RULE 5 (multi-hop access):0
862:netAccess(app_26,tcp,8080):0
895:networkServiceInfo(app_26,tomcat,tcp,8080,root):1
896:vulExists(app_26,vul_app,tomcat,remoteExploit,privEscalation):1
861:RULE 2 (remote exploit of a server program):0

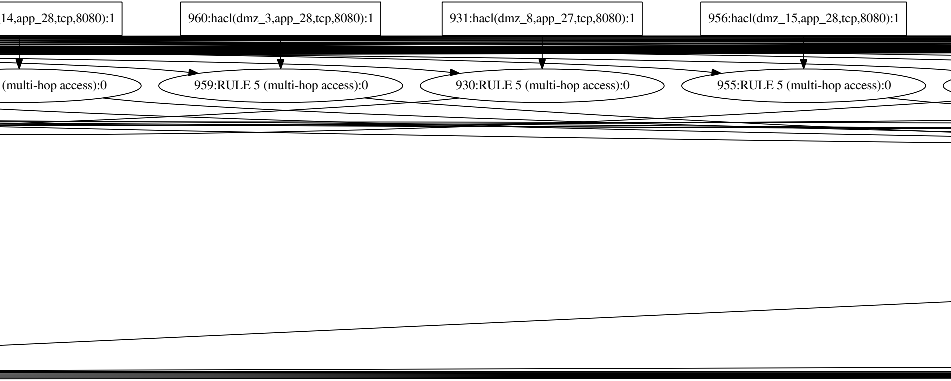

:hacl(dmz_9,app_25,tcp,8080):1
870:hacl(dmz_11,app_26,tcp,8080):1
919:hacl(dmz_2,app_27,tcp,8080):1
849:hacl(dmz_6,app_25,tcp,8080):1
905:hacl(dmz_1,app_27,tcp,8080):1
869:RULE 5 (multi-hop access):0
918:RULE 5 (multi-hop access):0
848:RULE 5 (multi-hop access):0
904:RULE 5 (multi-hop access):0
898:hacl(app_27,d
897:RULE 5 (mul

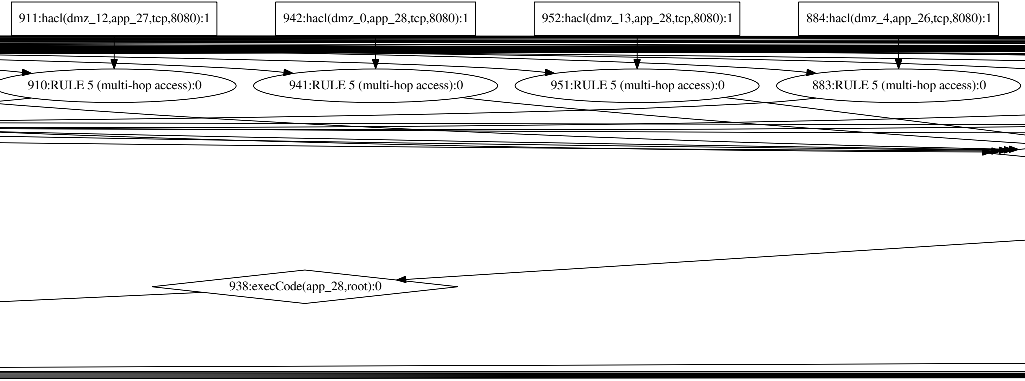

915:hacl(dmz_14,app_27,tcp,8080):1
921:hacl(dmz_3,app_27,tcp,8080):1
892:hacl(dmz_8,app_26,tcp,8080):1
917:hacl(dmz_15,app_27,tcp,8080):1
914:RULE 5 (multi-hop access):0
920:RULE 5 (multi-hop access):0
891:RULE 5 (multi-hop access):0
916:RULE 5 (multi-hop access):0
b_31,tcp,5432):1
lti-hop access):0

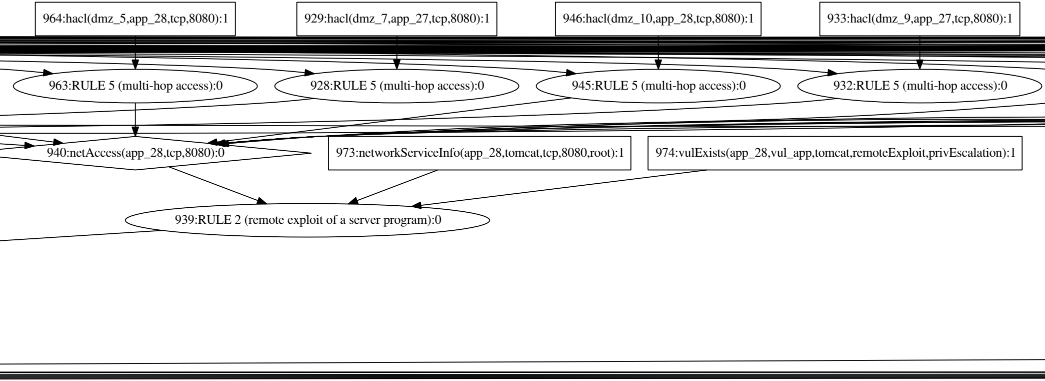

872:hacl(dmz_12,app_26,tcp,8080):1
903:hacl(dmz_0,app_27,tcp,8080):1
913:hacl(dmz_13,app_27,tcp,8080):1
845:hacl(dmz_4,app_25,tcp,8080):1
871:RULE 5 (multi-hop access):0
902:RULE 5 (multi-hop access):0
912:RULE 5 (multi-hop access):0
844:RULE 5 (multi-hop access):0
899:execCode(app_27,root):0

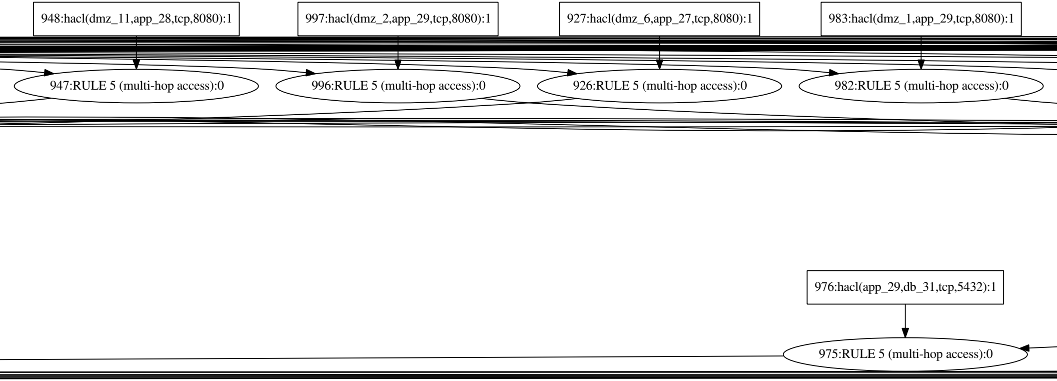

925:hacl(dmz_5,app_27,tcp,8080):1
890:hacl(dmz_7,app_26,tcp,8080):1
907:hacl(dmz_10,app_27,tcp,8080):1
894:hacl(dmz_9,app_26,tcp,8080):1
924:RULE 5 (multi-hop access):0
889:RULE 5 (multi-hop access):0
906:RULE 5 (multi-hop access):0
893:RULE 5 (multi-hop access):0
901:netAccess(app_27,tcp,8080):0
934:networkServiceInfo(app_27,tomcat,tcp,8080,root):1
935:vulExists(app_27,vul_app,tomcat,remoteExploit,privEscalation):1
900:RULE 2 (remote exploit of a server program):0

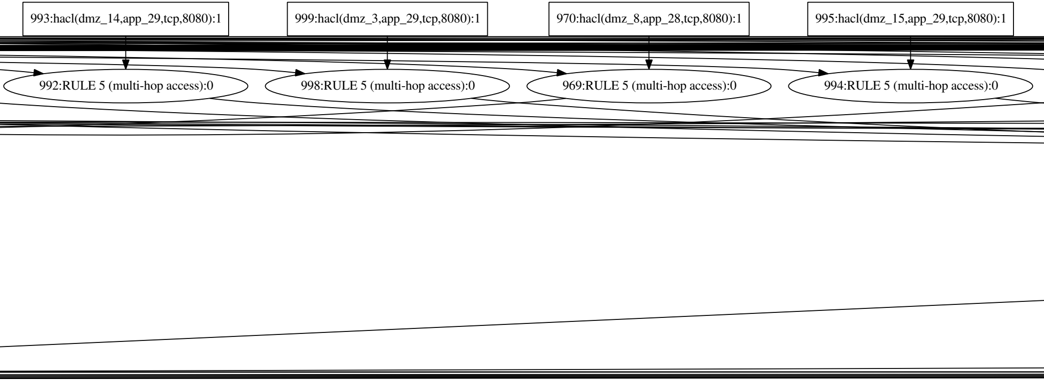

909:hacl(dmz_11,app_27,tcp,8080):1
958:hacl(dmz_2,app_28,tcp,8080):1
888:hacl(dmz_6,app_26,tcp,8080):1
944:hacl(dmz_1,app_28,tcp,8080):1
954:hacl(dmz_
908:RULE 5 (multi-hop access):0
957:RULE 5 (multi-hop access):0
887:RULE 5 (multi-hop access):0
943:RULE 5 (multi-hop access):0
953:RULE 5
937:hacl(app_28,db_31,tcp,5432):1
936:RULE 5 (multi-hop access):0

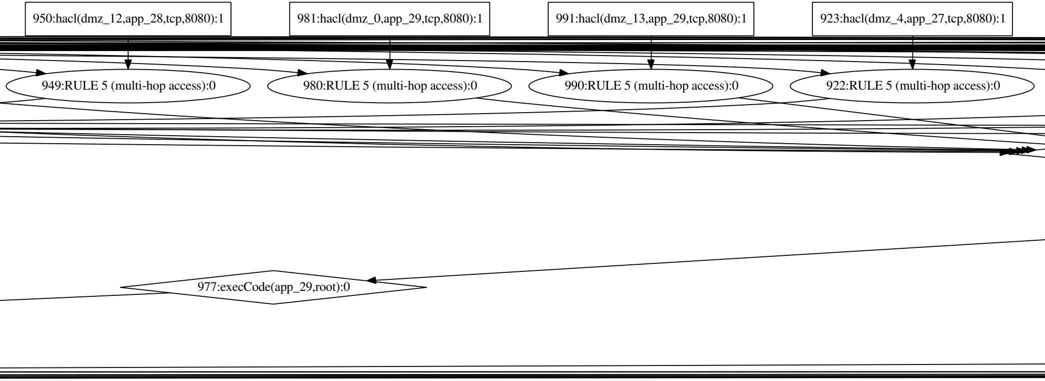

14,app_28,tcp,8080):1
960:hacl(dmz_3,app_28,tcp,8080):1
931:hacl(dmz_8,app_27,tcp,8080):1
956:hacl(dmz_15,app_28,tcp,8080):1
(multi-hop access):0
959:RULE 5 (multi-hop access):0
930:RULE 5 (multi-hop access):0
955:RULE 5 (multi-hop access):0

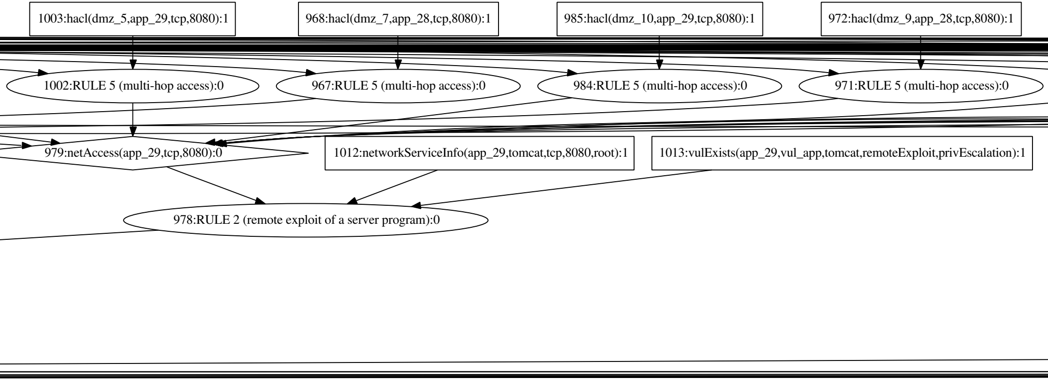

911:hacl(dmz_12,app_27,tcp,8080):1
942:hacl(dmz_0,app_28,tcp,8080):1
952:hacl(dmz_13,app_28,tcp,8080):1
884:hacl(dmz_4,app_26,tcp,8080):1
910:RULE 5 (multi-hop access):0
941:RULE 5 (multi-hop access):0
951:RULE 5 (multi-hop access):0
883:RULE 5 (multi-hop access):0
938:execCode(app_28,root):0

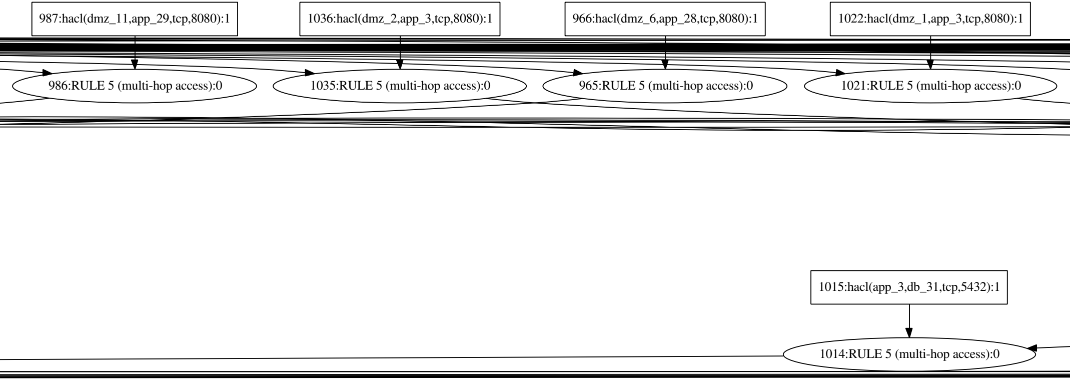

964:hacl(dmz_5,app_28,tcp,8080):1
929:hacl(dmz_7,app_27,tcp,8080):1
946:hacl(dmz_10,app_28,tcp,8080):1
933:hacl(dmz_9,app_27,tcp,8080):1
963:RULE 5 (multi-hop access):0
928:RULE 5 (multi-hop access):0
945:RULE 5 (multi-hop access):0
932:RULE 5 (multi-hop access):0
940:netAccess(app_28,tcp,8080):0
973:networkServiceInfo(app_28,tomcat,tcp,8080,root):1
974:vulExists(app_28,vul_app,tomcat,remoteExploit,privEscalation):1
939:RULE 2 (remote exploit of a server program):0

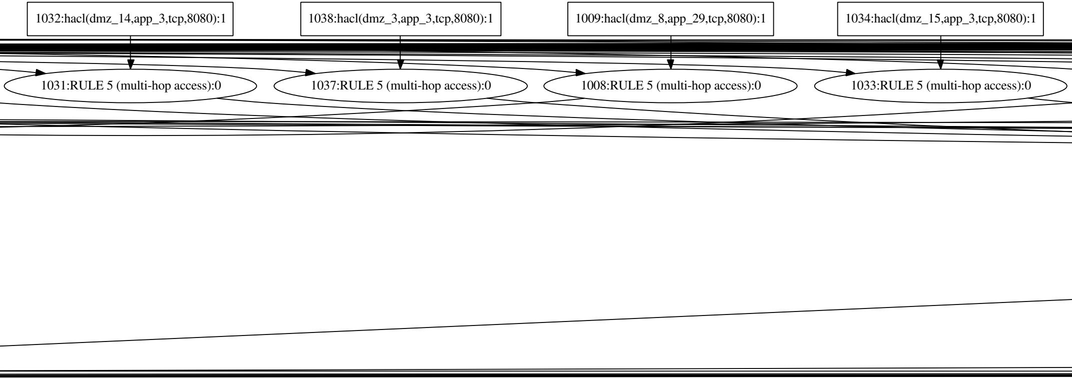

948:hacl(dmz_11,app_28,tcp,8080):1
997:hacl(dmz_2,app_29,tcp,8080):1
927:hacl(dmz_6,app_27,tcp,8080):1
983:hacl(dmz_1,app_29,tcp,8080):1
947:RULE 5 (multi-hop access):0
996:RULE 5 (multi-hop access):0
926:RULE 5 (multi-hop access):0
982:RULE 5 (multi-hop access):0
976:hacl(app_29,db_31,tcp,5432):1
975:RULE 5 (multi-hop access):0

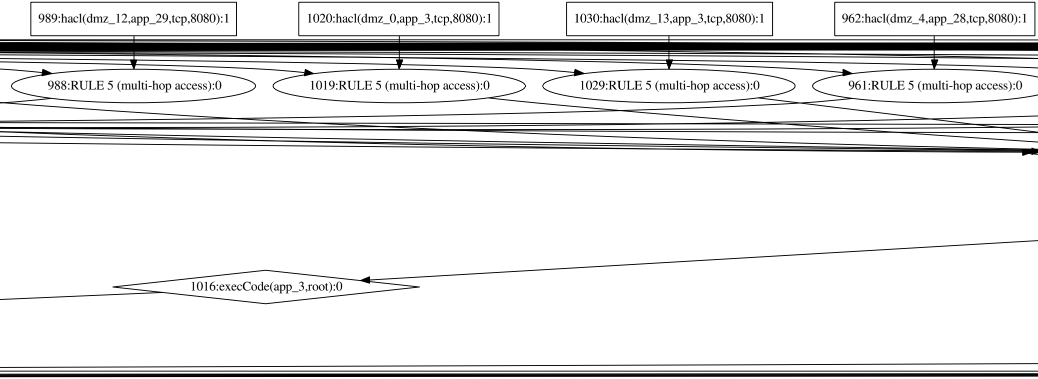

993:hacl(dmz_14,app_29,tcp,8080):1
999:hacl(dmz_3,app_29,tcp,8080):1
970:hacl(dmz_8,app_28,tcp,8080):1
995:hacl(dmz_15,app_29,tcp,8080):1
992:RULE 5 (multi-hop access):0
998:RULE 5 (multi-hop access):0
969:RULE 5 (multi-hop access):0
994:RULE 5 (multi-hop access):0

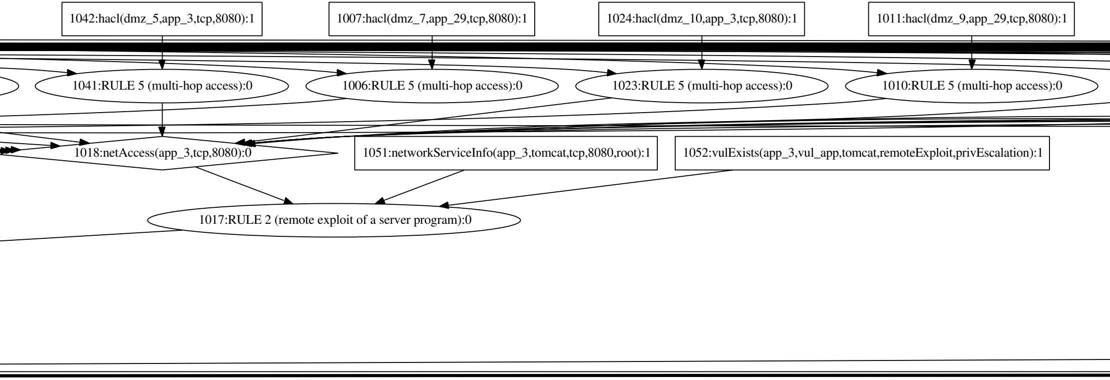

950:hacl(dmz_12,app_28,tcp,8080):1
981:hacl(dmz_0,app_29,tcp,8080):1
991:hacl(dmz_13,app_29,tcp,8080):1
923:hacl(dmz_4,app_27,tcp,8080):1
949:RULE 5 (multi-hop access):0
980:RULE 5 (multi-hop access):0
990:RULE 5 (multi-hop access):0
922:RULE 5 (multi-hop access):0
977:execCode(app_29,root):0

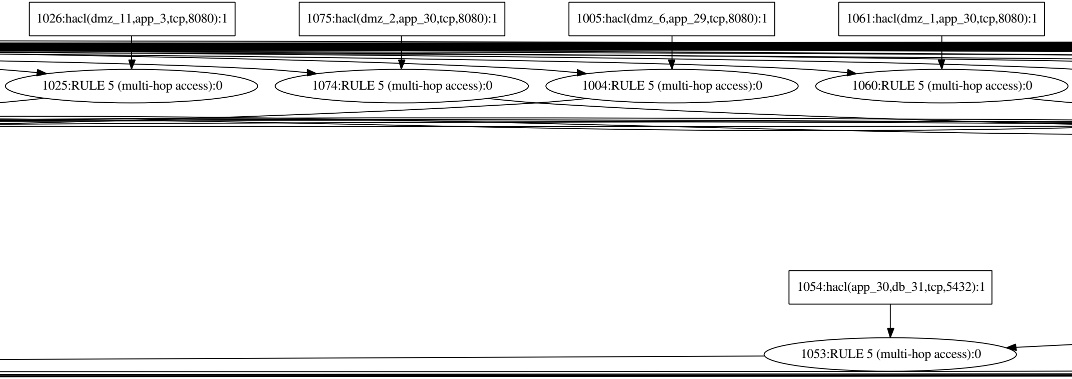

1003:hacl(dmz_5,app_29,tcp,8080):1
968:hacl(dmz_7,app_28,tcp,8080):1
985:hacl(dmz_10,app_29,tcp,8080):1
972:hacl(dmz_9,app_28,tcp,8080):1
1002:RULE 5 (multi-hop access):0
967:RULE 5 (multi-hop access):0
984:RULE 5 (multi-hop access):0
971:RULE 5 (multi-hop access):0
979:netAccess(app_29,tcp,8080):0
1012:networkServiceInfo(app_29,tomcat,tcp,8080,root):1
1013:vulExists(app_29,vul_app,tomcat,remoteExploit,privEscalation):1
978:RULE 2 (remote exploit of a server program):0

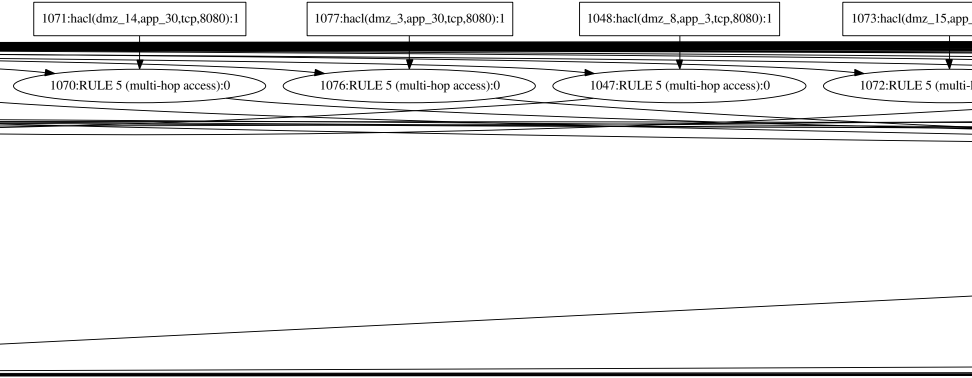

987:hacl(dmz_11,app_29,tcp,8080):1
1036:hacl(dmz_2,app_3,tcp,8080):1
966:hacl(dmz_6,app_28,tcp,8080):1
1022:hacl(dmz_1,app_3,tcp,8080):1
986:RULE 5 (multi-hop access):0
1035:RULE 5 (multi-hop access):0
965:RULE 5 (multi-hop access):0
1021:RULE 5 (multi-hop access):0
1015:hacl(app_3,db_31,tcp,5432):1
1014:RULE 5 (multi-hop access):0

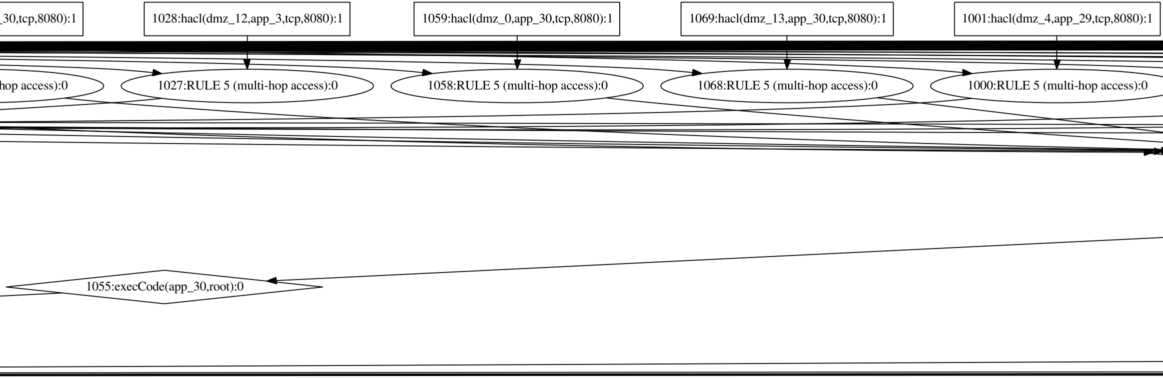

1032:hacl(dmz_14,app_3,tcp,8080):1
1038:hacl(dmz_3,app_3,tcp,8080):1
1009:hacl(dmz_8,app_29,tcp,8080):1
1034:hacl(dmz_15,app_3,tcp,8080):1
1031:RULE 5 (multi-hop access):0
1037:RULE 5 (multi-hop access):0
1008:RULE 5 (multi-hop access):0
1033:RULE 5 (multi-hop access):0

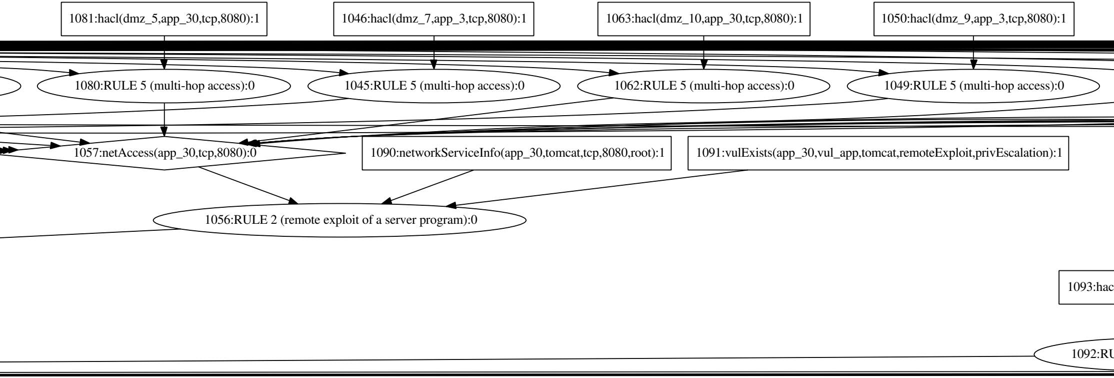

989:hacl(dmz_12,app_29,tcp,8080):1
1020:hacl(dmz_0,app_3,tcp,8080):1
1030:hacl(dmz_13,app_3,tcp,8080):1
962:hacl(dmz_4,app_28,tcp,8080):1
988:RULE 5 (multi-hop access):0
1019:RULE 5 (multi-hop access):0
1029:RULE 5 (multi-hop access):0
961:RULE 5 (multi-hop access):0
1016:execCode(app_3,root):0

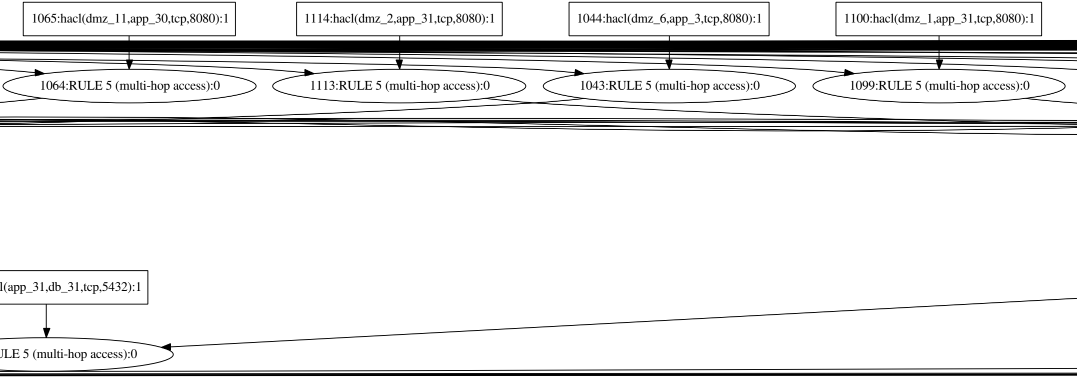

1042:hacl(dmz_5,app_3,tcp,8080):1
1007:hacl(dmz_7,app_29,tcp,8080):1
1024:hacl(dmz_10,app_3,tcp,8080):1
1011:hacl(dmz_9,app_29,tcp,8080):1
1041:RULE 5 (multi-hop access):0
1006:RULE 5 (multi-hop access):0
1023:RULE 5 (multi-hop access):0
1010:RULE 5 (multi-hop access):0
1018:netAccess(app_3,tcp,8080):0
1051:networkServiceInfo(app_3,tomcat,tcp,8080,root):1
1052:vulExists(app_3,vul_app,tomcat,remoteExploit,privEscalation):1
1017:RULE 2 (remote exploit of a server program):0

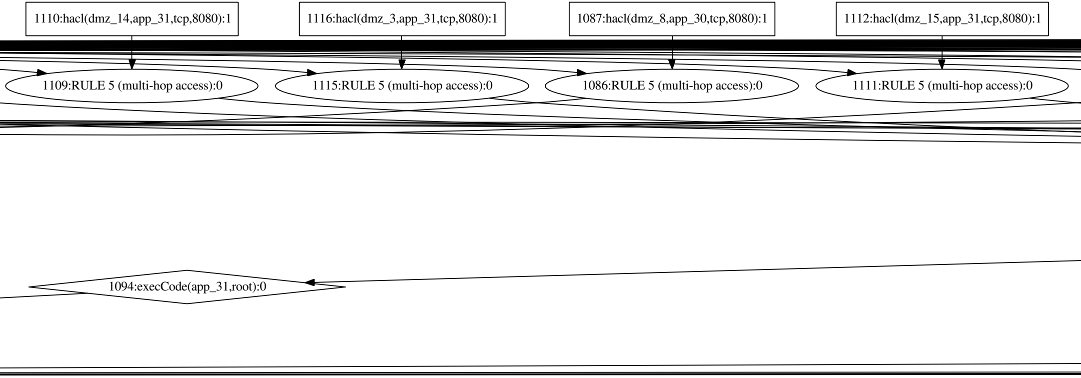

1026:hacl(dmz_11,app_3,tcp,8080):1
1075:hacl(dmz_2,app_30,tcp,8080):1
1005:hacl(dmz_6,app_29,tcp,8080):1
1061:hacl(dmz_1,app_30,tcp,8080):1
1025:RULE 5 (multi-hop access):0
1074:RULE 5 (multi-hop access):0
1004:RULE 5 (multi-hop access):0
1060:RULE 5 (multi-hop access):0
1054:hacl(app_30,db_31,tcp,5432):1
1053:RULE 5 (multi-hop access):0

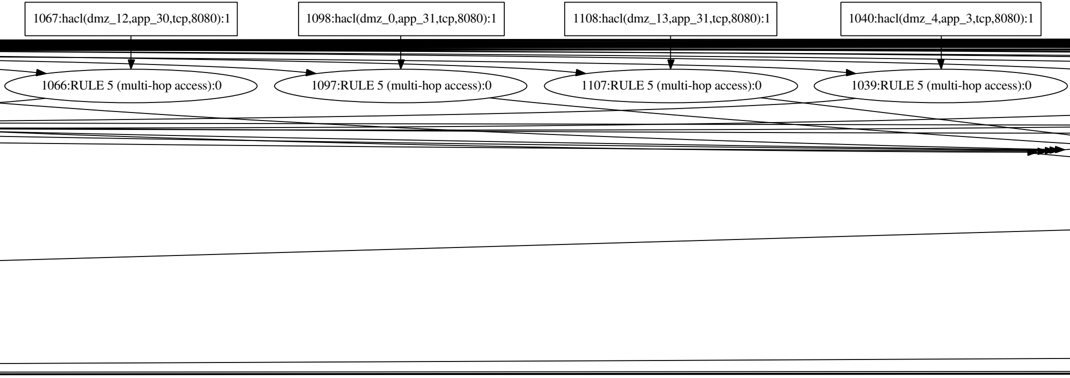

1071:hacl(dmz_14,app_30,tcp,8080):1
1077:hacl(dmz_3,app_30,tcp,8080):1
1048:hacl(dmz_8,app_3,tcp,8080):1
1073:hacl(dmz_15,app_
1070:RULE 5 (multi-hop access):0
1076:RULE 5 (multi-hop access):0
1047:RULE 5 (multi-hop access):0
1072:RULE 5 (multi-

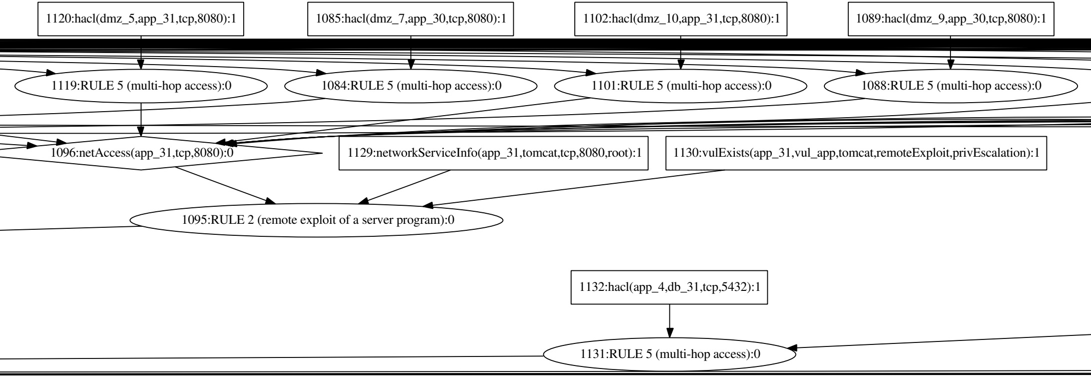

_30,tcp,8080):1
1028:hacl(dmz_12,app_3,tcp,8080):1
1059:hacl(dmz_0,app_30,tcp,8080):1
1069:hacl(dmz_13,app_30,tcp,8080):1
1001:hacl(dmz_4,app_29,tcp,8080):1
hop access):0
1027:RULE 5 (multi-hop access):0
1058:RULE 5 (multi-hop access):0
1068:RULE 5 (multi-hop access):0
1000:RULE 5 (multi-hop access):0
1055:execCode(app_30,root):0

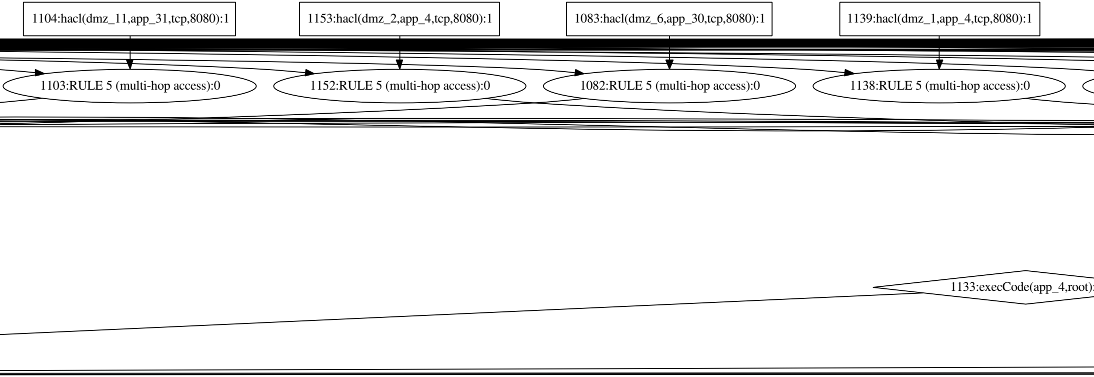

1081:hacl(dmz_5,app_30,tcp,8080):1
1046:hacl(dmz_7,app_3,tcp,8080):1
1063:hacl(dmz_10,app_30,tcp,8080):1
1050:hacl(dmz_9,app_3,tcp,8080):1
1080:RULE 5 (multi-hop access):0
1045:RULE 5 (multi-hop access):0
1062:RULE 5 (multi-hop access):0
1049:RULE 5 (multi-hop access):0
1057:netAccess(app_30,tcp,8080):0
1090:networkServiceInfo(app_30,tomcat,tcp,8080,root):1
1091:vulExists(app_30,vul_app,tomcat,remoteExploit,privEscalation):1
1056:RULE 2 (remote exploit of a server program):0
1093:hac
1092:RU

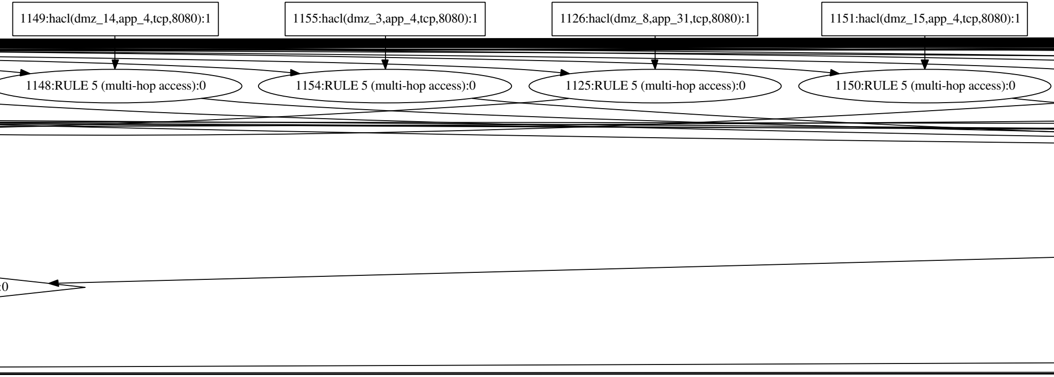

1065:hacl(dmz_11,app_30,tcp,8080):1
1114:hacl(dmz_2,app_31,tcp,8080):1
1044:hacl(dmz_6,app_3,tcp,8080):1
1100:hacl(dmz_1,app_31,tcp,8080):1
1064:RULE 5 (multi-hop access):0
1113:RULE 5 (multi-hop access):0
1043:RULE 5 (multi-hop access):0
1099:RULE 5 (multi-hop access):0
l(app_31,db_31,tcp,5432):1
ULE 5 (multi-hop access):0

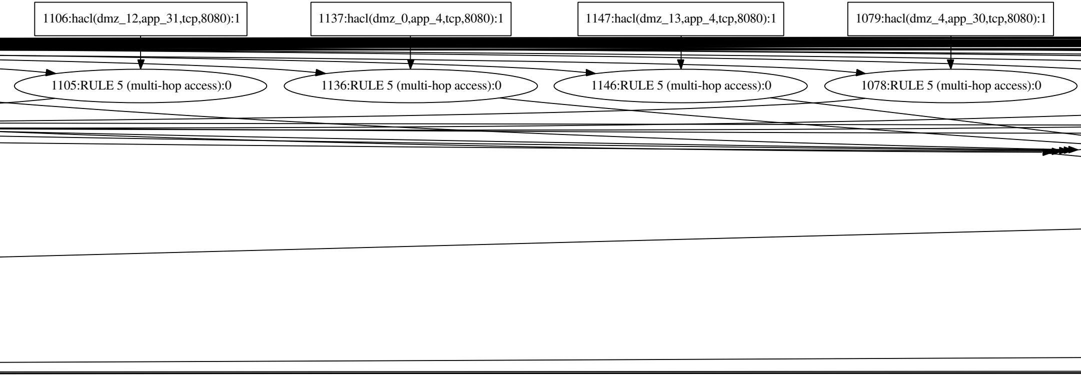

1110:hacl(dmz_14,app_31,tcp,8080):1
1116:hacl(dmz_3,app_31,tcp,8080):1
1087:hacl(dmz_8,app_30,tcp,8080):1
1112:hacl(dmz_15,app_31,tcp,8080):1
1109:RULE 5 (multi-hop access):0
1115:RULE 5 (multi-hop access):0
1086:RULE 5 (multi-hop access):0
1111:RULE 5 (multi-hop access):0
1094:execCode(app_31,root):0

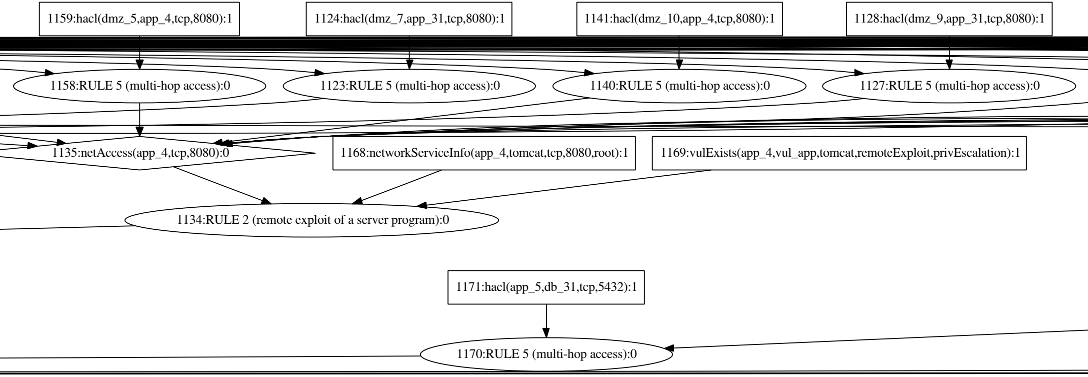

1067:hacl(dmz_12,app_30,tcp,8080):1
1098:hacl(dmz_0,app_31,tcp,8080):1
1108:hacl(dmz_13,app_31,tcp,8080):1
1040:hacl(dmz_4,app_3,tcp,8080):1
1066:RULE 5 (multi-hop access):0
1097:RULE 5 (multi-hop access):0
1107:RULE 5 (multi-hop access):0
1039:RULE 5 (multi-hop access):0

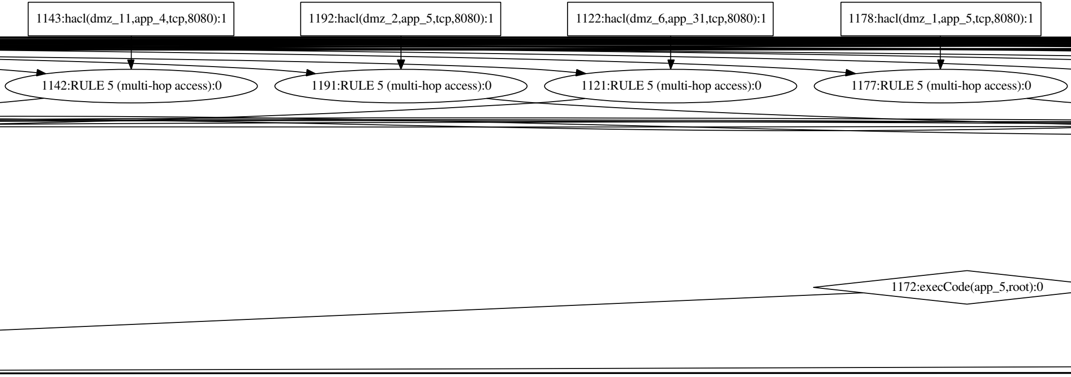

1120:hacl(dmz_5,app_31,tcp,8080):1
1085:hacl(dmz_7,app_30,tcp,8080):1
1102:hacl(dmz_10,app_31,tcp,8080):1
1089:hacl(dmz_9,app_30,tcp,8080):1
1119:RULE 5 (multi-hop access):0
1084:RULE 5 (multi-hop access):0
1101:RULE 5 (multi-hop access):0
1088:RULE 5 (multi-hop access):0
1096:netAccess(app_31,tcp,8080):0
1129:networkServiceInfo(app_31,tomcat,tcp,8080,root):1
1130:vulExists(app_31,vul_app,tomcat,remoteExploit,privEscalation):1
1095:RULE 2 (remote exploit of a server program):0
1132:hacl(app_4,db_31,tcp,5432):1
1131:RULE 5 (multi-hop access):0

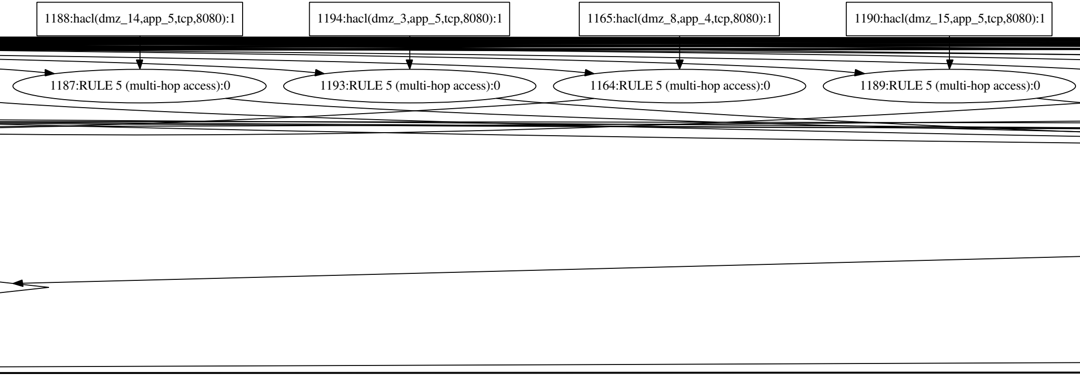

1104:hacl(dmz_11,app_31,tcp,8080):1
1153:hacl(dmz_2,app_4,tcp,8080):1
1083:hacl(dmz_6,app_30,tcp,8080):1
1139:hacl(dmz_1,app_4,tcp,8080):1
1103:RULE 5 (multi-hop access):0
1152:RULE 5 (multi-hop access):0
1082:RULE 5 (multi-hop access):0
1138:RULE 5 (multi-hop access):0
1133:execCode(app_4,root):

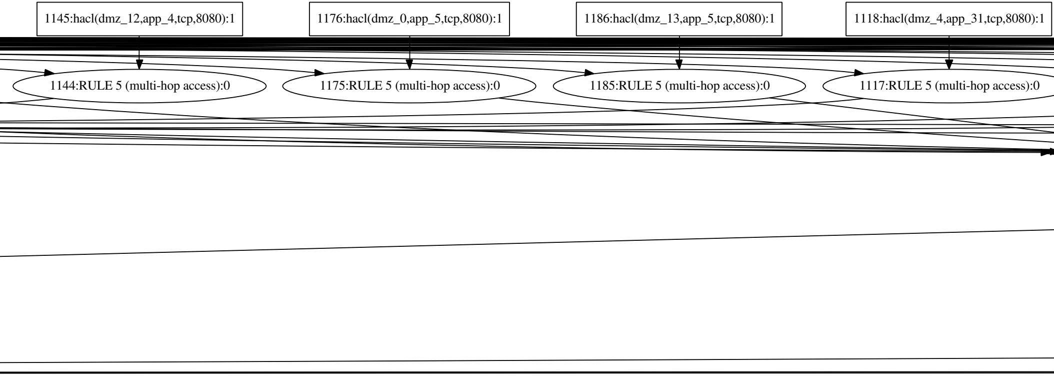

1149:hacl(dmz_14,app_4,tcp,8080):1
1155:hacl(dmz_3,app_4,tcp,8080):1
1126:hacl(dmz_8,app_31,tcp,8080):1
1151:hacl(dmz_15,app_4,tcp,8080):1
1148:RULE 5 (multi-hop access):0
1154:RULE 5 (multi-hop access):0
1125:RULE 5 (multi-hop access):0
1150:RULE 5 (multi-hop access):0
:0

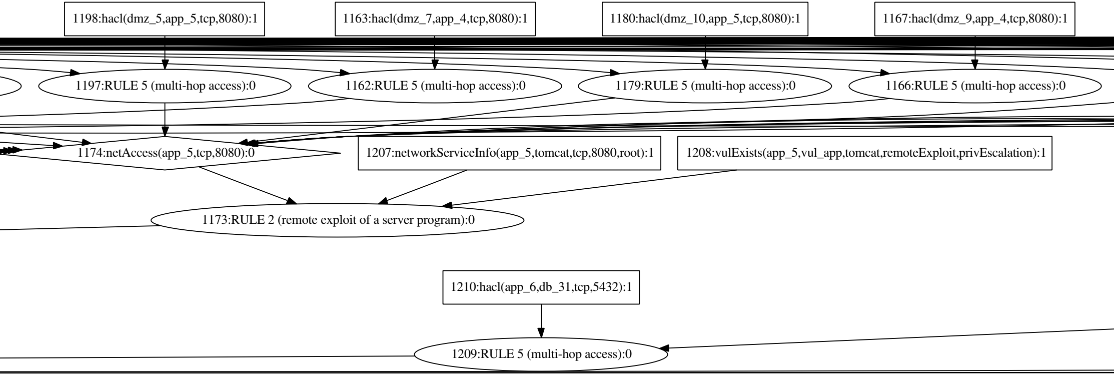

1106:hacl(dmz_12,app_31,tcp,8080):1
1137:hacl(dmz_0,app_4,tcp,8080):1
1147:hacl(dmz_13,app_4,tcp,8080):1
1079:hacl(dmz_4,app_30,tcp,8080):1
1105:RULE 5 (multi-hop access):0
1136:RULE 5 (multi-hop access):0
1146:RULE 5 (multi-hop access):0
1078:RULE 5 (multi-hop access):0

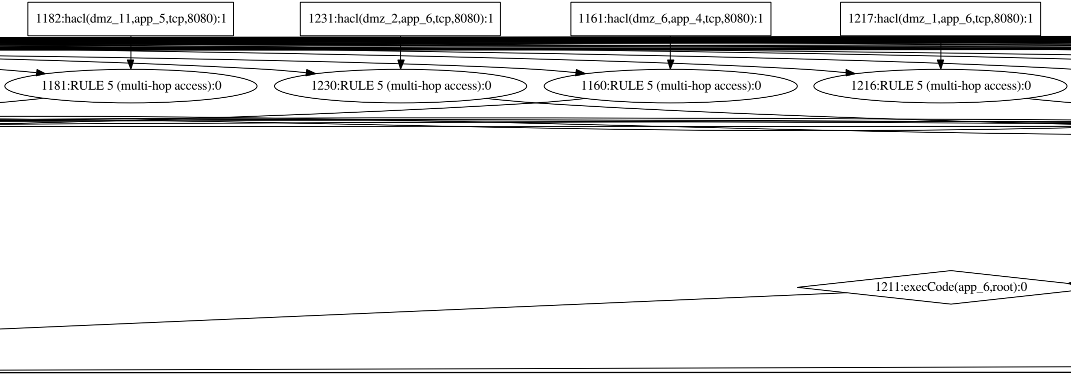

1159:hacl(dmz_5,app_4,tcp,8080):1
1124:hacl(dmz_7,app_31,tcp,8080):1
1141:hacl(dmz_10,app_4,tcp,8080):1
1128:hacl(dmz_9,app_31,tcp,8080):1
1158:RULE 5 (multi-hop access):0
1123:RULE 5 (multi-hop access):0
1140:RULE 5 (multi-hop access):0
1127:RULE 5 (multi-hop access):0
1135:netAccess(app_4,tcp,8080):0
1168:networkServiceInfo(app_4,tomcat,tcp,8080,root):1
1169:vulExists(app_4,vul_app,tomcat,remoteExploit,privEscalation):1
1134:RULE 2 (remote exploit of a server program):0
1171:hacl(app_5,db_31,tcp,5432):1
1170:RULE 5 (multi-hop access):0

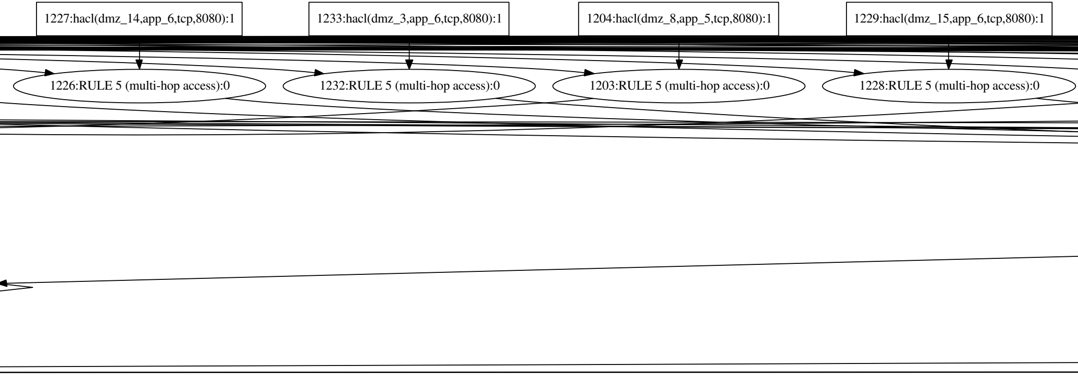

1143:hacl(dmz_11,app_4,tcp,8080):1
1192:hacl(dmz_2,app_5,tcp,8080):1
1122:hacl(dmz_6,app_31,tcp,8080):1
1178:hacl(dmz_1,app_5,tcp,8080):1
1142:RULE 5 (multi-hop access):0
1191:RULE 5 (multi-hop access):0
1121:RULE 5 (multi-hop access):0
1177:RULE 5 (multi-hop access):0
1172:execCode(app_5,root):0

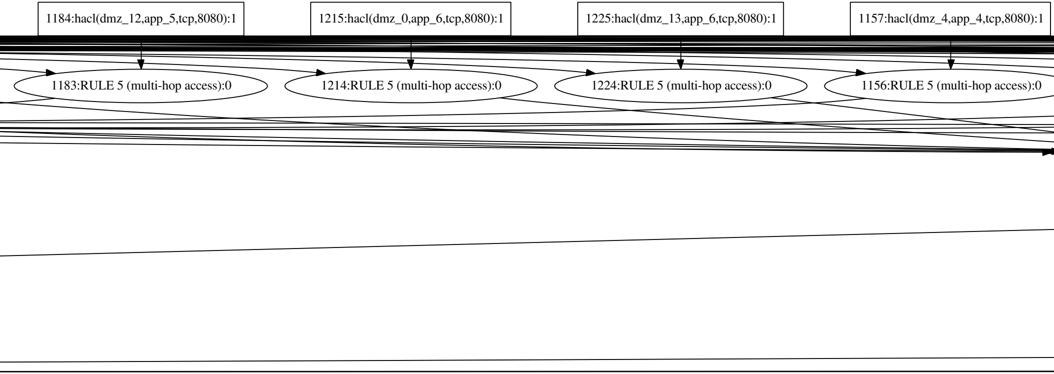

1188:hacl(dmz_14,app_5,tcp,8080):1
1194:hacl(dmz_3,app_5,tcp,8080):1
1165:hacl(dmz_8,app_4,tcp,8080):1
1190:hacl(dmz_15,app_5,tcp,8080):1
1187:RULE 5 (multi-hop access):0
1193:RULE 5 (multi-hop access):0
1164:RULE 5 (multi-hop access):0
1189:RULE 5 (multi-hop access):0

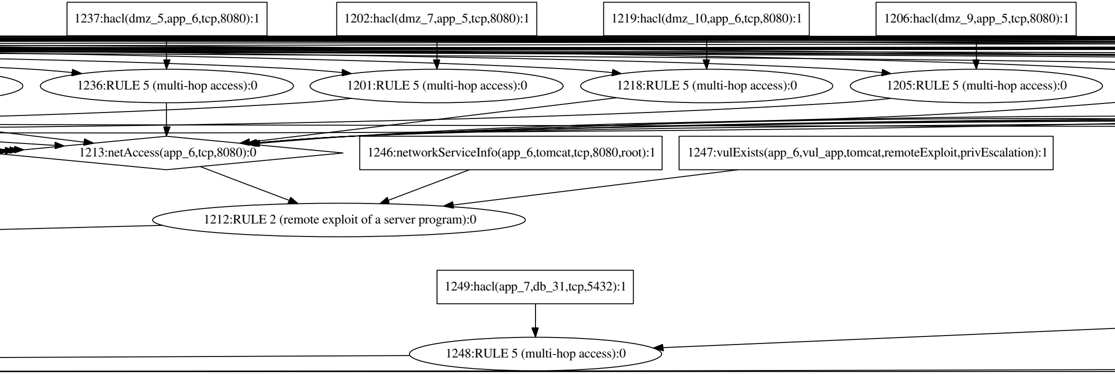

1145:hacl(dmz_12,app_4,tcp,8080):1
1176:hacl(dmz_0,app_5,tcp,8080):1
1186:hacl(dmz_13,app_5,tcp,8080):1
1118:hacl(dmz_4,app_31,tcp,8080):1
1144:RULE 5 (multi-hop access):0
1175:RULE 5 (multi-hop access):0
1185:RULE 5 (multi-hop access):0
1117:RULE 5 (multi-hop access):0

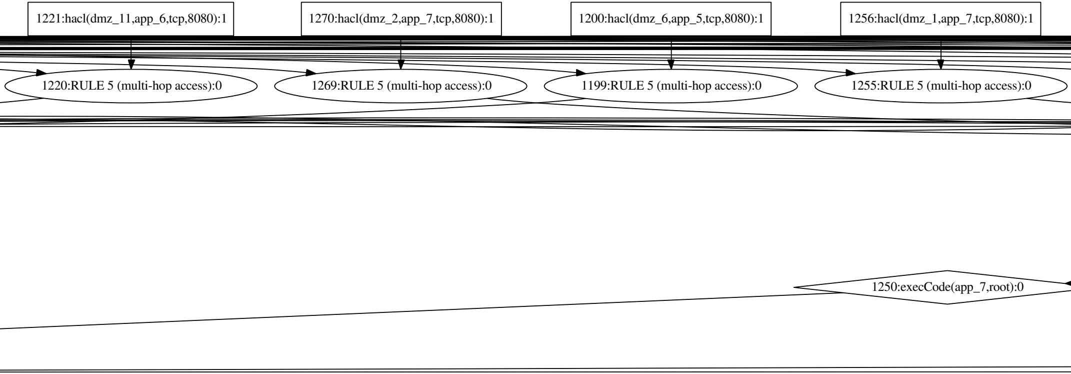

1198:hacl(dmz_5,app_5,tcp,8080):1
1163:hacl(dmz_7,app_4,tcp,8080):1
1180:hacl(dmz_10,app_5,tcp,8080):1
1167:hacl(dmz_9,app_4,tcp,8080):1
1197:RULE 5 (multi-hop access):0
1162:RULE 5 (multi-hop access):0
1179:RULE 5 (multi-hop access):0
1166:RULE 5 (multi-hop access):0
1174:netAccess(app_5,tcp,8080):0
1207:networkServiceInfo(app_5,tomcat,tcp,8080,root):1
1208:vulExists(app_5,vul_app,tomcat,remoteExploit,privEscalation):1
1173:RULE 2 (remote exploit of a server program):0
1210:hacl(app_6,db_31,tcp,5432):1
1209:RULE 5 (multi-hop access):0

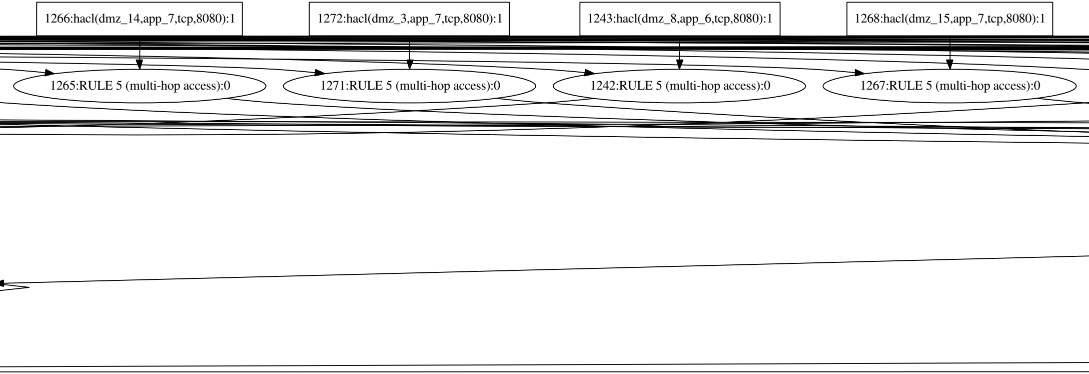

1182:hacl(dmz_11,app_5,tcp,8080):1
1231:hacl(dmz_2,app_6,tcp,8080):1
1161:hacl(dmz_6,app_4,tcp,8080):1
1217:hacl(dmz_1,app_6,tcp,8080):1
1181:RULE 5 (multi-hop access):0
1230:RULE 5 (multi-hop access):0
1160:RULE 5 (multi-hop access):0
1216:RULE 5 (multi-hop access):0
1211:execCode(app_6,root):0

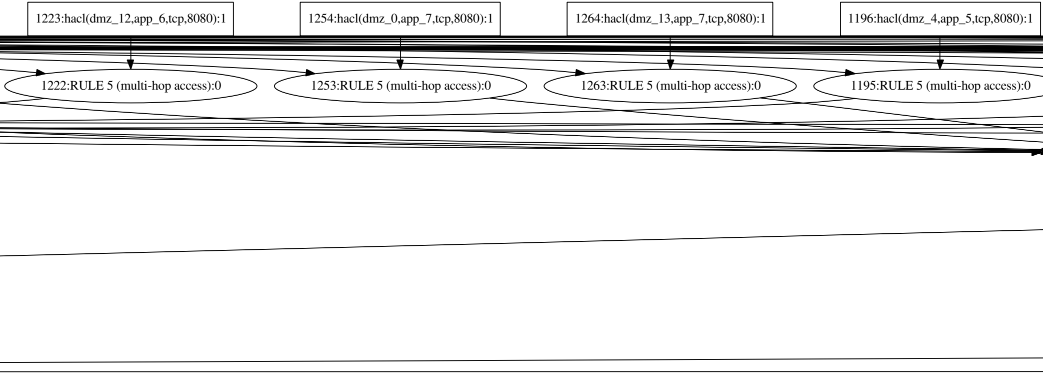

1227:hacl(dmz_14,app_6,tcp,8080):1
1233:hacl(dmz_3,app_6,tcp,8080):1
1204:hacl(dmz_8,app_5,tcp,8080):1
1229:hacl(dmz_15,app_6,tcp,8080):1
1226:RULE 5 (multi-hop access):0
1232:RULE 5 (multi-hop access):0
1203:RULE 5 (multi-hop access):0
1228:RULE 5 (multi-hop access):0

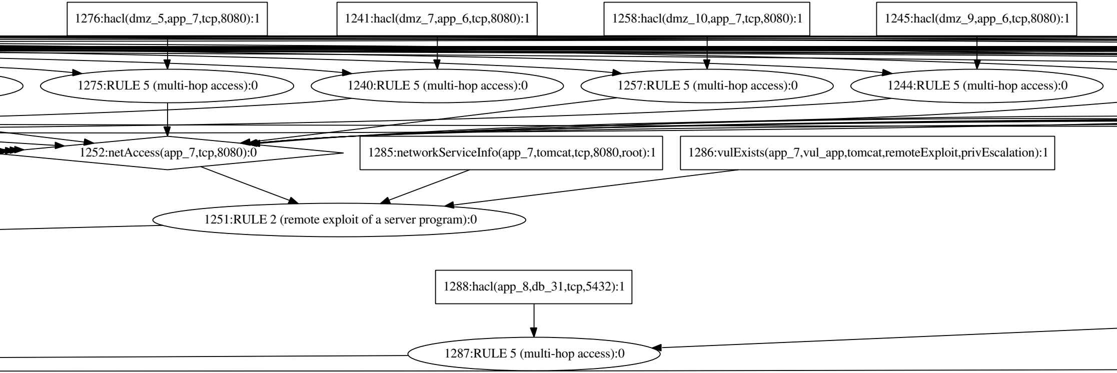

1184:hacl(dmz_12,app_5,tcp,8080):1
1215:hacl(dmz_0,app_6,tcp,8080):1
1225:hacl(dmz_13,app_6,tcp,8080):1
1157:hacl(dmz_4,app_4,tcp,8080):1
1183:RULE 5 (multi-hop access):0
1214:RULE 5 (multi-hop access):0
1224:RULE 5 (multi-hop access):0
1156:RULE 5 (multi-hop access):0

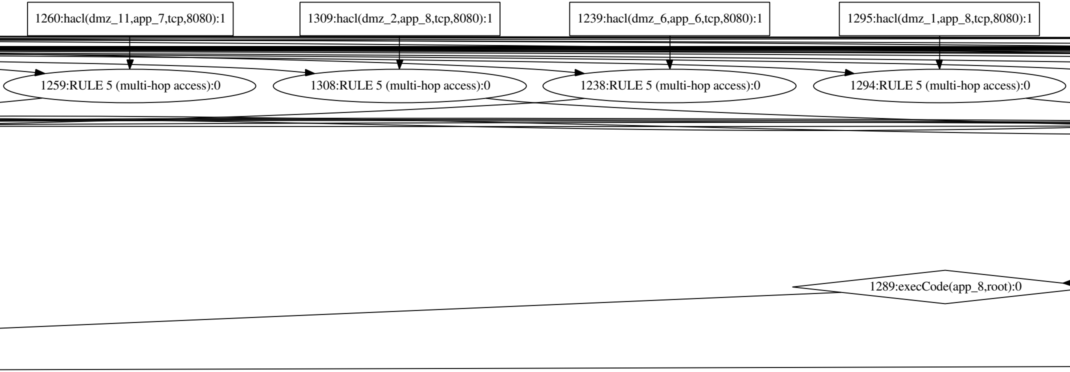

1237:hacl(dmz_5,app_6,tcp,8080):1
1202:hacl(dmz_7,app_5,tcp,8080):1
1219:hacl(dmz_10,app_6,tcp,8080):1
1206:hacl(dmz_9,app_5,tcp,8080):1
1236:RULE 5 (multi-hop access):0
1201:RULE 5 (multi-hop access):0
1218:RULE 5 (multi-hop access):0
1205:RULE 5 (multi-hop access):0
1213:netAccess(app_6,tcp,8080):0
1246:networkServiceInfo(app_6,tomcat,tcp,8080,root):1
1247:vulExists(app_6,vul_app,tomcat,remoteExploit,privEscalation):1
1212:RULE 2 (remote exploit of a server program):0
1249:hacl(app_7,db_31,tcp,5432):1
1248:RULE 5 (multi-hop access):0

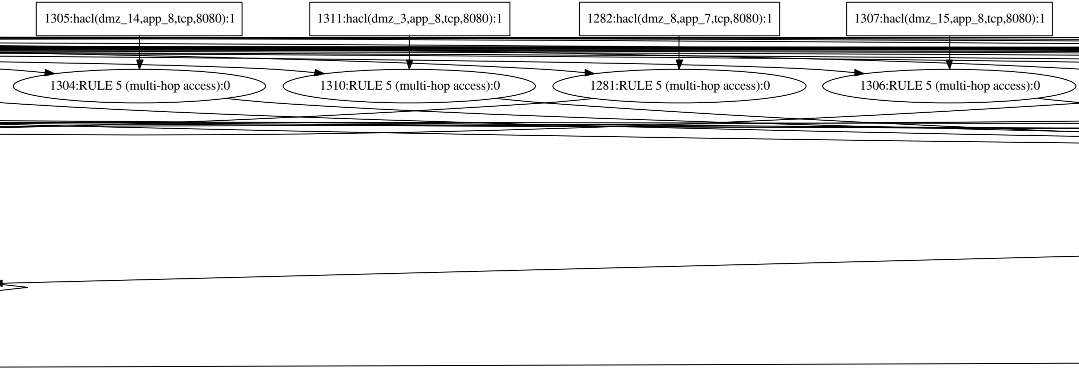

1221:hacl(dmz_11,app_6,tcp,8080):1
1270:hacl(dmz_2,app_7,tcp,8080):1
1200:hacl(dmz_6,app_5,tcp,8080):1
1256:hacl(dmz_1,app_7,tcp,8080):1
1220:RULE 5 (multi-hop access):0
1269:RULE 5 (multi-hop access):0
1199:RULE 5 (multi-hop access):0
1255:RULE 5 (multi-hop access):0
1250:execCode(app_7,root):0

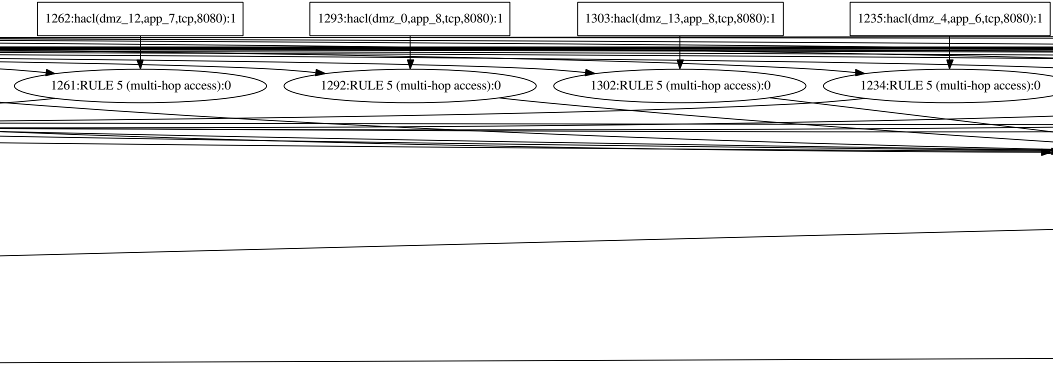

1266:hacl(dmz_14,app_7,tcp,8080):1
1272:hacl(dmz_3,app_7,tcp,8080):1
1243:hacl(dmz_8,app_6,tcp,8080):1
1268:hacl(dmz_15,app_7,tcp,8080):1
1265:RULE 5 (multi-hop access):0
1271:RULE 5 (multi-hop access):0
1242:RULE 5 (multi-hop access):0
1267:RULE 5 (multi-hop access):0

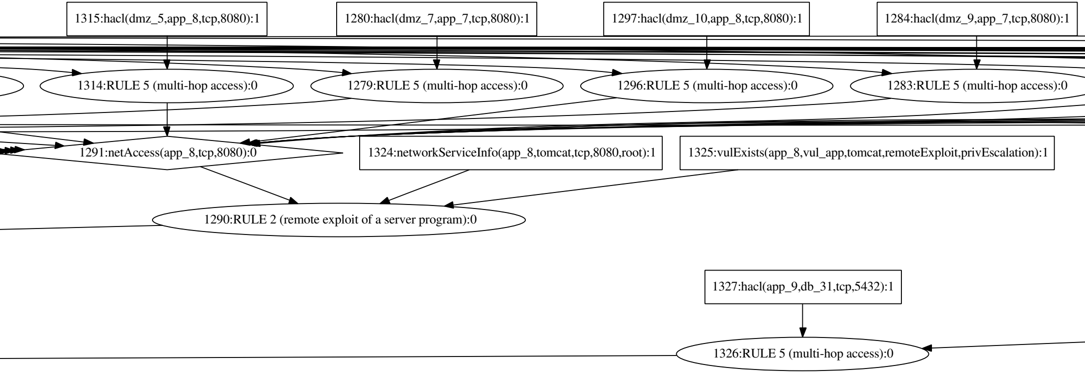

1223:hacl(dmz_12,app_6,tcp,8080):1
1254:hacl(dmz_0,app_7,tcp,8080):1
1264:hacl(dmz_13,app_7,tcp,8080):1
1196:hacl(dmz_4,app_5,tcp,8080):1
1222:RULE 5 (multi-hop access):0
1253:RULE 5 (multi-hop access):0
1263:RULE 5 (multi-hop access):0
1195:RULE 5 (multi-hop access):0

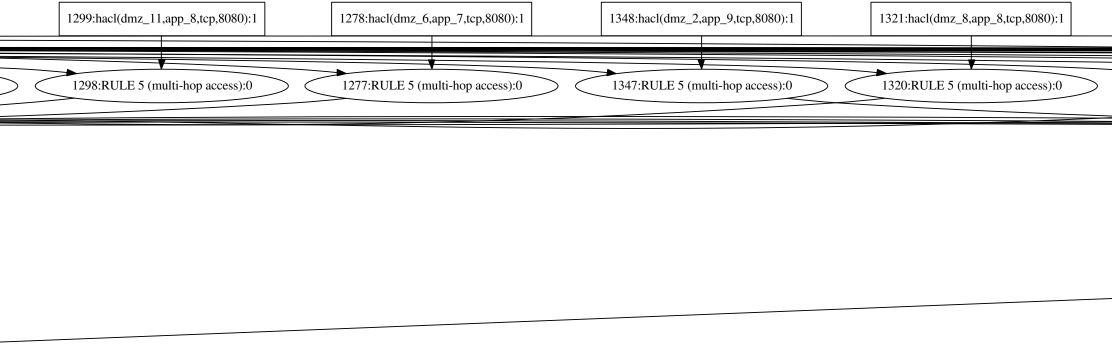

1276:hacl(dmz_5,app_7,tcp,8080):1
1241:hacl(dmz_7,app_6,tcp,8080):1
1258:hacl(dmz_10,app_7,tcp,8080):1
1245:hacl(dmz_9,app_6,tcp,8080):1
1275:RULE 5 (multi-hop access):0
1240:RULE 5 (multi-hop access):0
1257:RULE 5 (multi-hop access):0
1244:RULE 5 (multi-hop access):0
1252:netAccess(app_7,tcp,8080):0
1285:networkServiceInfo(app_7,tomcat,tcp,8080,root):1
1286:vulExists(app_7,vul_app,tomcat,remoteExploit,privEscalation):1
1251:RULE 2 (remote exploit of a server program):0
1288:hacl(app_8,db_31,tcp,5432):1
1287:RULE 5 (multi-hop access):0

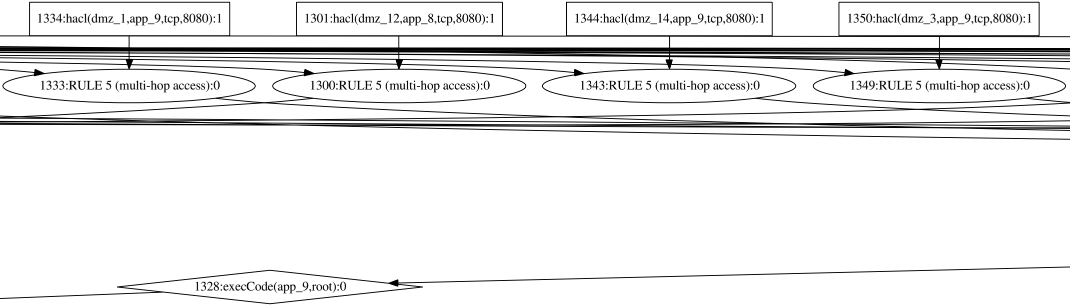

1260:hacl(dmz_11,app_7,tcp,8080):1
1309:hacl(dmz_2,app_8,tcp,8080):1
1239:hacl(dmz_6,app_6,tcp,8080):1
1295:hacl(dmz_1,app_8,tcp,8080):1
1259:RULE 5 (multi-hop access):0
1308:RULE 5 (multi-hop access):0
1238:RULE 5 (multi-hop access):0
1294:RULE 5 (multi-hop access):0
1289:execCode(app_8,root):0

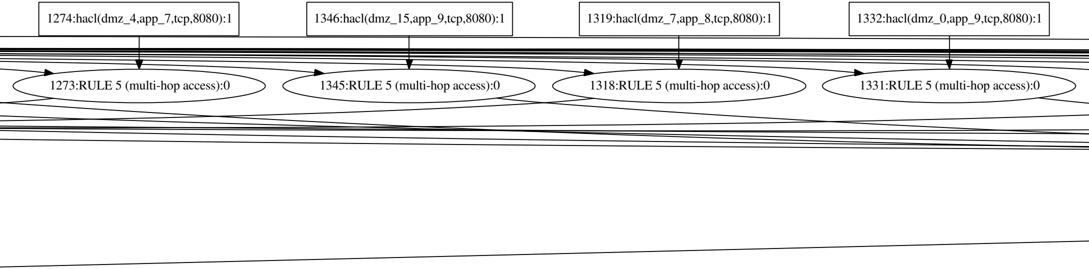

1305:hacl(dmz_14,app_8,tcp,8080):1
1311:hacl(dmz_3,app_8,tcp,8080):1
1282:hacl(dmz_8,app_7,tcp,8080):1
1307:hacl(dmz_15,app_8,tcp,8080):1
1304:RULE 5 (multi-hop access):0
1310:RULE 5 (multi-hop access):0
1281:RULE 5 (multi-hop access):0
1306:RULE 5 (multi-hop access):0

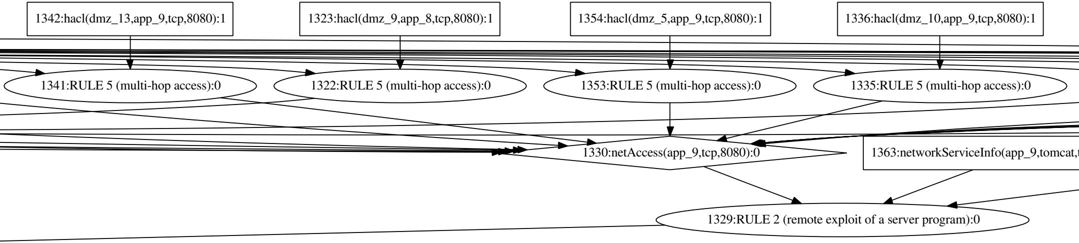

1262:hacl(dmz_12,app_7,tcp,8080):1
1293:hacl(dmz_0,app_8,tcp,8080):1
1303:hacl(dmz_13,app_8,tcp,8080):1
1235:hacl(dmz_4,app_6,tcp,8080):1
1261:RULE 5 (multi-hop access):0
1292:RULE 5 (multi-hop access):0
1302:RULE 5 (multi-hop access):0
1234:RULE 5 (multi-hop access):0

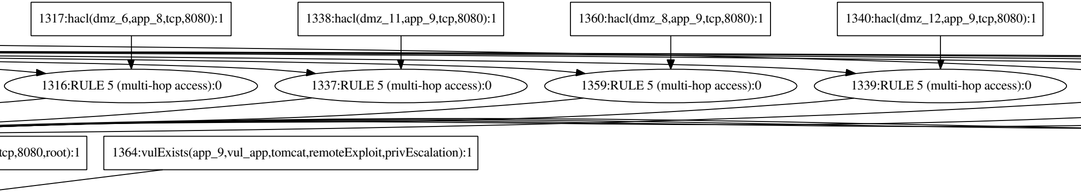

1315:hacl(dmz_5,app_8,tcp,8080):1
1280:hacl(dmz_7,app_7,tcp,8080):1
1297:hacl(dmz_10,app_8,tcp,8080):1
1284:hacl(dmz_9,app_7,tcp,8080):1
1314:RULE 5 (multi-hop access):0
1279:RULE 5 (multi-hop access):0
1296:RULE 5 (multi-hop access):0
1283:RULE 5 (multi-hop access):0
1291:netAccess(app_8,tcp,8080):0
1324:networkServiceInfo(app_8,tomcat,tcp,8080,root):1
1325:vulExists(app_8,vul_app,tomcat,remoteExploit,privEscalation):1
1290:RULE 2 (remote exploit of a server program):0
1327:hacl(app_9,db_31,tcp,5432):1
1326:RULE 5 (multi-hop access):0

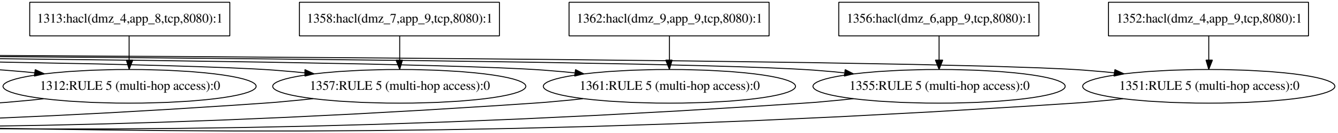

1299:hacl(dmz_11,app_8,tcp,8080):1
1278:hacl(dmz_6,app_7,tcp,8080):1
1348:hacl(dmz_2,app_9,tcp,8080):1
1321:hacl(dmz_8,app_8,tcp,8080):1
1298:RULE 5 (multi-hop access):0
1277:RULE 5 (multi-hop access):0
1347:RULE 5 (multi-hop access):0
1320:RULE 5 (multi-hop access):0


1334:hacl(dmz_1,app_9,tcp,8080):1
1301:hacl(dmz_12,app_8,tcp,8080):1
1344:hacl(dmz_14,app_9,tcp,8080):1
1350:hacl(dmz_3,app_9,tcp,8080):1
1333:RULE 5 (multi-hop access):0
1300:RULE 5 (multi-hop access):0
1343:RULE 5 (multi-hop access):0
1349:RULE 5 (multi-hop access):0
1328:execCode(app_9,root):0


1274:hacl(dmz_4,app_7,tcp,8080):1
1346:hacl(dmz_15,app_9,tcp,8080):1
1319:hacl(dmz_7,app_8,tcp,8080):1
1332:hacl(dmz_0,app_9,tcp,8080):1
1273:RULE 5 (multi-hop access):0
1345:RULE 5 (multi-hop access):0
1318:RULE 5 (multi-hop access):0
1331:RULE 5 (multi-hop access):0


1342:hacl(dmz_13,app_9,tcp,8080):1
1323:hacl(dmz_9,app_8,tcp,8080):1
1354:hacl(dmz_5,app_9,tcp,8080):1
1336:hacl(dmz_10,app_9,tcp,8080):1
1341:RULE 5 (multi-hop access):0
1322:RULE 5 (multi-hop access):0
1353:RULE 5 (multi-hop access):0
1335:RULE 5 (multi-hop access):0
1330:netAccess(app_9,tcp,8080):0
1363:networkServiceInfo(app_9,tomcat,t
1329:RULE 2 (remote exploit of a server program):0


1317:hacl(dmz_6,app_8,tcp,8080):1
1338:hacl(dmz_11,app_9,tcp,8080):1
1360:hacl(dmz_8,app_9,tcp,8080):1
1340:hacl(dmz_12,app_9,tcp,8080):1
1316:RULE 5 (multi-hop access):0
1337:RULE 5 (multi-hop access):0
1359:RULE 5 (multi-hop access):0
1339:RULE 5 (multi-hop access):0
tcp,8080,root):1
1364:vulExists(app_9,vul_app,tomcat,remoteExploit,privEscalation):1


1313:hacl(dmz_4,app_8,tcp,8080):1
1358:hacl(dmz_7,app_9,tcp,8080):1
1362:hacl(dmz_9,app_9,tcp,8080):1
1356:hacl(dmz_6,app_9,tcp,8080):1
1352:hacl(dmz_4,app_9,tcp,8080):1
1312:RULE 5 (multi-hop access):0
1357:RULE 5 (multi-hop access):0
1361:RULE 5 (multi-hop access):0
1355:RULE 5 (multi-hop access):0
1351:RULE 5 (multi-hop access):0